# 1.

## ANALYTICAL DEVELOPMENT OF FRESNEL'S OPTICAL THEORY OF CRYSTALS.

[*Philosophical Magazine*, XI. (1837), pp. 461—469, 537—541; XII. (1838), pp. 73—83, 341—345.]

THE following is, I believe, the first successful attempt to obtain the full development of Fresnel's Theory of Crystals by direct geometrical methods. Hitherto little has been done beyond finding and investigating the properties of the wave surface, a subject certainly curious and interesting, but not of chief importance for ordinary practical purposes. Mr Kelland, in a most valuable contribution to the *Cambridge Philosophical Transactions**, has incidentally obtained the difference of the squares of the velocities of a plane front in terms of the angles made by it with the optic axes. I have obtained each of the velocities *separately*, and in a form precisely the same for biaxal as for uniaxal crystals.

I have also assigned in my last proposition the place of the lines of vibration in terms of the like quantities, and *that* in a shape remarkably convenient for determining the *plane* of polarization when the ray is given. For at first sight there appears to be some ambiguity in selecting *which* of the *two* lines of vibration is to be chosen when the front is known. If $p$ be the perpendicular from the centre of the surface of elasticity let fall upon the front, $\iota_1$, $\iota_2$ the angles made by the front with the optic planes, $\epsilon_1$, $\epsilon_2$ the angles between its *due* line of vibration and the optic axes, I have shown that

$$\cos\epsilon_1 = \sqrt{\left(\frac{b^2-p^2}{a^2-c^2}\cdot\frac{\sin\iota_1}{\sin\iota_2}\right)},\quad \cos\epsilon_2 = \sqrt{\left(\frac{b^2-p^2}{a^2-c^2}\cdot\frac{\sin\iota_2}{\sin\iota_1}\right)},$$

so that all doubt is completely removed. The equation preparatory to obtaining the wave surface is found in Prop. 6 by common algebra, without any use of the properties of maxima and minima, and various other curious relations are discussed.

Without the most careful attention to preserve pure symmetry, the expressions could never have been reduced to their present simple forms.

* See *Lond. and Edinb. Phil. Mag.* Vol. x. p. 336.

## Analytical Reduction of Fresnel's Optical Theory of Crystals.

*Index of Contents.*

In Proposition 1, a plane front within a crystal being given, the two lines of vibration are investigated.

In Proposition 2 it is shown that the product of the cosines of the inclinations of one of the axes of elasticity to the two lines of vibration, is to the same for either other axis of elasticity in a constant ratio for the same crystal; and the two lines of vibration are proved to be perpendicular to each other.

In Proposition 3, a line of vibration being given, the front to which it belongs is determined; and it is proved that there is only one such, and consequently any line of vibration has but one other line conjugate to it.

In Proposition 4, certain relations are instituted between the positions of, and velocities due to, conjugate lines.

In Proposition 5, the angles made by the front with the planes of elasticity are found in terms of the velocities only.

In Proposition 6, the above is reversed.

In Proposition 7, the position of the planes in which the two velocities are equal (viz. the optic planes) is determined.

In Proposition 8, the position of a front in respect to the optic axes is expressed in terms of the velocities.

In Proposition 9, the problem is reversed, and it is shown that if $v_1$, $v_2$ be the two normal velocities with which any front can move perpendicular to itself, and $\iota_1$, $\iota_2$ the angles which it makes with the optic planes, then

$$v_1^2 = a^2 \left(\sin \frac{\iota_1 + \iota_2}{2}\right)^2 + c^2 \left(\cos \frac{\iota_1 + \iota_2}{2}\right)^2,$$

$$v_2^2 = a^2 \left(\sin \frac{\iota_1 - \iota_2}{2}\right)^2 + c^2 \left(\cos \frac{\iota_1 - \iota_2}{2}\right)^2.$$

In the 10th the angle made by a line of vibration with the axes of elasticity is expressed in terms of the two velocities of the front to which it belongs.

In the 11th Proposition the velocity due to any line of vibration is expressed in terms of the angles which it makes with the optic axes, viz.

$$v^2 - b^2 = (a^2 - c^2) \cos \epsilon_1 \cos \epsilon_2.$$

In the 12th Proposition $\epsilon_1$, $\epsilon_2$ are separately expressed in terms of $\iota_1$, $\iota_2$.

In the Appendix I have given the polar or rather radio-angular equation to the wave surface, from which the celebrated proposition of the ray flows as an immediate consequence.

PROPOSITION 1.

*If* $$lx + my + nz = 0 \qquad (a)$$

*be the equation to a given front, to determine the lines of vibration therein.*

It is clear that if $x, y, z$ be any point in one of these lines, the force acting on a particle placed there when resolved into the plane must tend to the centre. Consequently the line of force at $x, y, z$ must meet the perpendicular drawn upon the front from the origin. Now the equation to this perpendicular is

$$\frac{X}{l} = \frac{Y}{m} = \frac{Z}{n} \qquad (1)$$

and the forces acting at $x, y, z$ are $a^2x, b^2y, c^2z$ parallel to $x, y, z$, so that the equation to the line of force is

$$\frac{X-x}{a^2x} = \frac{Y-y}{b^2y} = \frac{Z-z}{c^2z}. \qquad (2)$$

From (2) we obtain

$$b^2yX - a^2xY = (b^2 - a^2)\,xy \qquad (3)$$

$$c^2zY - b^2yZ = (c^2 - b^2)\,yz \qquad (4)$$

$$a^2xZ - c^2zX = (a^2 - c^2)\,zx. \qquad (5)$$

Hence

$$(b^2 - a^2)\,xyn + (c^2 - b^2)\,yzl + (a^2 - c^2)\,zxm$$
$$= b^2y\,(nX - lZ) + c^2z\,(lY - mX) + a^2x\,(mZ - nY);$$

but by equations (1)

$$lZ - nX = 0,\quad mX - lY = 0,\quad nY - mZ = 0$$

therefore

$$(b^2 - a^2)\,\frac{n}{z} + (c^2 - b^2)\,\frac{l}{x} + (a^2 - c^2)\,\frac{m}{y} = 0. \qquad (b)$$

Also we have

$$nz + lx + my = 0 \qquad (a)$$

therefore

$$(b^2 - a^2)\,n^2 + (c^2 - b^2)\,l^2 + nl\left((c^2 - b^2)\,\frac{z}{x} + (b^2 - a^2)\,\frac{x}{z}\right) = (a^2 - c^2)\,m^2$$

or

$$(c^2 - b^2)\left(\frac{z}{x}\right)^2 + \frac{1}{nl}\,\{(c^2 - b^2)\,l^2 + (b^2 - a^2)\,n^2 - (a^2 - c^2)\,m^2\}\,\frac{z}{x} + (b^2 - a^2) = 0.$$

And in like manner interchanging $b, y, m$ with $c, z, n$

$$(b^2 - c^2)\left(\frac{y}{x}\right)^2 + \frac{1}{ml}\,\{(b^2 - c^2)\,l^2 + (c^2 - a^2)\,m^2 - (a^2 - b^2)\,n^2\}\,\frac{y}{x} + (c^2 - a^2) = 0.$$

Hence if $\left(\frac{y_1}{x_1}, \frac{z_1}{x_1}\right)\left(\frac{y_2}{x_2}, \frac{z_2}{x_2}\right)$ be the two systems of values of $\frac{y}{x}, \frac{z}{x}$, then

$$\left(\frac{Y}{X}=\frac{y_1}{x_1},\ \frac{Z}{X}=\frac{z_1}{x_1}\right)\left(\frac{Y}{X}=\frac{y_2}{x_2},\ \frac{Z}{X}=\frac{z_2}{x_2}\right)$$

are the two lines of vibration required.

PROPOSITION 2.

By last proposition it appears that

$$\frac{y_1y_2}{x_1x_2}=\frac{c^2-a^2}{b^2-c^2} \qquad (c)$$

and

$$\frac{z_1z_2}{x_1x_2}=\frac{b^2-a^2}{c^2-b^2} \qquad (d)$$

therefore

$$\frac{y_1y_2+z_1z_2}{x_1x_2}=\frac{c^2-b^2}{b^2-c^2}=-1$$

therefore

$$x_1x_2+y_1y_2+z_1z_2=0.$$

And therefore the two lines of vibration are perpendicular to each other.

N.B. Equations $(c)$ and $(d)$ must not be overlooked.

PROPOSITION 3.

*A line of vibration is given* $\left(\textit{that is } \frac{y_1}{x_1}, \frac{z_1}{x_1} \textit{ are given}\right)$ *and the position of the front is to be determined.*

Let $lx+my+nz=0$ be the front required, then $lx_1+my_1+nz_1=0$, and

$$(b^2-c^2)\frac{l}{x_1}+(c^2-a^2)\frac{m}{y_1}+(a^2-b^2)\frac{n}{z_1}=0.$$

Eliminating $n$ we get

$$l\left((a^2-b^2)\frac{x_1}{z_1}-(b^2-c^2)\frac{z_1}{x_1}\right)+m\left((a^2-b^2)\frac{y_1}{z_1}-(c^2-a^2)\frac{z_1}{y_1}\right)=0$$

therefore

$$\frac{l}{m}=\frac{x_1}{y_1}\frac{(a^2-b^2)\,y_1^2-(c^2-a^2)\,z_1^2}{(b^2-c^2)\,z_1^2-(a^2-b^2)\,x_1^2}$$

$$=\frac{x_1}{y_1}\frac{a^2(x_1^2+y_1^2+z_1^2)-(a^2x_1^2+b^2y_1^2+c^2z_1^2)}{b^2(x_1^2+y_1^2+z_1^2)-(a^2x_1^2+b^2y_1^2+c^2z_1^2)}.$$

If now we make $x_1^2 + y_1^2 + z_1^2 = 1$

$$a^2x_1^2 + b^2y_1^2 + c^2z_1^2 = v_1^2$$

and therefore

$$\frac{l}{m} = \frac{x_1}{y_1} \cdot \frac{a^2 - v_1^2}{b^2 - v_1^2}$$

and in like manner

$$\frac{l}{n} = \frac{x_1}{z_1} \cdot \frac{a^2 - v_1^2}{c^2 - v_1^2};$$

therefore

$$(a^2 - v_1^2)\, x_1x + (b^2 - v_1^2)\, y_1y + (c^2 - v_1^2)\, z_1z = 0$$

is the equation required.

## Proposition 4.

$\frac{l}{m}, \frac{l}{n}$ having each only one value, shows that only one front corresponds to the given line of vibration. Let $x_2, y_2, z_2, v_2$ correspond to $x_1, y_1, z_1, v_1$ for the conjugate line of vibration, then the equation to the front may be expressed likewise by

$$(a^2 - v_2^2)\, x_2x + (b^2 - v_2^2)\, y_2y + (c^2 - v_2^2)\, z_2z = 0,$$

so that

$$\frac{(a^2 - v_1^2)\, x_1}{(a^2 - v_2^2)\, x_2} = \frac{(b^2 - v_1^2)\, y_1}{(b^2 - v_2^2)\, y_2} = \frac{(c^2 - v_1^2)\, z_1}{(c^2 - v_2^2)\, z_2}.$$

## Proposition 5.

*To find $\omega$, $\phi$, $\psi$, the angles made by the front with the planes of elasticity in terms of $v_1$, $v_2$.*

By the last proposition

$$(\cos\omega)^2 = \frac{(a^2 - v_1^2)^2\, x_1^2}{(a^2 - v_1^2)^2\, x_1^2 + (b^2 - v_1^2)^2\, y_1^2 + (c^2 - v_1^2)^2\, z_1^2}$$

$$= \frac{(a^2 - v_1^2)(a^2 - v_2^2)\, x_1x_2}{(a^2 - v_1^2)(a^2 - v_2^2)\, x_1x_2 + (b^2 - v_1^2)(b^2 - v_2^2)\, y_1y_2 + (c^2 - v_1^2)(c^2 - v_2^2)\, z_1z_2}.$$

Now, by Proposition 2,

$$\frac{x_1x_2}{c^2 - b^2} = \frac{y_1y_2}{a^2 - c^2} = \frac{z_1z_2}{b^2 - a^2}$$

therefore $(\cos\omega)^2$

$$=\frac{(a^2-v_1^2)(a^2-v_2^2)(c^2-b^2)}{(a^2-v_1^2)(a^2-v_2^2)(c^2-b^2)+(b^2-v_1^2)(b^2-v_2^2)(a^2-c^2)+(c^2-v_1^2)(c^2-v_2^2)(b^2-a^2)}$$

$$=\frac{(a^2-v_1^2)(a^2-v_2^2)(c^2-b^2)}{a^4(c^2-b^2)+b^4(a^2-c^2)+c^4(a^2-b^2)}$$

$$=\frac{(a^2-v_1^2)(a^2-v_2^2)}{(a^2-b^2)(a^2-c^2)}.$$

Similarly,

$$(\cos\phi)^2=\frac{(b^2-v_1^2)(b^2-v_2^2)}{(b^2-a^2)(b^2-c^2)},$$

$$(\cos\psi)^2=\frac{(c^2-v_1^2)(c^2-v_2^2)}{(c^2-a^2)(c^2-b^2)}.$$

## Proposition 6.

*To find $v_1$, $v_2$ in terms of $\omega$, $\phi$, $\psi$.*

By the last proposition

$$\frac{(\cos\omega)^2}{a^2-v_1^2}=\frac{a^2}{(a^2-b^2)(a^2-c^2)}-v_2^2\cdot\frac{1}{(a^2-b^2)(a^2-c^2)}$$

$$\frac{(\cos\phi)^2}{b^2-v_1^2}=\frac{b^2}{(b^2-a^2)(b^2-c^2)}-v_2^2\cdot\frac{1}{(b^2-a^2)(b^2-c^2)}$$

$$\frac{(\cos\psi)^2}{c^2-v_1^2}=\frac{c^2}{(c^2-b^2)(c^2-a^2)}-v_2^2\cdot\frac{1}{(c^2-a^2)(c^2-b^2)}$$

therefore

$$\frac{(\cos\omega)^2}{a^2-v_1^2}+\frac{(\cos\phi)^2}{b^2-v_1^2}+\frac{(\cos\psi)^2}{c^2-v_1^2}=0.$$

Just in the same way

$$\frac{(\cos\omega)^2}{a^2-v_2^2}+\frac{(\cos\phi)^2}{b^2-v_2^2}+\frac{(\cos\psi)^2}{c^2-v_2^2}=0,$$

so that $v_1^2$, $v_2^2$ are the two roots of the equation

$$\frac{(\cos\omega)^2}{a^2-v^2}+\frac{(\cos\phi)^2}{b^2-v^2}+\frac{(\cos\psi)^2}{c^2-v^2}=0.$$

Cor. Hence the equation to the wave surface may be obtained by making

$$(\cos\omega)\,x+(\cos\phi)\,y+(\cos\psi)\,z=v,$$

or if we please to apply Prop. 5, we may make

$$\sqrt{\frac{(a^2-v_1^2)(a^2-v_2^2)}{(a^2-b^2)(a^2-c^2)}}\cdot x+\sqrt{\frac{(b^2-v_1^2)(b^2-v_2^2)}{(b^2-a^2)(b^2-c^2)}}\cdot y$$
$$+\sqrt{\frac{(c^2-v_1^2)(c^2-v_2^2)}{(c^2-a^2)(c^2-b^2)}}\cdot z=v_1,$$

or, if we please*,

$$\sqrt{\frac{(a^2u^2-1)(a^2-v^2)}{(a^2-b^2)(a^2-c^2)}}\cdot x+\sqrt{\frac{(b^2u^2-1)(b^2-v^2)}{(b^2-a^2)(b^2-c^2)}}\cdot y$$
$$+\sqrt{\frac{(c^2u^2-1)(c^2-v^2)}{(c^2-a^2)(c^2-b^2)}}\cdot z=1.$$

PROPOSITION 7.

*To find when* $v_1=v_2$.

By Prop. 4,

$$\frac{x_1(v_1^2-a^2)}{x_2(v_2^2-a^2)}=\frac{y_1(v_1^2-b^2)}{y_2(v_2^2-b^2)}=\frac{z_1(v_1^2-c^2)}{z_2(v_2^2-c^2)}. \qquad (\theta)$$

Hence when $v_1=v_2$ we have, generally speaking,

$$\frac{x_1}{x_2}=\frac{y_1}{y_2}=\frac{z_1}{z_2}.$$

Now $$x_1x_2+y_1y_2+z_1z_2=0;$$

therefore $x_1^2+y_1^2+z_1^2$ would $=0$, which is absurd.

The only case therefore when $v_1$ can $=v_2$ is when one of those terms of equation $(\theta)$ becomes $\frac{0}{0}$: thus suppose $v_1=b$, then we have $\frac{x_1}{x_2}=\frac{z_1}{z_2}=\frac{0}{0}$, and we can no longer infer $\frac{x_1}{x_2}=\frac{y_1}{y_2}$.

Let now $(\omega_1, \phi_1, \psi_1)(\omega_2, \phi_2, \psi_2)$ be the two systems of values which $\omega$, $\phi$, $\psi$ assume when $v_1=v_2=b$, then applying the equation of Prop. 5 we have

$$\cos\omega_1=\sqrt{\frac{a^2-b^2}{a^2-c^2}} \qquad \cos\omega_2=\sqrt{\frac{a^2-b^2}{a^2-c^2}}$$
$$\cos\phi_1=0 \qquad \cos\phi_2=0$$
$$\cos\psi_1=\sqrt{\frac{b^2-c^2}{a^2-c^2}} \qquad \cos\psi_2=\sqrt{\frac{b^2-c^2}{a^2-c^2}},$$

so that $b$ must correspond to the mean axis.

[* See below, p. 27. ED.]

### Proposition 8.

*$\iota_1$, $\iota_2$ being the angles made by the front with the optic planes, to find $\iota_1$, $\iota_2$ in terms of $v_1$, $v_2$.*

By analytical geometry

$$\cos\iota_1 = \cos\omega . \cos\omega_1 + \cos\phi . \cos\phi_1 + \cos\psi . \cos\psi_1$$

$$= \sqrt{\frac{(v_1^2-a^2)(v_2^2-a^2)}{(a^2-b^2)(a^2-c^2)}} . \sqrt{\frac{a^2-b^2}{a^2-c^2}}$$

$$+ \sqrt{\frac{(v_1^2-c^2)(v_2^2-c^2)}{(c^2-a^2)(c^2-b^2)}} . \sqrt{\frac{c^2-b^2}{c^2-a^2}}$$

$$= \frac{\sqrt{\{(v_1^2-a^2)(v_2^2-a^2)\}} + \sqrt{\{(v_1^2-c^2)(v_2^2-c^2)\}}}{a^2-c^2},$$

and similarly

$$\cos\iota_2 = \cos\omega . \cos\omega_2 + \cos\phi . \cos\phi_2 + \cos\psi . \cos\psi_2$$

$$= \frac{\sqrt{\{(v_1^2-a^2)(v_2^2-a^2)\}} - \sqrt{\{(v_1^2-c^2)(v_2^2-c^2)\}}}{a^2-c^2}.$$

### Proposition 9.

*To find $v_1$, $v_2$ in terms of $\iota_1$, $\iota_2$.*

By the last proposition

$$\cos\iota_1 . \cos\iota_2 = \frac{(v_1^2-a^2)(v_2^2-a^2) - (v_1^2-c^2)(v_2^2-c^2)}{(a^2-c^2)^2}$$

$$= \frac{(a^4-c^4) - (a^2-c^2)(v_1^2+v_2^2)}{(a^2-c^2)^2}$$

$$= \frac{(a^2+c^2) - (v_1^2+v_2^2)}{(a^2-c^2)}$$

therefore

$$v_1^2 + v_2^2 = a^2 + c^2 - (a^2-c^2)\cos\iota_1\cos\iota_2.$$

Again, $(\sin\iota_1)^2 . (\sin\iota_2)^2 = 1 - (\cos\iota_1)^2 - (\cos\iota_2)^2 + (\cos\iota_1)^2(\cos\iota_2)^2$

$$= 1 - 2 . \frac{(v_1^2-a^2)(v_2^2-a^2) + (v_1^2-c^2)(v_2^2-c^2)}{(a^2-c^2)^2}$$

$$+ \frac{(a^2+c^2)^2 - 2(a^2+c^2)(v_1^2+v_2^2) + (v_1^2+v_2^2)^2}{(a^2-c^2)^2}$$

$$= \frac{v_1^4 - 2v_1^2v_2^2 + v_2^4}{(a^2-c^2)^2}$$

therefore

$$v_1^2 - v_2^2 = (a^2 - c^2)\sin\iota_1 \,.\, \sin\iota_2$$

but

$$v_1^2 + v_2^2 = (a^2 + c^2) - (a^2 - c^2)\cos\iota_1\cos\iota_2$$

therefore

$$v_1^2 = \frac{a^2+c^2}{2} - \frac{a^2-c^2}{2}\cos(\iota_1+\iota_2)$$

$$= a^2\left(\sin\frac{\iota_1+\iota_2}{2}\right)^2 + c^2\left(\cos\frac{\iota_1+\iota_2}{2}\right)^2$$

$$v_2^2 = \frac{a^2+c^2}{2} - \frac{a^2-c^2}{2}\cos(\iota_1-\iota_2)$$

$$= a^2\left(\sin\frac{\iota_1-\iota_2}{2}\right)^2 + c^2\left(\cos\frac{\iota_1-\iota_2}{2}\right)^2.$$

Thus for uniaxal crystals where $\iota_1 + \iota_2 = 180^\circ$

$$v_1^2 = a^2$$

$$v_2^2 = a^2(\cos\iota)^2 + c^2(\sin\iota)^2.$$

Cor. Hence we may reduce the discovery of the two fronts into which a plane front is refracted on entering a crystal to the following trigonometrical problem.

Let a sphere be described about any point in the line in which the air front intersects the plane of incidence. Let the great circle $PI$ denote the latter plane, $IF$ the former, $OA$, $OC$ also great circles, the planes of single velocity. Suppose $IGH$ to be one of the refracted fronts intersecting $OA$, $OC$ in $G$ and $H$, then

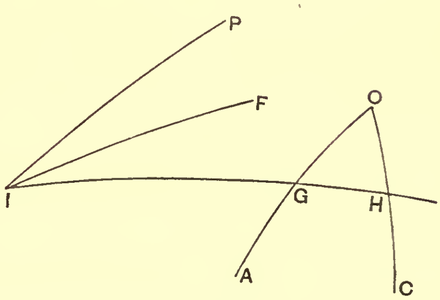


Fig. 1.

$$\frac{(a^2+c^2) - (a^2-c^2)\cos(G+H)}{2\,(\text{vel. in air})^2} = \frac{(\sin PIF)^2}{(\sin PIGH)^2}.$$

The double sign will give rise to two positions of the refracted front $IGH$.

The propositions which follow are perhaps more curious than immediately useful.

### Proposition 10.

*To determine the portion of a line of vibration in terms of the two velocities of its corresponding front.*

We have here to determine the quantities $\frac{y_1}{x_1}$, $\frac{z_1}{x_1}$ (of Prop. 1) in terms of $v_1$, $v_2$, or on putting $x_1^2+y_1^2+z_1^2=1$, $x_1, y_1, z_1$ are to be found in terms of $v_1$, $v_2$.

By Prop. 3

$$x_1 : y_1 : z_1 : : \frac{l}{a^2-v_1^2} : \frac{m}{b^2-v_1^2} : \frac{n}{c^2-v_1^2}$$

and by Prop. 5

$$\begin{aligned} l^2 : m^2 : n^2 : : &(b^2-c^2)(a^2-v_1^2)(a^2-v_2^2) \\ &:(c^2-a^2)(b^2-v_1^2)(b^2-v_2^2) \\ &:(a^2-b^2)(c^2-v_1^2)(c^2-v_2^2); \end{aligned}$$

therefore $\qquad x_1^2 \quad : \quad y_1^2 \quad : \quad z_1^2$

$$: : (b^2-c^2)\frac{a^2-v_2^2}{a^2-v_1^2} : (c^2-a^2)\frac{b^2-v_2^2}{b^2-v_1^2} : (a^2-b^2)\frac{c^2-v_2^2}{c^2-v_1^2}.$$

Let $\alpha$, $\beta$, $\gamma$ be the angles made by the given line of vibration with the elastic axes, then

$$\begin{aligned} (\cos\alpha)^2 &= \frac{x_1^2}{x_1^2+y_1^2+z_1^2} \\ &= (b^2-c^2)(a^2-v_2^2)(b^2-v_1^2)(c^2-v_1^2) \end{aligned}$$

divided by

$$(b^2-c^2)(a^2-v_2^2)(b^2-v_1^2)(c^2-v_1^2)+(c^2-a^2)(b^2-v_2^2)(c^2-v_1^2)(a^2-v_1^2)$$
$$+(a^2-b^2)(c^2-v_2^2)(a^2-v_1^2)(b^2-v_1^2)$$

and therefore

$$=\frac{(b^2-c^2)(a^2-v_2^2)(b^2-v_1^2)(c^2-v_1^2)}{(v_1^2-v_2^2)(a^2-b^2)(b^2-c^2)(c^2-a^2)}$$

(where it is to be observed that the reduction of the denominator is simply the effect of a vast heap of terms disappearing under the influence of contact with the magic circuit $(a^2-b^2)$, $(b^2-c^2)$, $(c^2-a^2)$, a simpler instance of which was seen in Proposition 5).

In fact the coefficient of $v^4 . v^2$

$$\begin{aligned} &= (b^2-c^2)+(c^2-a^2)+(a^2-b^2) \\ &= 0 \end{aligned}$$

that of $v_1^2 \,.\, v_2^2$

$$\begin{aligned} &= (c^2+b^2)\,.\,(c^2-b^2) \\ &\quad + (a^2+c^2)\,.\,(a^2-c^2) \\ &\quad + (b^2+a^2)\,.\,(b^2-a^2) \\ &= (c^4-b^4)+(a^4-c^4)+(b^4-a^4) \\ &= 0. \end{aligned}$$

The term in which neither $v_1$ nor $v_2$ enters

$$\begin{aligned} &= a^2b^2c^2\,\{(b^2-c^2)+(c^2-a^2)+(a^2-b^2)\} \\ &= 0. \end{aligned}$$

The coefficient of

$$-v_1^2 = a^2\,.\,(b^4-c^4)+b^2\,.\,(c^4-a^4)+c^2\,.\,(a^4-b^4)$$

and that of

$$v_2^2 = b^2c^2\,.\,(c^2-b^2)+c^2a^2\,.\,(a^2-c^2)+a^2b^2\,.\,(b^2-a^2)$$

each of which

$$= (a^2-b^2)\,.\,(b^2-c^2)\,.\,(c^2-a^2).$$

Hence

$$(\cos\alpha)^2 = \frac{v_1^2-b^2}{v_1^2-v_2^2}\cdot\frac{(a^2-v_2^2)(c^2-v_1^2)}{(a^2-b^2)(a^2-c^2)},$$

in like manner $(\cos\beta)^2 = \&c.$

and $$(\cos\gamma)^2 = \frac{v_1^2-b^2}{v_1^2-v_2^2}\cdot\frac{(c^2-v_2^2)(a^2-v_1^2)}{(c^2-b^2)(c^2-a^2)}.$$

## Proposition 11.

*$\epsilon_1$, $\epsilon_2$ being the angles between any line of vibration and the optic axes, required the velocity due to that line in terms of $\epsilon_1$, $\epsilon_2$.*

By analytical geometry,

$$\cos\epsilon_1 = \cos\alpha\,.\,\cos\phi_1 + \cos\gamma\,.\,\cos\psi_1$$

$$\cos\epsilon_2 = \cos\alpha\,.\,\cos\phi_1 - \cos\gamma\,.\,\cos\psi_1$$

therefore $$\cos\epsilon_1\,.\,\cos\epsilon_2 = (\cos\alpha)^2(\cos\phi_1)^2 - (\cos\gamma)^2(\cos\psi_1)^2$$

$$= \frac{v_1^2-b^2}{v_1^2-v_2^2}\cdot\left\{\frac{(a^2-v_2^2)\,.\,(c^2-v_1^2)-(c^2-v_2^2)\,.\,(a^2-v_1^2)}{(a^2-c^2)^2}\right\}$$

$$= \frac{v_1^2-b^2}{v_1^2-v_2^2}\cdot\frac{(a^2-c^2)(v_2^2-v_1^2)}{(a^2-c^2)^2}$$

$$= \frac{b^2-v_1^2}{a^2-c^2}.$$

Hence $$v_1^2 = b^2-(a^2-c^2)\cos\epsilon_1\cos\epsilon_2,$$

and in like manner, for the *conjugate* line of vibration

$$v_2^2 = b^2-(a^2-c^2)\cos\epsilon_1'\cos\epsilon_2'.$$

## Proposition 12.

*To find* $\epsilon_1$, $\epsilon_2$ *in terms of* $\iota_1$, $\iota_2$.

$$(\cos\epsilon_1)^2 + (\cos\epsilon_2)^2 = 2(\cos\alpha)^2 . (\cos\phi_1)^2 + 2(\cos\gamma)^2 . (\cos\psi_1)^2$$

$$= 2\frac{v_1^2 - b^2}{v_1^2 - v_2^2}\left\{\frac{(a^2 - v_2^2).(c^2 - v_1^2) + (c^2 - v_2^2).(a^2 - v_1^2)}{(a^2 - c^2)^2}\right\}$$

but by Prop. 9

$$v_1^2 = a^2\left(\sin\frac{\iota_1 + \iota_2}{2}\right)^2 + c^2\left(\cos\frac{\iota_1 + \iota_2}{2}\right)^2$$

$$v_2^2 = a^2\left(\sin\frac{\iota_1 - \iota_2}{2}\right)^2 + c^2\left(\cos\frac{\iota_1 - \iota_2}{2}\right)^2$$

therefore

$$(\cos\epsilon_1)^2 + (\cos\epsilon_2)^2 = \frac{b^2 - v_1^2}{(a^2 - c^2)\sin\iota_1 . \sin\iota_2}$$

multiplied by

$$\frac{2(a^2 - c^2)^2\left[\left(\cos\frac{\iota_1 + \iota_2}{2}\right)^2\left(\sin\frac{\iota_1 - \iota_2}{2}\right)^2 + \left(\cos\frac{\iota_1 - \iota_2}{2}\right)^2\left(\sin\frac{\iota_1 + \iota_2}{2}\right)^2\right]}{(a^2 - c^2)^2}$$

$$= \frac{b^2 - v_1^2}{(a^2 - c^2)\sin\iota_1 . \sin\iota_2}\{(\sin\iota_1)^2 + (\sin\iota_2)^2\}$$

and we have seen that

$$\cos\epsilon_1\cos\epsilon_2 = \frac{b^2 - v_1^2}{a^2 - c^2}$$

therefore

$$\cos\epsilon_1 + \cos\epsilon_2 = \sqrt{\left(\frac{b^2 - v_1^2}{a^2 - c^2}\right)} . \frac{\sin\iota_1 + \sin\iota_2}{\sqrt{(\sin\iota_1 . \sin\iota_2)}}$$

$$\cos\epsilon_1 - \cos\epsilon_2 = \sqrt{\left(\frac{b^2 - v_1^2}{a^2 - c^2}\right)} . \frac{\sin\iota_1 - \sin\iota_2}{\sqrt{(\sin\iota_1 . \sin\iota_2)}}$$

therefore

$$\cos\epsilon_1 = \sqrt{\left\{\frac{b^2 - v_1^2}{a^2 - c^2} . \frac{\sin\iota_1}{\sin\iota_2}\right\}}$$

$$\cos\epsilon_2 = \sqrt{\left\{\frac{b^2 - v_1^2}{a^2 - c^2} . \frac{\sin\iota_2}{\sin\iota_1}\right\}}$$

and in like manner

$$\cos\epsilon_1' = \sqrt{\left\{\frac{b^2 - v_2^2}{a^2 - c^2} . \frac{\sin\iota_1}{\sin\iota_2}\right\}}$$

$$\cos\epsilon_2' = \sqrt{\left\{\frac{b^2 - v_2^2}{a^2 - c^2} . \frac{\sin\iota_2}{\sin\iota_1}\right\}}$$

where $v_1$, $v_2$ for the sake of neatness are left *unexpressed* in terms of $\iota_1$, $\iota_2$.

This is the simplest form by which the position of the lines of vibration can be denoted.

Cor. From the last proposition it appears that

$$\frac{\cos \epsilon_1}{\cos \epsilon_2} = \frac{\sin \iota_1}{\sin \iota_2}.$$

Hence we may construct geometrically for the two planes of polarization.

Let $I$, $K$ be the projections of the two optic axes on a sphere, $E$ the projection of the normal to the front, $P$ the projection of one line of vibration; then

$$\frac{\cos PK}{\cos PI} = \frac{\sin KE}{\sin IE}.$$

Draw $FEG$ the circle of which $P$ is the pole, meeting $PK$, $PI$ produced in $G$ and $F$.

Then $\cos PK = \sin KG$,

and $\cos PI = \sin IF$,

therefore

$$\frac{\sin KG}{\sin FI} = \frac{\sin KE}{\sin IE}$$

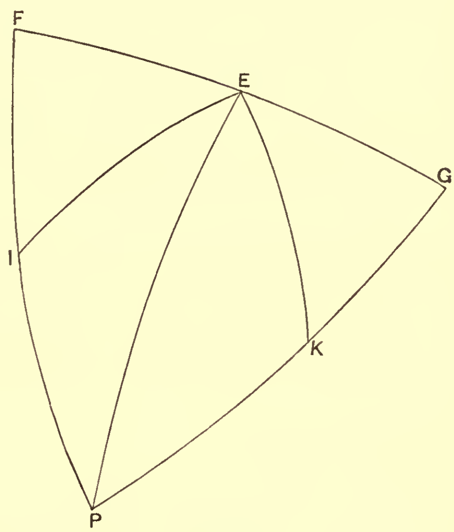


Fig. 2.

therefore

$$\frac{\sin KG}{\sin KE} = \frac{\sin IF}{\sin IE}$$

therefore

$$\sin KEG = \sin IEF$$

therefore $KEG = IEF$ or $180° - IEF$. But $PEF = PEG$, therefore $EP$ bisects either the angle $IEK$ or the supplement to it.

These two positions of $EP$ give the two planes of polarization. The construction is the same as that given in Mr Airy's tracts, and originally proposed, I believe, by Mr MacCullagh.

## ADDENDUM.

If in the equation of Prop. 6, viz.

$$\frac{(\cos\omega)^2}{a^2-v^2}+\frac{(\cos\phi)^2}{b^2-v^2}+\frac{(\cos\psi)^2}{c^2-v^2}=0$$

we change $a$, $b$, $c$, $v$ into $\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$, $\frac{1}{v}$, and consider $v$ to be the length of a line drawn perpendicular to the plane

$$\cos\omega \,.\, x+\cos\phi \,.\, y+\cos\psi \,.\, z=0,$$

the equation to the extremity thereof must be

$$\frac{a^2r^2(\cos\omega)^2}{a^2-r^2}+\frac{b^2r^2(\cos\phi)^2}{b^2-r^2}+\frac{c^2r^2(\cos\psi)^2}{c^2-r^2}$$

where $\omega$, $\phi$, $\psi$ denote the angles between the radius vector $r$, and the axes of $x$, $y$, $z$, so that the equation may be written

$$\frac{a^2x^2}{a^2-r^2}+\frac{b^2y^2}{b^2-r^2}+\frac{c^2z^2}{c^2-r^2}=0,$$

which is that of the wave surface.

But we have seen that

$$v^2=c^2\left\{\cos\left(\frac{\iota_1\pm\iota_2}{2}\right)\right\}^2+a^2\left\{\sin\left(\frac{\iota_1\pm\iota_2}{2}\right)\right\}^2,$$

therefore the equation to the wave surface may be written

$$\frac{1}{r^2}=\frac{\left(\cos\frac{\iota_1\pm\iota_2}{2}\right)^2}{c^2}+\frac{\left(\sin\frac{\iota_1\pm\iota_2}{2}\right)^2}{a^2},$$

where $\iota_1$, $\iota_2$ denote the angles between the radius vector $v$ and the two lines which would be the optic axes if $a$, $b$, $c$ were changed into $\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$ so that if $e$ be the inclination of either to the mean axis of elasticity

$$\cos e=\sqrt{\left(\frac{\frac{1}{a^2}-\frac{1}{b^2}}{\frac{1}{a^2}-\frac{1}{c^2}}\right)}=\frac{c}{b}\sqrt{\left(\frac{a^2-b^2}{a^2-c^2}\right)}$$

$$\sin e=\sqrt{\left(\frac{\frac{1}{b^2}-\frac{1}{c^2}}{\frac{1}{a^2}-\frac{1}{c^2}}\right)}=\frac{a}{b}\sqrt{\left(\frac{b^2-c^2}{a^2-c^2}\right)}.$$

These lines I shall call by way of distinction the prime radii*.

* Upon the authority of Professor Airy I have appropriated the term optic axes to the lines normal to the fronts of single velocity.

COR. 1. If $r_1$, $r_2$ be the two values of $r$ corresponding to the same values of $\iota_1$, $\iota_2$ we have

$$\frac{1}{r_1^2} - \frac{1}{r_2^2} = \frac{1}{c^2}\left\{\left(\cos\frac{\iota_1 - \iota_2}{2}\right)^2 - \left(\cos\frac{\iota_1 + \iota_2}{2}\right)^2\right\}$$
$$+ \frac{1}{a^2}\left\{\left(\sin\frac{\iota_1 - \iota_2}{2}\right)^2 - \left(\sin\frac{\iota_1 + \iota_2}{2}\right)^2\right\}$$
$$= \left(\frac{1}{c^2} - \frac{1}{a^2}\right)\sin\iota_1 . \sin\iota_2,$$

which proves the celebrated problem of *two rays* having a common direction in a crystal.

COR. 2. The intersection of any concentric sphere with the wave surface is found by making $r$ constant. Hence $\iota_1 \pm \iota_2$ becomes constant, and therefore $r\iota_1 \pm r\iota_2 = \text{constant}$. Hence the curve of intersection is the locus of points, the sum or difference of whose distances from two poles when measured by the arcs of great circles is constant; the poles being the points in which the prime radii pierce the sphere.

In three cases these spherico-ellipses or spherico-hyperbolas become great circles:

(1) When $\iota_1 \pm \iota_2 =$ the angle between the two poles, in which case the curve of intersection is the great circle which comprises the two poles.

(2) When $\iota_1 - \iota_2 = 0$, when the locus is a great circle perpendicular to the former and bisecting the angle between the optic axes.

(3) When $\iota_1 + \iota_2 = 180^\circ$, when the locus is a great circle perpendicular to the two above, and bisecting the supplemental angle between the two axes.

Various other properties may be with the greatest simplicity deduced from the radio-angular equation. The hurry of the press leaves me time only to subjoin the following

PROPOSITION.

*To find the inclination of the radius vector to the tangent plane, in terms of the angles which the radius vector makes with the prime radii.*

Let $O$ be the centre of the wave surface, $OA$, $OB$ the two prime radii, $OP$ any radius vector. Let $OP = v$, $POA = \iota_1$, $POB = \iota_2$, and let the inclination of the planes $POA$, $POB = \mu$;

then
$$\frac{1}{r^2} = \frac{\left(\sin\frac{\iota_1 + \iota_2}{2}\right)^2}{a^2} + \frac{\left(\cos\frac{\iota_1 + \iota_2}{2}\right)^2}{c^2},$$

(taking only the positive sign for the sake of brevity).

Let $OQ$, $OR$ be the two adjacent radii vectores, so assumed that

$$QOA = POA, \quad QOB = POB + \delta\iota_2,$$
$$ROB = POB, \quad ROA = POA + \delta\iota_1,$$

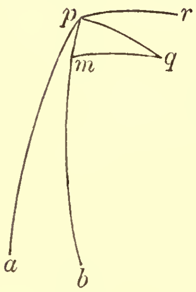

Fig. 3.

and let $p, q, r, a, b$ be the projections of $P, Q, R, A, B$ on a sphere of which $O$ is the centre, then it is clear that

$$qpa = 90°, \quad rpb = 90°,$$

draw $qm$ perpendicular to $pb$, then $pm = \delta\iota_2$, and therefore

$$pq = \frac{pm}{\sin pqm} = \frac{pm}{\sin apb} = \frac{\delta\iota_2}{\sin\mu}.$$

In like manner

$$pr = \frac{\delta\iota_1}{\sin\mu}.$$

Now the angle $QPO$

$$= \tan^{-1}.\frac{r.POQ}{OQ - OP} = \tan^{-1}.\frac{r.pq}{\frac{dr}{d\iota_2}.\delta\iota_2};$$

also

$$\frac{d.\frac{1}{r^2}}{d\iota_2} = d\iota_2\left\{\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\left(\cos\frac{\iota_1+\iota_2}{2}\right)^2\right\}$$
$$= -\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\left(\sin\frac{\iota_1+\iota_2}{2}\right)\left(\cos\frac{\iota_1+\iota_2}{2}\right);$$

therefore

$$\frac{dr}{rd\iota_2} = \tfrac{1}{4}r^2\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\sin(\iota_1+\iota_2),$$

therefore

$$\cot QPO = \frac{r^2}{4}\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\sin(\iota_1+\iota_2)\sin\mu.$$

In like manner

$$\cot RPO = \frac{r^2}{4}\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\sin(\iota_1+\iota_2)\sin\mu,$$

therefore

$$QPO = RPO.$$

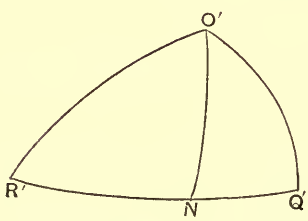

Fig. 4.

Also it is clear that $rpq = apb = \mu$. And to find the inclination of $OP$ to $RPQ$, we have only to describe a sphere of which $P$ is the centre, and intersecting $PQ$, $PR$, $PO$ in $Q'$, $R'$, $O'$.

Then $R'O'Q' = \mu$, and

$$O'Q' = O'R' = \cot^{-1}\left\{\frac{r^2}{4}\left(\frac{1}{c^2} - \frac{1}{a^2}\right)\sin(\iota_1+\iota_2)\sin\mu\right\}.$$

Draw $O'N$ perpendicular to $R'Q'$, then $O'N$ measures the inclination of the radius vector to the tangent plane*.

And $$Q'O'N = \frac{\mu}{2},$$

therefore $$\cos\frac{\mu}{2} = \tan O'N \, . \cot O'Q',$$

therefore $$\cot O'N = \frac{\cot O'Q'}{\cos\frac{\mu}{2}},$$

and therefore
$$\cot O'N = \tfrac{1}{2} r^2 \, . \left(\frac{1}{a^2} - \frac{1}{c^2}\right) \sin\frac{\mu}{2} \, . \sin(\iota_1 + \iota_2).$$

Let $AOB$ the angle between the optic axes $= 2e$, then by mere trigonometry
$$\sin\frac{\mu}{2} = \sqrt{\frac{\sin\left(e + \frac{\iota_1 - \iota_2}{2}\right)\sin\left(e - \frac{\iota_1 - \iota_2}{2}\right)}{\sin\iota_1 \, . \sin\iota_2}},$$
therefore the tangent of the inclination between the radius vector and the normal
$$= \tfrac{1}{2} r^2 \, . \left(\frac{1}{a^2} - \frac{1}{c^2}\right) \sin(\iota_1 + \iota_2) \, . \sqrt{\frac{\sin\left(e + \frac{\iota_1 - \iota_2}{2}\right)\sin\left(e - \frac{\iota_1 - \iota_2}{2}\right)}{\sin\iota_1 \, . \sin\iota_2}}.$$

In like manner the tangent of the inclination between the same radius vector and the normal at the other point of the wave-surface pierced by it
$$= \tfrac{1}{2} (r_1)^2 \left(\frac{1}{a^2} - \frac{1}{c^2}\right) \sin(\iota_1 - \iota_2) \, . \sqrt{\frac{\sin\left(e + \frac{\iota_1 + \iota_2}{2}\right)\sin\left(e - \frac{\iota_1 + \iota_2}{2}\right)}{\sin\iota_1 \, . \sin\iota_2}}.$$

We may, in the same way, find the inclination of the tangent plane to either of the prime radii, and to the plane which contains them both, in terms of $\iota_1$ and $\iota_2$; the former by a remarkably elegant construction; but the final expressions do not present themselves under the same simple aspect.

If we call $\phi$ the angle between the ray and the front, we may still further reduce by substituting for $r^2$ its values in terms of $\iota_1$, $\iota_2$ and we shall obtain
$$\cot\phi = \frac{2(c^2 - a^2)}{c^2 \tan\frac{\iota_1 \mp \iota_2}{2} + a^2 \cot\frac{\iota_1 \mp \iota_2}{2}}$$
$$\times \sqrt{\left\{\sin\left(e + \frac{\iota_1 \pm \iota_2}{2}\right)\sin\left(e - \frac{\iota_1 \pm \iota_2}{2}\right) . \operatorname{cosec}\iota_1 \, . \operatorname{cosec}\iota_2\right\}}.$$

* $O'$ is the projection of the ray and $R'O'$ of the tangent plane. Therefore $O'N$ being perpendicular to $R'Q'$ represents their inclination.

And if $\pi_1$, $\pi_2$ be the inclinations of the normal to the two prime radii, it may be shown that

$$\cos\pi_1 = \cos\phi \sin\iota_1 \mp \sin\phi\cos\iota_1 \sin\frac{\mu}{2},$$

$$\cos\pi_2 = \cos\phi \sin\iota_2 \pm \sin\phi\cos\iota_2 \sin\frac{\mu}{2}.$$

COR. 1. For uniaxal crystals $\frac{\mu}{2} = 90°$ and $\iota_1 + \iota_2 = 180°$, so that the tangent of the inclination of normal to radius vector

$$= \tfrac{1}{2} r^2 . \left(\frac{1}{a^2} - \frac{1}{c^2}\right) \sin 2\iota \text{ for one point,}$$

and $\qquad = 0$ for the other.

COR. 2. For every point in the circular section which passes through the poles $\sin\frac{\mu}{2} = 0$, and for the other two circular sections $\iota_1 \pm \iota_2 = 0$ or $180°$.

Therefore every point in the three circular sections is an apse.

COR. 3. When $a$ nearly $= c$, $\frac{1}{a^2} - \frac{1}{c^2}$ is very small; and therefore the normal and radius vector very nearly coincide.

COR. 4. Referring to fig. 4 we see that $O'N$ bisects the angle $R'O'Q'$. Now $R'O$, $Q'O$ are respectively perpendicular to the planes passing through $O'$ and the optic axes; and therefore the meridian plane as we may term it, that is, the plane containing both the ray and the normal, always bisects the angle formed by the two planes drawn through the ray and the two optic axes.

COR. 5. When

$$\iota_1 \text{ or } \iota_2 = 0,$$

$$\iota_2 \text{ or } \iota_1 = e.$$

And therefore $\phi$ assumes the form $\frac{0}{0}$, which indicates that the extremities of the four prime radii are singular points.

In concluding for the present it behoves me to state that one step has been omitted in the foregoing paper*, viz. the actual performance of the eliminations which lead to the rectilinear equation to the wave-surface. But Mr Archibald Smith's elegant and brief Memoir in the *Cambridge Philosophical Transactions*† of last year leaves nothing to be desired further on that head.

[* See below, p. 27. ED.] [† Vol. VI. Also *Phil. Mag.* April, 1838, p. 335. ED.]

That I have not exhibited it in its proper place (Prop. 6) arises only from my respect to the principle of literary propriety. With this important blank supplied the Analytical Theory may be pronounced to be complete.

For all errors and imperfections in what precedes my excuse must be press of time and a total want of the materials to be derived from consulting works of reference.

---

Since writing the above I have had an opportunity of reading the paper of our living Laplace inserted as part of the Third Supplement to his System of Rays in the *Transactions* of the Royal Irish Academy, in which the principal foregoing results are obtained by aid of a more refined and transcendental analysis.

The nature of the four singular points is there discussed and the existence of four circles of plane contact demonstrated.

The former may be very easily shown thus: when $\iota_1$ is very small $\iota_2 = 2e - \iota_1 \cos\psi$ very nearly, $\psi$ denoting the inclination of the plane in which $e$ is reckoned to the plane in which $\iota_1$ is reckoned.

Hence

$$\left(\frac{1}{r}\right)^2 = \frac{1}{2}\left(\frac{1}{a^2}+\frac{1}{c^2}\right) - \frac{1}{2}\left(\frac{1}{a^2}-\frac{1}{c^2}\right)\cos\{2e - \iota_1(\cos\psi \pm 1)\}$$

$$= \frac{1}{2}\left(\frac{1}{a^2}+\frac{1}{c^2}\right) - \frac{1}{2}\left(\frac{1}{a^2}-\frac{1}{c^2}\right)\cos 2e - \frac{1}{2}\left(\frac{1}{a^2}-\frac{1}{c^2}\right)\sin 2e\,(\cos\psi \pm 1)\,\iota_1$$

$$= \frac{1}{b^2} - \frac{1}{b^2ac}\sqrt{\{(a^2-b^2)(b^2-c^2)\}}\,(\cos\psi \pm 1)\,\iota_1,$$

therefore

$$r = b\left\{1 + \frac{1}{2}(\cos\psi \pm 1)\left(1 - \frac{b^2}{a^2}\right)^{\frac{1}{2}}\left(\frac{b^2}{c^2} - 1\right)^{\frac{1}{2}}\iota_1\right\}.$$

Take $\psi$ constant and let the abscissæ and ordinates be reckoned respectively along and perpendicular to the prime ray.

Then $$\iota_1 = \frac{y}{x} \text{ nearly, and } r = \sqrt{(y^2+x^2)} = x,$$

or, if we change the origin to the other extremity of the prime ray,

$$\iota_1 = \frac{y}{b}, \quad r = b - x,$$

so that the equation becomes

$$-\frac{x}{y} = \frac{1}{2}(\cos\psi \pm 1)\sqrt{\left\{\left(1 - \frac{b^2}{a^2}\right)\left(\frac{b^2}{c^2} - 1\right)\right\}}.$$

Hence at each singular point the surface is touched by a cone, the equation to the generating line of which is given by the above, the extreme angle between it and the prime ray being

$$\cot^{-1}\left[\sqrt{\left\{\left(1-\frac{b^2}{a^2}\right)\left(\frac{b^2}{c^2}-1\right)\right\}}\right].$$

When $b=a$, $\psi$ always $=\frac{\pi}{2}$ and the cone returns into a plane.

Again, let us suppose that the position of any perpendicular from the centre is given, and that of the corresponding radius vector required.

Let $OA$, $OB$* denote what we have termed the optic axes, but which it will be more agreeable to analogy to term the prime perpendiculars from centre, and let $OP$ be the given normal. Take $OQ$, $OR$ contiguous perpendiculars from centre in planes $POQ$, $ROP$, perpendicular to $POA$, $POB$ respectively, then the inclination of the two former will be the same as that of the two latter, and may be termed $\mu$.

Let $\iota_1$, $\iota_2$ now denote the angles $POA$, $POB$ respectively, then

$$QOA=\iota_1, \qquad QOB=\iota_2+\delta\iota_2,$$

$$ROA=\iota_1+\delta\iota_1, \quad ROB=\iota_2.$$

The ray will be found by joining $O$ with the intersection of three planes drawn at $P$, $Q$, $R$, perpendicular to $OP$, $OQ$, $OR$, respectively.

Now from Prop. 9 it appears that

$$OP=\sqrt{\left\{a^2\left(\sin\frac{\iota_1+\iota_2}{2}\right)^2+c^2\left(\cos\frac{\iota_1+\iota_2}{2}\right)^2\right\}},$$

using only one sign for the sake of simplicity, which we may do by throwing the ambiguity upon the *way* in which $\iota_1$ or $\iota_2$ is measured, also

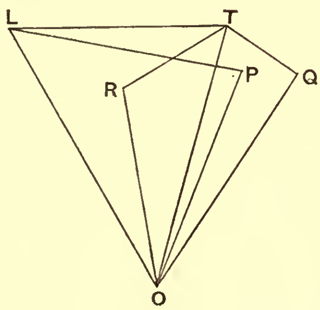

Fig. 5.

$$OQ=OP+\frac{d\,.\,OP}{d\iota_2}\delta\iota_2,$$

$$OR=OP+\frac{d\,.\,OP}{d\iota_1}\delta\iota_1.$$

Let $\delta\iota_1=\delta\iota_2$, then it is clear that $OQ=OR$, and the intersection of the two planes perpendicular to $OQ$, $OR$ is therefore a line perpendicular to the plane $QOR$, and to the line which bisects the angle $QOR$.

In fact if we draw $QT$, $RT$ perpendicular to $OQ$, $OR$ respectively in the plane $QOR$, the intersection in question passes through $T$ and is perpendicular to $OT$; also

$$OT=OQ\,.\sec\left(\tfrac{1}{2}ROQ\right)=OQ$$

to the first order of smallness.

* $OA$, $OB$ are not expressed in the figure.

Now it is easy to see (just as on p. 16) that

$$ROP = \frac{\delta\iota_1}{\sin\mu},$$

and also

$$QOP = \frac{\delta\iota_2}{\sin\mu},$$

therefore $ROP = QOP$ and therefore $POT$ is perpendicular to $QOR$.

Hence the problem is reduced to finding $L$ the intersection of two lines $TL$, $PL$ drawn in the *same* plane $POT$.

Now because $OTL$, $OPL$ are each right angles, a circle may be made to pass through $L$, $T$, $P$, $O$.

Hence the angle

$$PLO = PTO = \tan^{-1}\frac{OP \times POT}{OT - OP}$$

$$= \tan^{-1}\frac{OP \times POR \,.\, \cos\frac{1}{2}\mu}{\frac{d\,.\,OP}{d\iota_2}\,\delta\iota_2} = \tan^{-1}\frac{OP \times \frac{\delta\iota_2}{\sin\mu}\cos\frac{1}{2}\mu}{\frac{d\,.\,OP}{d\iota_2}\,\delta\iota_2},$$

and

$$OL = OP \,.\, \sec POL.$$

Also the position of the plane $POL$ is known, and therefore the radius is completely determined in magnitude and position.

It may be worth while also to remark that the above constructions enable us to form a series of equations between the magnitude of the radius and its inclinations to the two prime perpendiculars.

In fact, if we call $\pi_1$, $\pi_2$ the two inclinations in question

$$\cos\pi_1 = \cos POL \cos\iota_1 \pm \sin POL \sin\iota_1 \,.\, \sin\frac{\mu}{2},$$

$$\cos\pi_2 = \cos POL \cos\iota_2 \mp \sin POL \sin\iota_2 \,.\, \sin\frac{\mu}{2},$$

and of course if we call the angle between the two prime normals $2E$

$$\sin\frac{\mu}{2} = \sqrt{\frac{\sin\left(E + \frac{\iota_1+\iota_2}{2}\right)\sin\left(E - \frac{\iota_1+\iota_2}{2}\right)}{\sin\iota_1 \sin\iota_2}}.$$

Cor. 1. When $\iota_1$ or $\iota_2 = 0$, $\tan POL$ assumes the form $\frac{0}{0}$ which may be interpreted analogously to the method used in the reverse problem, but may be more elegantly illustrated by

Cor. 2. Which is that the meridian plane $POT$ (that is, the plane in which both normal and radius lie) bisects the angle formed by $ROP$, $QOP$, and therefore that formed by the planes drawn through the normal and the two prime normals to which these two are perpendicular.

Now we have found (Cor. 4, page 18), that it also bisects the angle formed by the two planes passing through the radius and the two prime radii. Hence when the ray is given, we may find by the easiest geometry the normal and the tangent plane, and *vice versâ*.

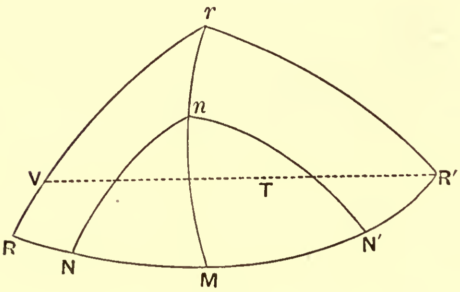


Fig. 6.

Thus suppose $(N, N')$ $(R, R')$ to be the projections of the prime perpendiculars and prime radii on a sphere concentric with the wave surface.

Let $n$ be the projection of any given perpendicular on the same sphere; join $nN$, $nN'$; bisect $NnN'$ by $nM$, which will be the meridian plane.

Draw from $R'$, $R'TV$ perpendicular to $nM$ and make $R'T = TV$. Produce $RV$ to meet $Mn$ in $r$, then $RrM = R'rM$, and therefore $r$ is the projection of the radius. Just in the same way when $r$ is given we may find $n$.

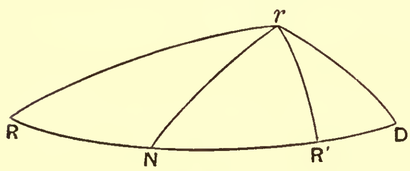


Fig. 7.

Now suppose $n$ to come to $N$, then the position of the meridian plane $nM$ becomes indeterminate, and $r$ from a point becomes a locus, subject to the condition that $R'rN = RrN$. From $r$ draw $rD$ perpendicular to $rN$.

Then it is clear that because $rN$ bisects $RrR'$

$$\frac{\sin RD}{\sin R'D} = \frac{\sin Rr}{\sin R'r} = \frac{\sin RN}{\sin R'N},$$

and therefore $D$ is a fixed point and $ND$ a fixed length, and

$$\cos rND = \tan rN \,.\, \cot ND\,;$$

therefore the projection of the locus of $r$ upon a plane drawn at $N$ perpendicular to the line joining $N$ with the centre $O$ is given by the equation

$$\rho = ON \,.\, \cot ND \,.\, \cos\theta,$$

$N$ being the origin and the projection of $ND$ the prime radius; which is the equation to a circle passing through $N$, and whose diameter $= ON \cot ND$.

Hence at the extremity of each prime perpendicular the tangent plane meets the surface in a circle passing through that extremity and whose radius $= \frac{1}{2}b \cot \alpha$, $\alpha$ being to be found from the equation

$$\frac{\sin(2E+\alpha)}{\sin\alpha} = \frac{\sin(E+e)}{\sin(E-e)},$$

that is $$\tan(E+\alpha) = (\tan E)^2 \cot e.$$

Just in the same way it may be shown that the trace of the perpendiculars to the tangent planes of the surface at the point where it is pierced by any prime radius upon a plane perpendicular to that radius at its extremity, is also a circle passing through it, and curved in an opposite direction from the circle of plane contact nearest to it.

Hence the enveloping cone at these points may be described as being perpendicular to the circular cone, formed by drawing lines from the centre to the above described circle; that is every generating line of the one will be perpendicular to the generating line which it meets of the other.

More generally it easily appears from fig. 6 that if a series of great circles (representing meridian planes) be taken intersecting the great circle $NRR'N'$ in a fixed point, a plane perpendicular to the radius passing through that point, will intersect the cone of rays as well as the cone of perpendiculars corresponding to those meridian planes, in two *circles*. So that there exist an indefinite number of *circular* cones of rays corresponding to *circular* cones of perpendiculars touching each other in a line lying in the plane containing the extreme axes, and having their circular sections perpendicular to that line.

The cusps are explained by the cone of *rays* degenerating into a right line, and the circles of plane contact by the cone of perpendiculars so degenerating.

Furthermore I observe in conclusion that when a ray is given it follows from the general geometrical construction above that there will be two meridian planes according as we take $R$ with $R'$, or with a point 180° from $R'$, and consequently these two planes will be perpendicular to each other.

And similarly when a *normal* is given there will be two meridian planes perpendicular to each other.

Thus the planes passing through any radius and the two normals at the points where it pierces the wave surface, are *perpendicular* to each other, as are also the two planes passing through any normal and *its* two corresponding radii.

Moreover a glance at fig. 2 will show that the two lines of vibration corresponding to any front lie respectively in the two meridian planes passing through the perpendicular to that front or, in other words, the intersection of a plane drawn through either ray belonging to a front perpendicular thereunto is *always* a line of vibration in that front.

This has been noticed, I think, by Sir William Hamilton for the particular case of the singular points.

As two fronts belong to every ray, so two rays pertain to every front. And from what has been said above it appears that the two lines of vibration in any front are the projections of its two rays upon its own plane.

## Note 1.

In the paper above, it is shown that the meridian plane, that is, the plane containing the ray and normal, always passes through a line of vibration in the corresponding point. Now the line of force called into action by a displacement in the line of vibration clearly lies in this very plane; for the resolved part of it lies in the line of vibration itself.

Harmony and analogy concur in suggesting that as two of these four lines are perpendicular to each other, so are also the other two, or in other words, that the ray is always perpendicular to the direction of unresolved force.

The following investigation verifies this conjecture.

Let $x$, $y$, $z$ be the coordinates of a point taken at distance unity from the origin and in any line of vibration; then the cosines of the angles made by the line of force with the axes are as $a^2x : b^2y : c^2z$ respectively.

Let $a$ be the inclination between the line of vibration and the line of force, then

$$\cos\omega = \frac{a^2x \,.\, x + b^2y \,.\, y + c^2z \,.\, z}{\sqrt{\{(a^4x^2 + b^4y^2 + c^4z^2)(x^2 + y^2 + z^2)\}}} = \frac{a^2x^2 + b^2y^2 + c^2z^2}{\sqrt{(a^4x^2 + b^4y^2 + c^4z^2)}}.$$

Let $$\sqrt{(a^4x^2 + b^4y^2 + c^4z^2)} = P,$$

then $$P^2 = v^4 (\sec\omega)^2.$$

Now let $\alpha$, $\beta$, $\gamma$ be the angles of inclination between the coordinate planes and the front in which the line of vibration lies, and $\lambda$ some quantity to be determined. I have shown in Prop. 3 that if

$$\lambda\cos\alpha = (a^2 - v^2)\,x,$$

then will $$\lambda\cos\beta = (b^2 - v^2)\,y,$$

and $$\lambda\cos\gamma = (c^2 - v^2)\,z\,;$$

therefore $$\lambda^2 = a^4x^2 + b^4y^2 + c^4z^2 - 2v^2(a^2x^2 + b^2y^2 + c^2z^2) + v^4 = P^2 - v^4.$$

Again,

$$\lambda^2 \,.\left(\frac{(\cos\alpha)^2}{(a^2 - v^2)^2} + \frac{(\cos\beta)^2}{(b^2 - v^2)^2} + \frac{(\cos\gamma)^2}{(c^2 - v^2)^2}\right) = x^2 + y^2 + z^2 = 1\,;$$

therefore $$\frac{1}{P^2 - v^4} = \frac{(\cos\alpha)^2}{(a^2 - v^2)^2} + \frac{(\cos\beta)^2}{(b^2 - v^2)^2} + \frac{(\cos\gamma)^2}{(c^2 - v^2)^2}.$$

Now $$\frac{1}{P^2 - v^4} = \frac{1}{v^4(\sec\omega)^2 - v^4} = \frac{1}{v^4}(\cot\omega)^2.$$

And in Mr Smith's investigation of the form of the wave surface (already alluded to*) by great good fortune I find ready to my hand

$$\frac{(\cos\alpha)^2}{(a^2-v^2)^2}+\frac{(\cos\beta)^2}{(b^2-v^2)^2}+\frac{(\cos\gamma)^2}{(c^2-v^2)^2}=\frac{1}{v^2(r^2-v^2)},$$

$r$ being the radius vector to the point whose tangent plane is parallel to the point in question.

Hence

$$(\cot\omega)^2=\frac{v^4}{v^2(r^2-v^2)}=\frac{v^2}{r^2-v^2}=\frac{p^2}{r^2-p^2},$$

$p$ being the length of the perpendicular from the centre upon the tangent plane, for $p=v$.

Hence $(\cot\omega)^2$ = the square of the cotangent of the angle between radius vector and normal.

Or, in other words, the line of force is as much inclined to the line of vibration as the ray is to the normal.

Now the normal is perpendicular to the line of vibration, and all four lines lie in one plane.

Therefore the ray is perpendicular to the line of force. Q. E. D.

I may be allowed to conclude this long paper with a summary of some of the most remarkable consequences which I have extricated from Fresnel's hypothesis.

(1) The two meridian planes corresponding to any given radius are perpendicular to each other†.

(2) So are the two corresponding to any given normal.

(3) Every meridian plane bisects the angle formed by two planes drawn through the radius and the two prime radii.

(4) It also bisects the angle formed by two planes drawn through the normal and the two prime normals.

(5) Each meridian plane contains one line of vibration and the corresponding line of force.

(6) The ray is perpendicular to the line of force.

All these conclusions, except the fourth, are, I believe, original.

* See above, p. 18.

† I have defined the meridian plane to be that which contains radius vector and normal belonging to the same point.

The theory of external and internal conical refraction follows immediately as a particular consequence from the third and fourth combined as already shown; the same propositions also enable us to draw a tangent plane to any point of the wave surface by mere Euclidean geometry. May not some of these conclusions serve to suggest to physical inquirers the question, Has the theory been started from the most natural point of view?

## NOTE 2. *Investigation* of the Wave Surface.*

Since the appearance of the preceding parts, I have succeeded in completing the self-sufficiency of my method by deducing the equation to the wave surface from the expressions given in Prop. 5 for the angles between a front and the principal planes in terms of its two velocities. If these angles be $\omega$, $\phi$, $\psi$, and the two velocities $v_1$, $v_2$ we found

$$\cos\omega = \sqrt{\frac{(a^2-v_1^2)(a^2-v_2^2)}{(a^2-b^2)(a^2-c^2)}},$$

$$\cos\phi = \sqrt{\frac{(b^2-v_1^2)(b^2-v_2^2)}{(b^2-a^2)(b^2-c^2)}},$$

$$\cos\psi = \sqrt{\frac{(c^2-v_1^2)(c^2-v_2^2)}{(c^2-a^2)(c^2-b^2)}}.$$

Let the tangent plane to the wave surface be written

$$\frac{\cos\omega}{v_1}.x + \frac{\cos\phi}{v_1}.y + \frac{\cos\psi}{v_1}.z = 1, \qquad (\alpha)\dagger$$

then
$$\frac{d\dfrac{\cos\omega}{v_1}\ddagger}{d\left(\dfrac{1}{v_1}\right)^2}x + \frac{d\dfrac{\cos\phi}{v_1}}{d\left(\dfrac{1}{v_1}\right)^2}y + \frac{d\dfrac{\cos\psi}{v_1}}{d\left(\dfrac{1}{v_1}\right)^2}z = 0, \qquad (\beta)$$

$$\frac{d\cos\omega}{d(v_2)^2}x + \frac{d\cos\phi}{d(v_2)^2}y + \frac{d\cos\psi}{d(v_2)^2}z = 0. \qquad (\gamma)$$

Let
$$\frac{1}{v_1}\sqrt{\frac{(a^2-v_1^2)}{(a^2-v_2^2)}} = \xi, \qquad \sqrt{\{(a^2-b^2)(a^2-c^2)\}} = \frac{1}{A},$$

$$\frac{1}{v_1}\sqrt{\frac{(b^2-v_1^2)}{(b^2-v_2^2)}} = \eta, \qquad \sqrt{\{(b^2-a^2)(b^2-c^2)\}} = \frac{1}{B},$$

$$\frac{1}{v_1}\sqrt{\frac{(c^2-v_1^2)}{(c^2-v_2^2)}} = \zeta, \qquad \sqrt{\{(c^2-a^2)(c^2-b^2)\}} = \frac{1}{C},$$

* This investigation supplies the step which Mr Tovey was desirous should appear in the *Magazine*. [*Phil. Mag.* March, 1838, p. 261. ED.]

† In lieu of $v_1$ we might write $v_2$ in the denominator without affecting the result.

‡ Observe, that $\dfrac{\cos\omega}{v_1} = \dfrac{\sqrt{\left\{\left(\dfrac{a^2}{v_1^2}-1\right)(a^2-v_2^2)\right\}}}{\sqrt{\{(a^2-b^2)(a^2-c^2)\}}}$, and so on for the rest.

then equation ($\gamma$) becomes

$$A\xi x + B\eta y + C\zeta z = 0, \tag{1}$$

and equation ($\beta$)

$$\frac{Aa^2}{\xi} x + \frac{Bb^2}{\eta} y + \frac{Cc^2}{\zeta} z = 0, \tag{2}$$

and equation ($\alpha$) may be written under two forms, viz.

$$(a^2 - v_2^2) A\xi x + (b^2 - v_2^2) B\eta y + (c^2 - v_2^2) C\zeta z = 1, \tag{3}$$

or

$$\left(\frac{a^2}{v_1^2} - 1\right) \frac{A}{\xi} x + \left(\frac{b^2}{v_1^2} - 1\right) \frac{B}{\eta} y + \left(\frac{c^2}{v_1^2} - 1\right) \frac{C}{\zeta} z = 1. \tag{4}$$

From (1)

$$A\xi x + B\eta y = - C\zeta z. \tag{5}$$

From (2)

$$\frac{Aa^2}{\xi} x + \frac{Bb^2}{\eta} y = - \frac{Cc^2}{\zeta} z. \tag{6}$$

From (3) and (1)

$$A (a^2 - c^2) \xi x + B (b^2 - c^2) \eta y = 1. \tag{7}$$

From (2) and (4)

$$A (a^2 - c^2) \frac{x}{\xi} + B (b^2 - c^2) \frac{y}{\eta} = c^2. \tag{8}$$

From (5) and (6)

$$C^2c^2z^2 - B^2b^2y^2 - A^2a^2x^2 = ABxy \left(a^2 \frac{\eta}{\xi} + b^2 \frac{\xi}{\eta}\right). \tag{9}$$

From (7) and (8)

$$c^2 - B^2 (b^2 - c^2)^2 y^2 - A^2 (a^2 - c^2)^2 x^2 = ABxy \left(\frac{\eta}{\xi} + \frac{\xi}{\eta}\right) \times (a^2 - c^2)(b^2 - c^2). \tag{10}$$

From (9) and (10)

$$\begin{aligned} AB (a^2 - b^2) (a^2 - c^2) (b^2 - c^2) xy \frac{\xi}{\eta} &= a^2c^2 - (a^2 - c^2) (b^2 - c^2) C^2c^2z^2 \\ &\quad - \{a^2 (b^2 - c^2)^2 - b^2 (a^2 - c^2) (b^2 - c^2)\} B^2y^2 \\ &\quad - \{a^2 (a^2 - c^2)^2 - a^2 (a^2 - c^2) (b^2 - c^2)\} A^2x^2 = a^2c^2 - c^2z^2 - c^2y^2 - a^2x^2. \end{aligned} \tag{11}$$

From (11), interchanging ($a$, $x$, $\xi$) with ($b$, $y$, $\eta$) we have

$$AB (b^2 - a^2) (b^2 - c^2) (a^2 - c^2) xy \frac{\eta}{\xi} = b^2c^2 - c^2z^2 - c^2x^2 - b^2y^2. \tag{12}$$

Finally, from (11) and (12) we have

$$\begin{aligned} \{a^2c^2 - (a^2 - c^2) x^2 - c^2 (x^2 + y^2 + z^2)\} \{b^2c^2 - (b^2 - c^2) y^2 - c^2 (x^2 + y^2 + z^2)\} \\ = (a^2 - c^2) (b^2 - c^2) x^2y^2, \end{aligned}$$

that is

$$\begin{aligned} (x^2 + y^2 + z^2) (a^2x^2 + b^2y^2 + c^2z^2) - a^2 (b^2 + c^2) x^2 \\ - b^2 (a^2 + c^2) y^2 - c^2 (b^2 + a^2) z^2 + a^2b^2c^2 = 0 \end{aligned}$$

the equation required.

# 2.

## ON THE MOTION AND REST OF FLUIDS.

[*Philosophical Magazine*, XIII. (1838), pp. 449—453.]

M. OSTROGRADSKY'S memoir on this subject inserted in the *Scientific Memoirs* seems to have excited much attention, and has been made the occasion of some annotations* by a distinguished writer in the *Philosophical Magazine.* Mr Ivory's recent papers in the same periodical must still more tend to invest with a new interest all such speculations. It seems to me desirable therefore to present the theory of fluids in all the simplicity of which it is susceptible.

I consider a fluid as a collection of particles subject to some law of relative position other than that of rigidity. These particles by their mutual actions maintain the connections of the system. As to the law of force between them we know nothing; but I assume it is a general principle of nature, that for each instant of time the sum of the internal actions (reckoned by the product of each particle into the square of the space due to the internal force acting on it) is a minimum. This in fact is Gauss's principle of least restraint. We may if we please split this principle into two parts; that is to say, assume that the internal system of forces is always such as if acting alone would keep the fluid at rest; and then again assume that any equilibrating system of forces must be subject to the law of virtual velocities. I say *assume,* because it is impossible *à priori* to prove this.

Lagrange's so-called demonstration is unworthy of his name, and (albeit sanctioned by the powerful oral authority of an ex-Cambridge Professor) contrary alike to sense and honesty. It is better therefore at once to proceed upon Gauss's principle. It might easily be shown that this is in effect tantamount in all cases to D'Alembert's and Lagrange's principles combined.

Before entering upon the investigation I may call attention to one point of great analytical interest, and relating to the difficult subject of the algebraical sign, viz. that if the density of a point $(x, y)$ in any circumscribed space be expressed by the quantity $\frac{du}{dx}+\frac{dv}{dy}$ so that the mass is

$$\iint dx\,dy\left(\frac{du}{dx}\right)+\iint dx dy\left(\frac{dv}{dy}\right),$$

[* *Phil. Mag.* May, 1838, p. 385. ED.]

that is not equivalent to

$$\int (udy + vdx),$$

that is if we please

$$\int \left(u\frac{dy}{ds} + v\frac{dx}{ds}\right) ds,$$

(where $s$ is for clearness' sake and to avoid double limits taken an element of the bounding curve) as at first sight it might appear to be, but is in fact equal to

$$\int \left(u\frac{dy}{ds} - v\frac{dx}{ds}\right) ds.$$

I shall demonstrate this point in the next number* of the *Magazine*. It at first caused me some trouble in conducting the annexed inquiry. I shall also take occasion at some other time to revert to a new species (as I believe) of partial differential equations; that is to say, where there are fewer of them than of the principal variables, which may be called therefore Indeterminate Partial Differential Equations. A complete solution of one of these appears in the subjoined

*Investigation.*

For the sake of simplicity I take an incompressible fluid. The method is nowise different for a fluid of varying density.

Let $\Delta x$, $\Delta y$, $\Delta z$ be any displacement undergone by a particle at the point $x, y, z$ parallel to the axes $x, y, z$ respectively; it is easily shown that to satisfy the condition of invariability of mass we must have

$$\frac{d\Delta x}{dx} + \frac{d\Delta y}{dy} + \frac{d\Delta z}{dz} = 0. \qquad (\alpha)$$

One relation between $u$, $v$, $w$ the velocities parallel to $x$, $y$, $z$ is obtained immediately by putting $u\delta t$, $v\delta t$, $w\delta t$, for $\Delta x$, $\Delta y$, $\Delta z$, which gives

$$\frac{du}{dx} + \frac{dv}{dy} + \frac{dw}{dz} = 0, \qquad (1)$$

as usual.

Again, if $X$, $Y$, $Z$ be the impressed forces, and $X_1$, $Y_1$, $Z_1$ the internal forces acting on any particle parallel to the axes, we have

$$X_1 + X = \frac{du}{dt} + \frac{du}{dx}u + \frac{du}{dy}v + \frac{du}{dz}w, \qquad (2)$$

$$Y_1 + Y = \frac{dv}{dt} + \frac{dv}{dx}u + \frac{dv}{dy}v + \frac{dv}{dz}w, \qquad (3)$$

$$Z_1 + Z = \frac{dw}{dt} + \frac{dw}{dx}u + \frac{dw}{dy}v + \frac{dw}{dz}w, \qquad (4)$$

from the mere geometry of the question.

[* p. 36, below. Ed.]

Finally, Gauss's principle teaches us that

$$\iiint dxdydz\,\{X_1\Delta X_1+Y_1\Delta Y_1+Z_1\Delta Z_1\}=0. \qquad (\beta)$$

Now
$$\frac{d(X+X_1)}{dx}+\frac{d(Y+Y_1)}{dy}+\frac{d(Z+Z_1)}{dz}$$

$$=\left(\frac{du}{dx}\right)^2+\left(\frac{dv}{dy}\right)^2+\left(\frac{dw}{dz}\right)^2+2\left\{\frac{dv}{dz}\frac{dw}{dy}+\frac{dw}{dx}\frac{du}{dz}+\frac{du}{dy}\frac{dv}{dx}\right\},$$

as appears from the equations (1), (2), (3), (4); and hence

$$\frac{d\Delta X_1}{dx}+\frac{d\Delta Y_1}{dy}+\frac{d\Delta Z_1}{dz}=0,$$

the complete solution of which, free from the sign of integration, is

$$\Delta X_1=\frac{d\psi}{dy}-\frac{d\phi}{dz},$$

$$\Delta Y_1=\frac{d\omega}{dz}-\frac{d\psi}{dx},$$

$$\Delta Z_1=\frac{d\phi}{dx}-\frac{d\omega}{dy},$$

$\omega$, $\phi$, $\psi$ being any three independent functions of $x$, $y$, $z$.

On substituting these values in equation ($\beta$) we obtain

$$\iiint dxdydz\left\{X_1\frac{d\psi}{dy}-Y_1\frac{d\psi}{dx}\right\}+\iiint dxdydz\left\{Y_1\frac{d\omega}{dz}-Z_1\frac{d\omega}{dy}\right\}$$
$$+\iiint dxdydz\left\{Z_1\frac{d\phi}{dx}-X_1\frac{d\phi}{dz}\right\}=0.$$

This may be put under the form

$$\int dz\iint dxdy\left\{\frac{d}{dy}(\psi X_1)-\frac{d}{dx}(\psi Y_1)\right\}$$
$$+\int dx\iint dydz\left\{\frac{d}{dz}(\omega Y_1)-\frac{d}{dy}(\omega Z_1)\right\}$$
$$+\int dy\iint dzdx\left\{\frac{d}{dx}(\phi Z_1)-\frac{d}{dz}(\phi X_1)\right\}$$
$$-\iiint dxdydz\,.\,\psi\left(\frac{dX_1}{dy}-\frac{dY_1}{dx}\right)$$
$$-\iiint dxdydz\,.\,\omega\left(\frac{dY_1}{dz}-\frac{dZ_1}{dy}\right)$$
$$-\iiint dxdydz\,.\,\phi\left(\frac{dZ_1}{dx}-\frac{dX_1}{dz}\right)=0.$$

Here it must be remembered that $\omega$, $\phi$, $\psi$ are perfectly independent of each other. Also the values of the three first written quantities depend upon the values of $X_1$, $Y_1$, $Z_1$ at the bounding surface; the values of the three last-written depend upon the general values of $X_1$, $Y_1$, $Z_1$. It is clear therefore that each system of three equations and each member of each system must be separately zero.

The three latter equations give

$$\left.\begin{aligned} \frac{dX_1}{dy} - \frac{dY_1}{dx} = 0 \\ \frac{dY_1}{dz} - \frac{dZ_1}{dy} = 0 \\ \frac{dZ_1}{dx} - \frac{dX_1}{dz} = 0 \end{aligned}\right\}. \qquad (\gamma)$$

The three former require that for each section of the surface parallel to the plane $xy$

$$\int \psi\,(X_1 dx + Y_1 dy) = 0,$$

for each section parallel to $yz$

$$\int \omega\,(Y_1 dy + Z_1 dz) = 0, \qquad (\delta)^*$$

for each section parallel to $zx$

$$\int \phi\,(Z_1 dz + X_1 dx) = 0$$

and these equations are to hold good whatever $\psi$, $\phi$, $\omega$ may be. From the equations ($\gamma$) we derive

$$X_1 dx + Y_1 dy + Z_1 dz = df, \qquad (5)$$

from equations ($\delta$) we obtain

$f$ = constant for all points in any section of the bounding surface parallel to the plane of $xy$,

$f$ = constant for all points in any section of the bounding surface parallel to the plane of $yz$,

$f$ = constant for all points in any section of the bounding surface parallel to the plane of $zx$.

Now by drawing through all the points in a plane parallel to $xy$, planes parallel to $yz$, we may cover the whole surface; hence $f$ is constant all over the surface bounding the fluid.

* See remark at introduction.

Therefore $$X_1 dx + Y_1 dy + Z_1 dz = 0, \tag{6}$$
for all variations of $dx$, $dy$, $dz$ taken upon the surface.

The equations (1, 2, 3, 4, 5, 6) are coincident with those obtained by the usual method; with this difference, that $X_1$, $Y_1$, $Z_1$ here take the place of

$$-\frac{dp}{dx}, \; -\frac{dp}{dy}, \; -\frac{dp}{dz}.$$

Thus then we have obtained all the conditions requisite for determining the motion of fluids from the universal principle of least constraint conjoined with the specific character of the system in question.

### *General Remarks.*

In the case of equilibrium, that is in the case where no particle moves, we have $X_1 + X = 0$, $Y_1 + Y = 0$, $Z_1 + Z = 0$. Hence $Xdx + Ydy + Zdz$ is a complete differential always and zero for the surface.

The above results have been obtained upon the principles of the differential calculus, and the continuity of the forces has been tacitly assumed. If now we were to suppose forces of finite magnitude (as compared with the *whole sum* acting upon the entire system) to be applied to a layer of single particles or to a layer of a thickness of the same order of magnitude as the distances between the particles themselves, (which has been treated as an infinitesimal) it would appear that our results would be no longer applicable, just in the same manner as it would be erroneous to apply the principle of *vis-viva* (for example) without modification, to the case of impulsive forces, because we had deduced it by the calculus in the case of the motion being continuous. Hence the above equations ought not strictly to apply to the motion or rest of a fluid *contained between physical surfaces*; for the pressure afforded by these surfaces, whatever its actual value may be, we know *à priori* is commensurable with the whole amount of force acting on the fluid; but the immediate application of this pressure (*alias* repulsive force) is confined to the bounding layer of fluid particles, or at most extends to a distance bearing a low ratio to the distances between the particles themselves.

Accordingly, to the non-applicability of the equations for free fluids to the case of fluids confined at the boundaries, and to an independent investigation upon the minimum principle for this class of problems, it is, that I look for the true explanation of the phenomena of capillary attraction (vulgarly so called).

# 3.

## ON THE MOTION AND REST OF RIGID BODIES.

[*Philosophical Magazine*, XIV. (1839), pp. 188—190.]

IN the subjoined investigation, which, as far as I know, is my own, I apply the same method to rigid as in the preceding paper I applied to fluid systems.

Let $x$, $y$, $z$ be the coordinates of any particle in a rigid body; $x'$, $y'$, $z'$ the coordinates of some other particle, and let

$$x' = x + h, \quad y' = y + k, \quad z' = z + l.$$

Call $\Delta x$, $\Delta y$, $\Delta z$ the increments which $x$, $y$, $z$ receive after the lapse of a small interval of time; so that terms in which they enter in two or more dimensions may be neglected.

Then
$$\Delta(x') = \Delta x + \frac{d\Delta x}{dx} h + \frac{d\Delta x}{dy} k + \frac{d\Delta x}{dz} l + P,$$

$$\Delta(y') = \Delta y + \frac{d\Delta y}{dx} h + \frac{d\Delta y}{dy} k + \frac{d\Delta y}{dz} l + Q,$$

$$\Delta(z') = \Delta z + \frac{d\Delta z}{dx} h + \frac{d\Delta z}{dy} k + \frac{d\Delta z}{dz} l + R,$$

$P$, $Q$, $R$ containing binary and higher combinations of $h$, $k$, $l$, which we shall have no occasion to express.

At the commencement of the interval the squared distance of the two particles was $(x'-x)^2 + (y'-y)^2 + (z'-z)^2$; at the end of the interval the distance squared is

$$(x' - x + \Delta(x') - \Delta x)^2 + (y' - y + \Delta(y') - \Delta y)^2 + (z' - z + \Delta(z') - \Delta z)^2,$$

and these two expressions must be the same by the conditions of rigidity whatever $h$, $k$, and $l$ may be; that is

$$h^2 + k^2 + l^2 = \left(h + \frac{d\Delta x}{dx} h + \frac{d\Delta x}{dy} k + \frac{d\Delta x}{dz} l + P\right)^2$$
$$+ \left(k + \frac{d\Delta y}{dx} h + \frac{d\Delta y}{dy} k + \frac{d\Delta y}{dz} l + Q\right)^2$$
$$+ \left(l + \frac{d\Delta z}{dx} h + \frac{d\Delta z}{dy} k + \frac{d\Delta z}{dz} l + R\right)^2,$$

for all values of $h$, $k$, and $l$.

Hence rejecting infinitesimals of the second order and equating to zero separately the coefficients of $h^2$, $k^2$, $l^2$, and of $kl$, $lh$, $hk$, we have

$$\frac{d\Delta x}{dx} = 0. \quad (a) \qquad \frac{d\Delta y}{dz} + \frac{d\Delta z}{dy} = 0. \quad (d)$$

$$\frac{d\Delta y}{dy} = 0. \quad (b) \qquad \frac{d\Delta z}{dx} + \frac{d\Delta x}{dz} = 0. \quad (e)$$

$$\frac{d\Delta z}{dz} = 0. \quad (c) \qquad \frac{d\Delta x}{dy} + \frac{d\Delta y}{dx} = 0. \quad (f)$$

By differentiating $(d)$, $(e)$, $(f)$ with respect to $z$, $x$, $y$ respectively, and substituting from $(a)$, $(b)$, $(c)$, we obtain

$$\frac{d^2\Delta y}{dz^2} = 0, \quad \frac{d^2\Delta z}{dx^2} = 0, \quad \frac{d^2\Delta x}{dy^2} = 0.$$

By differentiating the same with respect to $y$, $z$, $x$ respectively, and proceeding as before, we have

$$\frac{d^2\Delta z}{dy^2} = 0, \quad \frac{d^2\Delta x}{dz^2} = 0, \quad \frac{d^2\Delta y}{dx^2} = 0.$$

Thus, then, we have

$$\frac{d\Delta x}{dx} = 0, \quad \frac{d^2\Delta x}{dy^2} = 0, \quad \frac{d^2\Delta x}{dz^2} = 0,$$

$$\frac{d\Delta y}{dy} = 0, \quad \frac{d^2\Delta y}{dz^2} = 0, \quad \frac{d^2\Delta y}{dx^2} = 0,$$

$$\frac{d\Delta z}{dz} = 0, \quad \frac{d^2\Delta z}{dx^2} = 0, \quad \frac{d^2\Delta z}{dy^2} = 0,$$

therefore

$$\Delta x = A + By + Cz, \qquad (o)$$

$$\Delta y = D + Ez + Fx, \qquad (p)$$

$$\Delta z = G + Hx + Ky, \qquad (q)$$

$A$, $B$, $C$, $D$, $E$, $F$, being constant for a *given instant* of time; between which by virtue of the equations $(d)$, $(e)$, $(f)$, we have the relations

$$E + K = 0, \qquad H + C = 0, \qquad B + F = 0.$$

If we call $u$, $v$, $w$ the three component velocities of the particles at $x$, $y$, $z$ parallel to the three axes, and $X_1$, $Y_1$, $Z_1$, the three internal forces, it is at once seen that $u$, $v$, $w$, as also $\Delta X_1$, $\Delta Y_1$, $\Delta Z_1$ must be subject to the same equations as limit $\Delta x$, $\Delta y$, $\Delta z$; so that

$$u = a + \gamma y - \beta z, \qquad (1)$$

$$v = b + \alpha z - \gamma x, \qquad (2)$$

$$w = c + \beta x - \alpha y, \qquad (3)$$

$$\Delta X_1 = a_1 + \gamma_1 y - \beta_1 z, \qquad (h)$$

$$\Delta Y_1 = b_1 + \alpha_1 z - \gamma_1 x, \qquad (j)$$

$$\Delta Z_1 = c_1 + \beta_1 x - \alpha_1 y. \qquad (k)$$

Also if $X$, $Y$, $Z$ be the impressed forces, we have

$$X_1 + X = \frac{du}{dt}, \tag{4}$$

$$Y_1 + Y = \frac{dv}{dt}, \tag{5}$$

$$Z_1 + Z = \frac{dw}{dt}. \tag{6}$$

And by Gauss's principle, calling $m$ the mass of the particle at $x$, $y$, $z$,

$$\Delta \Sigma m (X_1^2 + Y_1^2 + Z_1^2) = 0.$$

Hence equating separately to zero the coefficients of $a_1$, $b_1$, $c_1$ and of $\alpha_1$, $\beta_1$, $\gamma_1$ in the quantity $\Sigma m (X_1 \Delta X_1 + Y_1 \Delta Y_1 + Z_1 \Delta Z_1)$ we have

$$\left.\begin{aligned} \Sigma m\, X_1 &= 0 \\ \Sigma m\, Y_1 &= 0 \\ \Sigma m\, Z_1 &= 0 \\ \Sigma m\, (Z_1 y - Y_1 z) &= 0 \\ \Sigma m\, (X_1 z - Z_1 x) &= 0 \\ \Sigma m\, (Y_1 x - X_1 y) &= 0 \end{aligned}\right\}. \tag{7—12}$$

Lastly, we have the equations

$$u = \frac{dx}{dt}, \tag{13}$$

$$v = \frac{dy}{dt}, \tag{14}$$

$$w = \frac{dz}{dt}. \tag{15}$$

From the fifteen equations marked (1) to (15), the motion may be determined by assigning the position of each particle at the end of the time $t$ in terms of its three initial coordinates, its three initial velocities, and the initial values of the nine quantities

$$\begin{array}{lll} \Sigma mx, & \Sigma myz, & \Sigma mx^2, \\ \Sigma my, & \Sigma mzx, & \Sigma my^2, \\ \Sigma mz, & \Sigma mxy, & \Sigma mz^2. \end{array}$$

In the case of rest $X_1 = -X$, $Y_1 = -Y$, $Z_1 = -Z$, and the equations (7) to (12) inclusively taken, express the conditions of equilibrium.

The equations ($o$), ($p$), ($q$), which have been obtained from conditions *purely geometrical,* establish the well-known but interesting and *not obvious* fact, that any *small* motion of a rigid body may be conceived as made up of a motion of translation and a motion about *one* axis.

# 4.

## ON DEFINITE DOUBLE INTEGRATION, SUPPLEMENTARY TO A FORMER PAPER ON THE MOTION AND REST OF FLUIDS.

[*Philosophical Magazine*, XIV. (1839), pp. 298—300.]

IN a paper on Fluids which appeared in the December Number of this *Magazine*, I had occasion to remark, that the mass of an area having at the point $(x, y)$ a density $\frac{du}{dx} + \frac{dv}{dy}$ could be expressed by the simple formula

$$\int_l^0 \left\{ u \frac{dy}{ds} - v \frac{dx}{ds} \right\} ds;$$

$l$ being the length, and $ds$ an element of the bounding curve: this may be thought to require some explanation.

(1) Let $APBq$ represent any oval; $PpL$, $QqM$ any two contiguous ordinates cutting the curve in $P$, $p$, $Q$, $q$ respectively, $AC$, $BD$ the two extreme tangents parallel to $Oy$, and $\rho$ the density at any point $(x, y)$. The expression $\iint \rho dx dy$ will serve to denote the mass of the oval area $APBq$, and the limits may be twice taken, that is, (i) the two values of $y$ corresponding to any one of $x$; and (ii) the two values of $x$ corresponding to $C$ and $D$. This method is in fact tantamount to taking the sum of the columns $Pp$ $qQ$; but this is not necessary, for $APBq$ may be considered as the *algebraical* sum of the mixtilinear area $APQBDC$, and the mixtilinear area $BDCApq$, or (if any line $O'C'D'$ be drawn parallel to $OCLMD$) of $APQBD'C'$ and $BD'C'Apq$.

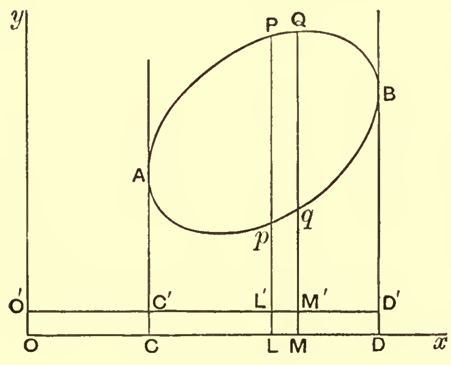


Fig. 1.

Thus then the mass $= \int dx \left(\int \rho dy\right)$, $\int \rho dy$ being left indeterminate, and the extremity of $x$ travelled round from $C$ to $D$, and back again from $D$ to $C$.

This will be better expressed by transforming the variable, and summing with respect to some quantity, such as the arc of the curve, which continuously increases, or if we please, with respect to $\theta$, the angle subtending any point taken within the curve.

The mass is then

$$= \pm \int_{2\pi}^{0} d\theta \left\{(\int \rho dy) \frac{dx}{d\theta}\right\};$$

always remembering that *no* constant need be added to $\int \rho dy$, and that the doubtful sign arises from the choice of ways in which $\theta$ may be measured round. If the area be not included by one line; but by several, as for example, by a curve and a right line, the above integral, if broken up into as many parts as there are breaches of continuity, will still apply.

(2) Let us suppose that we have two areas exactly coinciding with, and overlapping one another; but the density of the one at $(x, y)$ to be $\rho$, and of the other $\rho'$.

Let the mass of the first be treated as the sum of columns parallel to $Oy$, and that of the second as the sum of columns parallel to $Ox$.

The one will be represented by

$$\pm \int_{2\pi}^{0} d\theta \, (\int \rho dy) \frac{dx}{d\theta},$$

the other will be represented by

$$\pm \int_{2\pi}^{0} d\theta \, (\int \rho' dx) \frac{dy}{d\theta},$$

and the sum of the two, or the joint mass, by

$$\pm \int_{2\pi}^{0} d\theta \, \{\int \rho dy\} \frac{dx}{d\theta} \pm \int_{2\pi}^{0} d\theta \, \{\int \rho' dx\} \frac{dy}{d\theta}.$$

So long as these two operations are performed separately, the doubtful signs may be preserved in each term, because $s$ need not be travelled round in the same direction for the two summations; but if we perform the second integration conjointly for the two masses, their sum

$$= \pm \int_{2\pi}^{0} d\theta \left\{(\int \rho dy) \frac{dx}{d\theta} \pm ? \, (\int \rho' dx) \frac{dy}{d\theta}\right\},$$

the mark of interrogation denoting that *one or the other*, but not *either* of the signs $\pm$ must be used, and the question is, which ?

This will be answered by taking different points in the bounding line which may be continuous or not. Now every line returning into itself, whether continuous or not, will naturally divide with respect of any given system of axes, into at most four parts, or sets of parts; two in which $dx$ and $dy$ both increase or both decrease, and two in which one increases and the other decreases.

Take $P_1$, $P_2$, $P_3$, $P_4$, any points in the four quadrants respectively, it will be observed that,

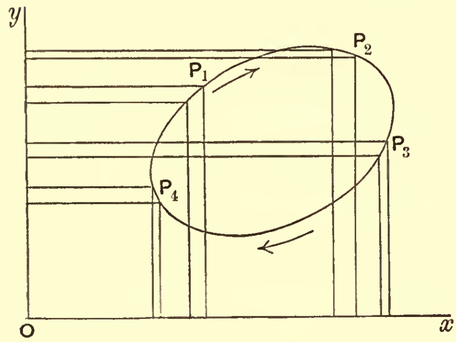


Fig. 2.

At $P_1$ the $\rho$ column enters additively, and the $\rho'$ column subtractively.

At $P_2$ both columns are additive.

At $P_3$ the $\rho'$ column is additive and the $\rho$ column subtractive.

At $P_4$ both columns enter subtractively.

Again, reckoning round in the direction of the arrows,

At $P_1$, $x$ and $y$ are both increasing.

At $P_2$, $x$ is increasing and $y$ decreasing.

At $P_3$, $x$ and $y$ both decrease.

At $P_4$, $x$ is decreasing and $y$ increasing.

Thus when $\int\rho dy$ and $\int\rho' dx$ are affected with the same signs, $dx$ and $dy$ are of opposite signs; and when $\int\rho dy$, $\int\rho' dx$ are of opposite signs, $dx$ and $dy$ are of the same sign.

Hence it appears that the mass of the area, whose density at $(x, y)$ is $\rho + \rho'$, is capable of being represented by

$$\pm \int_{2\pi}^{0} d\theta \left\{ (\int\rho dy) \frac{dx}{d\theta} - (\int\rho' dx) \frac{dy}{d\theta} \right\}.$$

# 5.

## ON AN EXTENSION OF SIR JOHN WILSON'S THEOREM TO ALL NUMBERS WHATEVER.

[*Philosophical Magazine,* XIII. (1838), p. 454.]

THE annexed original theorem in numbers will serve as a pendant to the elegant discovery announced by the ever-to-be-lamented and commemorated Horner*, with his dying voice, in your valued pages†.

THEOREM.

If $N$ be any number whatever and

$$p_1, p_2, p_3 \cdots\cdots p_c$$

be all the numbers less than $N$ and prime to it, then either

$$p_1 \cdot p_2 \cdot p_3 \cdots\cdots p_c + 1,$$

or else
$$p_1 \cdot p_2 \cdot p_3 \cdots\cdots p_c - 1,$$
is a multiple of $N$.

# 6.

## NOTE TO THE FOREGOING.

[*Philosophical Magazine,* XIV. (1839), pp. 47, 48.]

I HAVE to apologize for calling "original" (in the last Number of the *Magazine*) the theorem of numbers which I termed "a pendant to Horner's theorem." This Mr Ivory has done me the honour to inform me may be found in Gauss's *Disquisitiones Arithmeticæ,* p. 76. As Horner's extension of Fermat's theorem suggested this extension of Sir John Wilson's to me, so I concluded that had this extension of Wilson's been known to the world it would naturally have suggested his to Horner. No acknowledgment of this kind having been made, I took it for granted that the theorem I gave was new. Undoubtedly had Mr Horner been aware of Gauss's theorem he would have made mention of it.

I take this opportunity of adding that my acquaintance with Gauss's principle‡ has not been derived from the study of his works, but from a casual statement of it in an English work, *Dynamics*, by Mr Earnshaw, of St John's College, Cambridge.

* Horner's proof is highly valuable as a novel and highly ingenious form of reasoning, but his theorem may be deduced with infinitely more ease and brevity from Fermat's than he seems to have been aware of.

[† *Phil. Mag.* Vol. XI. p. 456. ED.]

[‡ See p. 28 above. ED.]

# 7.

## ON RATIONAL DERIVATION FROM EQUATIONS OF COEXISTENCE, THAT IS TO SAY, A NEW AND EXTENDED THEORY OF ELIMINATION*. PART I.

[*Philosophical Magazine*, XV. (1839), pp. 428—435.]

ANY number of equations existing at the same time and having the same quantities repeated, may be termed equations of coexistence: in the *present* paper we consider only the case of *two* algebraical equations:

$$x^m + a_1 x^{m-1} + a_2 x^{m-2} + \ldots\ldots + a_m = 0,$$

$$x^n + b_1 x^{n-1} + b_2 x^{n-2} + \ldots\ldots + b_n = 0.$$

The above being "equations of coexistence," $x$ is called "the repeating term."

If we suppose the equation

$$c_0 x^r + c_1 x^{r-1} + c_2 x^{r-2} + \ldots\ldots + c_r = 0$$

to be capable of being deduced from the two above, and, therefore, necessarily implied by them, this will be called "a Particular Derivative" from the equations of coexistence, of the $r$th degree, ($r$ being supposed less than $m$ and $n$†, and the coefficients being *rational* functions of the coefficients of the equations of coexistence).

There will be an *indefinite* number in general of such derivatives, and the form involving arbitrary quantities which includes them *all* is called "the general derivative of the $r$th degree."

Any "Particular Derivative," in which the terms are all integral, numerically as well as literally speaking, is called an "Integral Derivative."

That "Integral Derivative" of any given degree in which the literal parts of the coefficients are of the *lowest possible dimensions*‡, and the numerical parts as low as they can be made, is called the "Prime Derivative"

[* The results of this and some following papers were repeated, with demonstrations, in the paper "On a Theory of the Syzygetic Relations of two rational integral functions comprising an application to the Theory of Sturm's Functions, and that of the greatest Algebraical Common Measure," *Phil. Trans. Royal Soc.* Vol. CXLIII., Part I. pp. 407—548, 1853. See below Section II. Art. (16) of that paper. ED.]

† This restriction upon the value of $r$ is not essentially requisite, and is only introduced to keep the attention fixed upon the particular objects of this first Part.

‡ Of course the dimensions of the coefficients in the equations of coexistence are to be understood as denoted by the indices subscribed.

of that degree. So that there is nothing left ambiguous in the prime derivative save the sign.

The "Derivative by succession" is that particular derivative which is obtained by performing upon the equations of coexistence the process commonly employed for the discovery of the greatest common measure, and equating the *successive remainders* to zero.

To express the product of the sums formed by adding each of one row of quantities to each of another row, we simply write the one row above the other; a notation clearly capable of extension to any number of rows, which would not be the case if we spoke of *differences* instead of *sums**.

THEOREM 1.

Let $h_1, h_2, \ldots h_m$, be the roots of one equation of coexistence, $k_1, k_2, \ldots k_n$, the roots of the other. The general derivative of the $r$th degree is represented by

$$\Sigma\left(SR(h_1, h_2, h_3 \ldots h_r)\{(x-h_1)(x-h_2)\ldots(x-h_r)\} \times \begin{Bmatrix} h_{r+1}, h_{r+2} \ldots h_m \\ -k_1, -k_2 \ldots -k_n \end{Bmatrix}\right) = 0,$$

$SR(h_1, h_2, h_3 \ldots h_r)$ denoting any *symmetrical rational* (integral or fractional) function of $h_1, h_2 \ldots h_r$;

$$\begin{Bmatrix} h_{r+1}, h_{r+2} \ldots h_m \\ -k_1, -k_2 \ldots -k_n \end{Bmatrix}$$

being to be interpreted as above explained, and $\Sigma$ of course including as many terms as there are ways of putting $n$ things $r$ and $r$ together†.

A form tantamount to the above, and which may be substituted for it, is its analogue,

$$\Sigma\left(SR(k_1, k_2 \ldots k_r)\{(x-k_1)(x-k_2)\ldots(x-k_r)\} \times \begin{Bmatrix} k_{r+1}, k_{r+2} \ldots k_n \\ -h_1, -h_2 \ldots -h_m \end{Bmatrix}\right) = 0.$$

When $r = 0$ the theorem gives simply

$$\begin{Bmatrix} h_1, h_2 \ldots h_m \\ -k_1, -k_2 \ldots -k_n \end{Bmatrix} = 0,$$

and is coincident with that given by Bezout in his Theory of *Elimination*.

* The wider views which I have attained since writing the above, and which will be developed in a future paper, lead me to request that this notation may be considered only as temporary. It would have been more in accordance with these views to have used the two rows to denote products of differences than of sums. But a change now in the text would be very apt to cause errors in printing.

† The general derivative may clearly be expressed also by the sum of any two particular derivatives affected respectively with arbitrary rational coefficients. The equivalency of an arbitrary *function* to two arbitrary *multipliers* is very remarkable, and analogous to what occurs in the solution of certain differential equations.

*Subsidiary Theorem* (A).

If $h_1, h_2 \ldots h_m$ be the roots of the equation

$$x^m + a_1 x^{m-1} + a_2 x^{m-2} + \ldots\ldots + a_m = 0,$$

and if

$$e^m + a_1 e^{m-1} + a_2 e^{m-2} + \ldots\ldots + a_m - u = 0,$$

then

$$\Sigma \frac{h_1^r}{(h_1 - h_2)(h_1 - h_3) \ldots (h_1 - h_m)} = \frac{1}{r+1} \frac{d}{du} \Sigma (e^{r+1}),$$

$u$ being made zero after differentiation.

COR. If $R(h_1)$ denote any integral rational function of $h_1$, then

$$\Sigma \frac{R(h_1)}{(h_1 - h_2)(h_1 - h_3) \ldots (h_1 - h_m)}$$

is *always* integral and is *zero* when the dimensions of $R(h_1)$ fall short of $(m-1)$.

*Subsidiary Theorem* (B).

$$\Sigma \frac{SR(h_1, h_2 \ldots h_r)}{\begin{Bmatrix} h_1, h_2 \ldots h_r \\ -h_{r+1}, -h_{r+2} \ldots -h_m \end{Bmatrix}}$$

can be expressed by the sum of terms, each of which is the product of series of the form

$$\Sigma \frac{R(h_1)}{(h_1 - h_2)(h_1 - h_3) \ldots (h_1 - h_m)},$$

it is *always* integral, and when the dimensions of the numerator fall short of $(m-r)r$ it vanishes*.

*Subsidiary Theorem* (C).

The *only* modes of satisfying the equation

$$\Sigma \{f(h_1, h_2 \ldots h_r) \times SR(h_1, h_2 \ldots h_r)\} = 0,$$

for all forms of the latter factors short of $(m-r)(n-r)$ dimensions, are to put $f(h_1, h_2 \ldots h_r) = 0$, or else

$$f(h_1, h_2 \ldots h_r) = \frac{\text{constant}}{\begin{pmatrix} h_1, h_2 \ldots h_r \\ -h_{r+1}, -h_{r+2} \ldots -h_m \end{pmatrix}}.$$

* It may be remarked also in passing, that any term in the numerator which contains *any* one power not greater than $m - 2r$ may be neglected and thrown out of calculation. Moreover, an analogous proposition may be stated of fractions in the denominators of which *any number* of rows are written one under the other; see the first note, page 41.

THEOREM 2.

By virtue of the subsidiary theorem (B), the two equations

$$\pm\Sigma\left((x-h_1)(x-h_2)\dots(x-h_r)\times\frac{\begin{Bmatrix}h_{r+1}, h_{r+2}\dots h_m\\ -k_1, -k_2\dots -k_n\end{Bmatrix}}{\begin{Bmatrix}h_{r+1}, h_{r+2}\dots h_m\\ -h_1, -h_2\dots -h_r\end{Bmatrix}}\right)=0,$$

$$\pm\Sigma\left((x-k_1)(x-k_2)\dots(x-k_r)\times\frac{\begin{Bmatrix}k_{r+1}, k_{r+2}\dots k_n\\ -h_1, -h_2\dots -h_m\end{Bmatrix}}{\begin{Bmatrix}k_{r+1}, k_{r+2}\dots k_n\\ -k_1, -k_2\dots -k_r\end{Bmatrix}}\right)=0,$$

are each integer derivatives of the $r$th degree.

THEOREM 3.

And by virtue of the subsidiary theorem (C), the two above equations are the "Prime Integer Derivatives," and are exactly identical with each other.

COR. 1. The leading coefficient of the "prime derivative" of the $r$th degree is always of $(m-r)(n-r)$ dimensions.

COR. 2. If $P_r$ be the prime derivative of the $r$th degree and if $(X=0, Y=0)$ be the two equations of coexistence, and $\lambda_r$, $\mu_r$ the two "prime constituents of multiplication" to the said derivative, that is if $\lambda_r$ and $\mu_r$ satisfy the equation $\lambda_r X+\mu_r Y=P_r$, then the coefficient of the leading terms in $\lambda_r$ and in $\mu_r$ is of $(m-r-1)(n-r-1)$ dimensions.

THEOREM 4.

The "Prime Derivative" of any given degree is an exact factor of the "derivative by succession," of the same degree. The quotient resulting from striking out this factor is called "the quotient of succession."

THEOREM 5.

If $L_1$, $L_2$, $L_3$, &c., be the leading coefficients of the derivatives occurring first, second, third, &c., in order after the equations of coexistence, and if $Q_1$, $Q_2$, $Q_3$, &c., represent the first, second, third, "quotients of succession" reckoned in the same order, then

$$Q_1=1,$$

$$Q_2=\frac{1}{L_1^2},$$

$$Q_3=\frac{L_1^4}{L_2^2},$$

$$Q_4=\frac{L_2^4}{L_1^4L_3^2},$$

and in general

$$Q_{2n} = \frac{L_2^4 L_4^4 \ldots L_{2n-4}^4 L_{2n-2}^4}{L_1^4 L_3^4 \ldots L_{2n-3}^4 L_{2n-1}^2} *,$$

$$Q_{2n+1} = \frac{L_1^4 L_3^4 \ldots L_{2n-3}^4 L_{2n-1}^4}{L_2^4 L_4^4 \ldots L_{2n-2}^4 L_{2n}^2} \dagger.$$

COR. Hence, in place of Sturm's auxiliary functions, we may substitute the functions derived from the equations of coexistence $\left(fx = 0, \ \frac{dfx}{dx} = 0\right)$ according to Theorem 2, due regard being had to the sign.

*Scholium.* Hitherto it has been supposed that the values of the coefficients in the equations of coexistence are independent of one another, but particular relations may be supposed to exist which shall cause the leading terms given by Theorem 2 to vanish, giving rise to anormal or singular primes, as they may be called, of the degree $r$ of fewer than $(m-r)(n-r)$ dimensions. The theory of this, the failing case (so to say), is highly interesting, and I have already discovered the law of formation for the quotients of succession on the supposition of *any number* of primes vanishing consecutively; but I forbear to vex the patience of my reader further, the more so, as I hope soon to be able to present a complete memoir, with all the steps here indicated filled up, and numerous important additions, (the perfect image of which this is but a rough mould), as homage to the learned and illustrious society which has lately done me the honour of admitting me into its ranks.

Why this has not already been done must be excused, by the fact of the theory having suggested itself abroad in the intervals of sickness‡. Yet thus much will I add in general terms, namely, that as many primes as vanish consecutively, so many units must be added to the index 2 of the accessions

* That the appearance of the index 4 may not startle, let my reader bear in mind that there are what may be termed secondary derivatives of succession for every degree appearing in the process of successive division.

† The prime derivatives must be capable of yielding an *internal* evidence of the truth of Sturm's theorem. In fact, for the case of all the roots being possible, a little consideration will serve to show that the leading term of each prime derivative of the equation $\left\{fx \frac{dfx}{dx}\right\} = 0$ will consist of a series of fractions, each of which fractions is, *numerically* speaking, of the *same sign.*

‡ The reflections which Sturm's memorable theorem had originally excited, were revived by happening to be present at a sitting of the French Institute, where a letter was read from the Minister of Public Instruction, requesting an opinion upon the expediency of forming tables of elimination between two equations as high as the 5th or 6th degree containing one repeating term. The offer was rejected, on the ground of the excessive labour that would be required. I think that this has been very much overrated; and probably many will be of the same opinion who have dwelt upon the fact that no numerical quantity will occur in the result higher than the highest index of the repeating term. Would it not redound to the honour of British science that some painstaking ingenious person should gird himself to the task? and would not this be a proper object to meet with encouragement from the Scientific Association of Great Britain?

received in the numerator and denominator of the subsequent quotient; and in the quotient after that, it is not the square of the leading term of the penultimate prime,—but the product of this term by the leading term of that anormal prime of the same degree which has the lowest dimensions,—that finds its way into the numerator. The rest of the formation remaining undisturbed, *unless* and *until* a new failure have taken place.

## Note on Sturm's Theorem.

When one of the equations of coexistence is the differential coefficient with respect to the repeated term of the other, the prime derivatives given in Theorem 2 which coincide in this case with Sturm's auxiliary functions *reduced* to their lowest terms, may be exhibited under an integral *aspect*.

Let $SPD$ intimate that the squared product of the differences is to be taken of the quantities which follow it.

Let $S_1$ indicate the sum of the quantities to which it is prefixed.

$S_2$ the sum of the binary products.

$S_3$ the sum of the ternary products, and so on

Let $h_1, h_2 \dots h_n$ be the roots of any equation.

Then Sturm's last auxiliary function may be *replaced* by

$$SPD\,(h_1, h_2 \dots h_n).$$

The last but one may be replaced by

$$\Sigma SPD\,(h_1, h_2 \dots h_{n-1})\,x + \Sigma S_1\,(h_2, h_3 \dots h_{n-1})\,SPD\,(h_1, h_2 \dots h_{n-1}).$$

The one preceding by

$$\Sigma SPD\,(h_1, h_2 \dots h_{n-2})\,x^2 + \Sigma S_1\,(h_1, h_2 \dots h_{n-2})\,SPD\,(h_1, h_2 \dots h_{n-2})\,x + \Sigma S_2\,(h_1, h_2 \dots h_{n-2})\,SPD\,(h_1, h_2 \dots h_{n-2}),$$

and so on.

Thus then Sturm's rule for determining the absolute number of real roots in an equation is based wholly and solely upon the following

## Algebraical Proposition.

If there be $n$ quantities, real and imaginary, the imaginary ones entering in pairs, as many changes of sign as there are in the terms

$$\Sigma SPD\,(h_1, h_2),$$
$$\Sigma SPD\,(h_1, h_2, h_3),$$
$$\dots\dots\dots\dots$$
$$\Sigma SPD\,(h_1, h_2 \dots h_{n-1}),$$
$$\Sigma SPD\,(h_1, h_2 \dots h_n),$$

so many in number are these pairs.

Query (1). Is there no proposition applicable to any $n$ quantities *whatever* ?

Query (2). Is there no faintly analogous proposition applicable to higher powers than the squares ?

Query (3). Seeing that in forming the coefficients in the equation of the squares of the differences, we pass from $n$ functions of the roots to $n\frac{n-1}{2}$ and not $n$ functions, of their squared differences, does not a natural passage to the former lie through $n$ functions of the squared differences ?

In other words, may not the quantities $\Sigma SPD(h_1, h_2 \dots h_n)$, &c., serve as natural and valuable intermediaries between the coefficients of an equation involving simple quantities and the coefficients of the equation involving the squares of their differences ?

P.S. In the next part I trust to be able to present the readers of this *Magazine* with a *direct* and *symmetrical* method of eliminating any number of unknown quantities between any number of equations of any degree, by a newly invented process of symbolical multiplication, and the use of *compound* symbols of notation.

I must not omit to state that the constituents of multiplication $\lambda_r$ and $\mu_r$ explained in Cor. 2 to Theorem 3 are equal to the expression

$$\Sigma\,(x-k_1)(x-k_2)\dots(x-k_{n-r-1})\,\frac{\begin{pmatrix}k_1,\ k_2\dots k_{n-r-1}\\ -h_1,\ -h_2\dots -h_m\end{pmatrix}}{\begin{pmatrix}k_1,\ k_2\dots k_{n-r-1}\\ -k_{n-r}\dots -k_n\end{pmatrix}},$$

and its analogue respectively.

# 8.

## ON DERIVATION OF COEXISTENCE. PART II. BEING THE THEORY OF SIMULTANEOUS SIMPLE HOMOGENEOUS EQUATIONS.

[*Philosophical Magazine*, XVI. (1840), pp. 37—43.]

Art. (1). We shall have constant occasion in this paper to denote different quantities by the same letter affected with different subscribed numerical indices.

Such a letter is to be termed a "Base."

Every character consisting of a base and an inferior index, this index is called an argument of the base, namely, the first, second, or $n$th argument, according as 1, 2, or in general $n$, be the number subscribed.

Art. (2). I use the symbol $PD$ to denote the product of the differences of the quantities to which it is prefixed (each being to be subtracted *from* each that follows); thus

$PD\,(a, b, c)$ indicates $(b-a)(c-a)(c-b)$.

$PD\,(0, a, b, c)$ indicates $abc\,(b-a)(c-a)(c-b)$.

$PD\,(0, a, b, c \ldots l)$ indicates $abc \ldots l \times PD\,(a, b, c \ldots l)$.

Art. (3). For want of a better symbol I use the Greek letter $\zeta$ to denote that the product of factors to which it is prefixed is to be effected after a certain symbolical manner. This I shall distinguish as the zeta-ic product.

The symbol $\zeta$ will never be prefixed except to factors, each of which is made up of one or more terms, consisting solely of linear arguments of different bases, that is, characters bearing indices below but none above.

I am thereby enabled to give this short rule for zeta-ic multiplication: "Imagine all the inferior indices to become superior, so that each argument is transformed into a *power* of its base; multiply according to the rules of ordinary algebra; after the multiplication has been *done fully out* depress all the indices into their original position; the result is the zeta-ic product*."

* It is scarcely necessary to add that an analogous interpretation may be extended to any zeta-ic function whatever. Thus

$$\zeta\,(a_1+b_1)^2 = a_2 + 2a_1b_1 + b_2,$$

$$\zeta \cos\,(a_1) = 1 - \frac{a_2}{1\,.\,2} + \frac{a_4}{1\,.\,2\,.\,3\,.\,4}, \text{ \&c.}$$

Thus for example $\zeta(a_r, b_s)$ is the same as simply $a_r b_s$, but $\zeta(a_r, a_s)$ represents not $a_r a_s$ but $a_{r+s}$.

So in like manner

$$\zeta\{(a_h - b_k)(a_l - b_m)\}$$
$$= a_{h+l} - a_h b_m - b_k a_l + b_{m+k},$$
$$\zeta\{(a_1 - b_1)(a_1 - c_1)(b_1 - c_1)\}$$
$$= \text{the } \textit{depressed product} \text{ of } (a-b)(a-c)(b-c)$$
$$= \text{the } \textit{depressed} \text{ value of } a^2(b-c) + b^2(c-a) + c^2(a-b),$$

that is, $\quad = a_2 b_1 - a_2 c_1 + b_2 c_1 - b_2 a_1 + c_2 a_1 - c_2 b_1.$

Art. (4). We shall have occasion in this part to combine the two symbols $\zeta$, $PD$: thus we shall use

$$\zeta PD(a_1 b_1) \text{ to denote } \zeta(b_1 - a_1),$$
$$\zeta PD(a_1 b_1 c_1) \text{ to denote } \zeta\{(b_1 - a_1)(c_1 - a_1)(c_1 - b_1)\}.$$

Art. (5). For the sake of elegance of diction I shall in future sometimes omit to insert the inferior index when it is unity; but the reader must always bear in mind that it is to be *understood* though not expressed.

I shall thus be able to speak of the zeta-ic product of such and such bases mentioned by name.

Art. (6). We are not yet come to the limit of the powers of our notation. The zeta-ic product of the sum of arguments will consist of the sum of products of arguments, each argument being (as I have defined) made up of a base and an inferior index. Now we may imagine each index of every term of the zeta-ic product *after it is fully expanded* to be increased or diminished by unity, or each at the same time to be increased or diminished by 2, or each in general to be increased or diminished by $r$. I shall denote this alteration by affixing an $r$ with the positive or negative sign to the $\zeta$. Thus

$$\zeta(a_1 - b_1)(a_1 - c_1) \text{ being equal to } a_2 - a_1 c_1 + b_1 c_1 - b_1 a_1,$$
$$\zeta_{+1}(a_1 - b_1)(a_1 - c_1) \text{ is equal to } a_3 - a_2 c_2 + b_2 c_2 - b_2 a_2,$$
$$\zeta_{-1}(a_1 - b_1)(a_1 - c_1) \text{ is equal to } a_1 - a_0 c_0 + b_0 c_0 - b_0 a_0.$$

In like manner $\zeta PD(a, b, c)$ indicating

$$b_2 a_1 - b_2 c_1 + c_2 b_1 - c_2 a_1 + a_2 c_1 - a_2 b_1,$$

$\zeta_{\pm r} PD(a, b, c)$ indicates

$$b_{2\pm r} a_{1\pm r} - b_{2\pm r} c_{1\pm r} + c_{2\pm r} b_{1\pm r} - c_{2\pm r} a_{1\pm r} + a_{2\pm r} c_{1\pm r} - a_{2\pm r} b_{1\pm r}.$$

I shall in general denote $\zeta_{+r} PD(a, b, c \ldots l)$ *actually expanded* as the zeta-ic product of $a, b, c, \ldots l$ in its $r$th *phase.*

Art. (7). *General Properties of Zeta-ic Products of Differences.*

If there be made one interchange in the order of the bases to which $\zeta$ is prefixed, the zeta-ic product, in whatever phase it be taken, remains unaltered in magnitude, but changes its sign.

If in any *phase* of a zeta-ic product two of the bases be made to coincide, the expansion vanishes.

Let $\int_1$ be used, agreeably to the ordinary notation, to denote the sum of the quantities to which it is prefixed, $\int_2$ to denote the sum of the binary products, $\int_3$ of the ternary ones, and so on.

Thus let $\int_1 (a_1 b_1 c_1)$ or $\int_1 (a, b, c)$ indicate $a_1 + b_1 + c_1$,

and $\int_2 (a_1 b_1 c_1)$ or $\int_2 (a, b, c)$ indicate $a_1 b_1 + a_1 c_1 + b_1 c_1$,

and $\int_3 (a_1 b_1 c_1)$ or $\int_3 (a, b, c)$ indicate $a_1 b_1 c_1$,

we shall be able now to state the following remarkable proposition connecting the several phases of certain the same zeta-ic products.

Art. (8). Let $a, b, c, \ldots l$, denote any number of independent bases, say $(n-1)$; but let the arguments of each base be periodic, and the number of terms in each period the same for every base, namely $n$, so that

$$\begin{array}{ll}
a_r = a_{r+n} = a_{r-n}, & a_n = a_0 = a_{-n}, \\
b_r = b_{r+n} = b_{r-n}, & b_n = b_0 = b_{-n}, \\
c_r = c_{r+n} = c_{r-n}, & c_n = c_0 = c_{-n}, \\
\cdots\cdots\cdots\cdots & \cdots\cdots\cdots\cdots \\
l_r = l_{r+n} = l_{r-n}, & l_n = l_0 = l_{-n},
\end{array}$$

$r$ being any number whatever. Then

$$\zeta_{-1} PD\,(0, a, b, c \ldots l) = \zeta\,\{\textstyle\int_1 (a, b, c \ldots l)\ \zeta PD\,(0, a, b, c \ldots l)\},$$

$$\zeta_{-2} PD\,(0, a, b, c \ldots l) = \zeta\,\{\textstyle\int_2 (a, b, c \ldots l)\ \zeta PD\,(0, a, b, c \ldots l)\},$$

$$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$\zeta_{-r} PD\,(0, a, b, c \ldots l) = \zeta\,\{\textstyle\int_r (a, b, c \ldots l)\ \zeta PD\,(0, a, b, c \ldots l)\}.$$

This proposition admits of a great generalization*, but we have now all that is requisite for enabling us to arrive at a proposition exhibiting under one *coup d'œil* every combination and every effect of every combination that can possibly be made with any number of coexisting equations of the first degree, containing any number of *repeated*, or to use the ordinary language of analysts, (variable or) unknown quantities.

* See the Postscript to this paper for *one* specimen.

For the sake of symmetry I make every equation homogeneous; so that to eliminate $n$ repeated terms, no more than $n$ equations will be required.

In like manner the problem of determining $n$ quantities from $n$ equations will be here represented by the case in which we have to determine the *ratios* of $(n+1)$ quantities from $n$ equations.

### Art. (9). *Statement of the Equations of Coexistence.*

Let there be any number of bases $(a, b, c \dots l)$, and as many repeated terms $(x, y, z \dots t)$, and let the number of equations be any whatever, say $n$. The system may be represented by the *type* equation

$$a_r x + b_r y + c_r z + \dots + l_r t = 0,$$

in which $r$ can take up all integer values from $-\infty$ to $+\infty$. The specific number of equations given will be represented by making the arguments of each base *periodic*, so that

$$a_r = a_{\mu n+r}, \quad b_r = b_{\mu n+r}, \quad c_r = c_{\mu n+r}, \dots l_r = l_{\mu n+r},$$

$\mu$ being any integer whatever.

### Art. (10). *Combination of the given Equations.—Leading Theorem.*

Take $f, g, \dots k$ as the *arbitrary* bases of new and absolutely independent but periodic arguments, having the same index of periodicity $(n)$ as $a, b, c \dots l$, and being in number $(n-1)$, that is, one fewer than there are units in that index.

The number of *differing* arbitrary constants thus *manufactured* is $n(n-1)$.

Let $Ax + By + Cz + \dots + Lt = 0$ be the general *prime* derivative from the given equations, then we may make

$$\begin{aligned} A &= \zeta PD(0, a, f, g \dots k), \\ B &= \zeta PD(0, b, f, g \dots k), \\ C &= \zeta PD(0, c, f, g \dots k), \\ &\dots\dots\dots\dots\dots\dots \\ L &= \zeta PD(0, l, f, g \dots k). \end{aligned}$$

### Art. (11). Cor. 1. *Inferences from the Leading Theorem.*

Let the number of equations, or, which is the same thing, the index of periodicity $(n)$, be the same as the number of repeated terms $(x, y, z \dots t)$, then one relation exists between the coefficients: this is found by making the $(n-1)$ new bases coincide with $(n-1)$ out of the old bases. We get accordingly, as the result of elimination,

$$\zeta PD(0, a, b, c \dots l) = 0.$$

Art. (11). Cor. 2. Let the number of equations be one more than that of the given bases, there will then be two equations of condition. These are represented by preserving one new arbitrary base, as $\lambda$. The result of elimination being in this case

$$\zeta PD\,(0, a, b, c \ldots l, \lambda) = 0.$$

*Example.* The result of eliminating between

$$a_1 x + b_1 y = 0,$$

$$a_2 x + b_2 y = 0,$$

$$a_3 x + b_3 y = 0,$$

is $\zeta PD\,(0, a, b, \lambda) = 0$, that is

$$\lambda_3 b_2 a_1 - \lambda_3 b_1 a_2 + \lambda_1 b_3 a_2 - \lambda_1 b_2 a_3 + \lambda_2 b_1 a_3 - \lambda_2 b_3 a_1 = 0,$$

from which we infer, seeing that $\lambda_3, \lambda_2, \lambda_1$ are independent,

$$b_2 a_1 - b_1 a_2 = 0,$$

$$b_3 a_2 - b_2 a_3 = 0,$$

$$b_1 a_3 - b_3 a_1 = 0,$$

any *two* of which imply the third.

In like manner, in general, if the number of equations exceed in any manner the number of bases or repeated terms, the rule is to introduce so many *new* and *arbitrary* bases as together with the old bases shall make up the number of equations, and then equate the zeta-ic product of the differences of zero, the old bases and the new bases, to nothing.

Art. (12). Cor. 3. Let the number of equations be *one* fewer than the number ($n$) of bases or repeated terms; the number of introduced bases in the general theorem is here $(n-2)$. Make these $(n-2)$ bases equal severally to the bases which in the type equation are affixed to $z, u \ldots t$, then

$$C = 0,$$

$$D = 0,$$

$$\ldots\ldots$$

$$L = 0,$$

and we have left simply

$$\zeta PD\,(0, a, c, d \ldots kl)\, x + \zeta PD\,(0, b, c, d \ldots kl)\, y = 0.$$

In like manner we may make to vanish all but $A$ and $C$, and thus get

$$\zeta PD\,(0, a, b, d \ldots kl)\, x + \zeta PD\,(0, c, b, d \ldots kl)\, z = 0,$$

and similarly

$$\zeta PD(0, a, b \ldots k)\, x + \zeta PD(0, b, c \ldots l)\, t = 0.$$

Hence

$$\left.\begin{array}{r} x \\ y \\ z \\ . \\ . \\ . \\ t \end{array}\right\} \text{are severally as} \left\{\begin{array}{l} \zeta PD(0, b, c \ldots l) \\ \zeta PD(a, 0, c \ldots l) \\ \zeta PD(a, b, 0 \ldots l) \\ \cdots\cdots\cdots\cdots \\ \cdots\cdots\cdots\cdots \\ \cdots\cdots\cdots\cdots \\ \zeta PD(a, b, c \ldots 0). \end{array}\right.$$

This is the symbolical representation as a *formula* of the remarkable *method* discovered by Cramer, perfected by Bezout and demonstrated by Laplace for the solution of simultaneous simple equations.

Art. (13). Cor. 4. In like manner if the number of repeated terms be two greater than the number of equations, we have for the relation between any *three* of them, taken at pleasure, for instance, $x, y, z$,

$$\zeta PD(0, a, d \ldots l)\, x + \zeta PD(0, b, d \ldots l)\, y + \zeta PD(0, c, d \ldots l)\, z = 0.$$

And in like manner we may proceed, however much in excess the number of repeated terms (unknown quantities) is over the number of equations.

Art. (14). *Subcorollary to Corollary* 3.

If there be any number of bases $(a, b, c \ldots l)$, and any other two fewer in number $(f, g \ldots k)$

$$\left.\begin{array}{l} \zeta PD(a, f, g \ldots k) \times \zeta PD(b, c \ldots l) \\ + \zeta PD(b, f, g \ldots k) \times \zeta PD(a, c \ldots l) \\ + \zeta PD(c, f, g \ldots k) \times \zeta PD(b, c \ldots l) \\ \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ + \zeta PD(l, f, g \ldots k) \times \zeta PD(a, b, c \ldots\ \ ) = 0 \end{array}\right\}^*,$$

a formula that from its very nature suggests and *proves* a wide extension of itself.

In conclusion I feel myself bound to state that the principal substance of *Corollaries* (1), (2) and (3) may be found in Garnier's *Analyse Algébrique*, in the chapter headed "Dévéloppement de la Théorie donnée par M. Laplace, &c." But I am not aware of having been anticipated either in the fertile notation which serves to express them nor in the general theorems to which it has given birth.

P.S. I shall content myself for the present with barely enunciating a theorem, one of a class destined it seems to the author to play no secondary part in the development of some of the most curious and interesting points of analysis.

* The cross is used to denote *ordinary* algebraical multiplication.

Let there be $(n-1)$ bases $a, b, c \ldots l$, and let the arguments of each be "recurrents of the $n$th order*," that is to say let

$$a_\iota = \phi\left(\cos\frac{2\pi\iota}{n}\right), \; b_\iota = \psi\left(\cos\frac{2\pi\iota}{n}\right), \; c_\iota = \chi\left(\cos\frac{2\pi\iota}{n}\right), \; \ldots\ldots l_\iota = \omega\left(\cos\frac{2\pi\iota}{n}\right).$$

Let $R_r$ denote that any symmetrical function of the $r$th degree is to be taken of the quantities in a parenthesis which come after it, and let $\vartheta$ indicate any function whatever. Then the zeta-ic product

$$\zeta\{\zeta R_r(a, b, c \ldots l) \times \zeta_\rho \vartheta PD(0, a, b, c \ldots l)\}$$

is equal to the product of the *number*

$$R_r\left\{\left(\cos\frac{2\pi}{n} + \sqrt{(-1)}\sin\frac{2\pi}{n}\right), \left(\cos\frac{4\pi}{n} + \sqrt{(-1)}\sin\frac{4\pi}{n}\right), \left(\cos\frac{6\pi}{n} + \sqrt{(-1)}\sin\frac{6\pi}{n}\right),\right.$$

$$\left.\ldots\ldots, \left(\cos\frac{2(n-1)\pi}{n} + \sqrt{(-1)}\sin\frac{2(n-1)\pi}{n}\right)\right\},$$

multiplied by the zeta-ic phase

$$\zeta_{\rho-r}\vartheta PD(0, a, b, c \ldots l)!!$$

* I am indebted for this term to Professor De Morgan, whose pupil I may boast to have been. I have the sanction also of his authority, and that of another profound analyst, my colleague Mr Graves, for the use of the arbitrary terms zeta-ic, zeta-ically. I take this opportunity of retracting the symbol *SPD* used in my last paper, the letter *S* having no meaning except for English readers. I substitute for it *QDP*, where *Q* represents the Latin word Quadratus. On some future occasion I shall enlarge upon a new method of notation, whereby the language of analysis may be rendered much more expressive, depending essentially upon the use of similar figures inserted within one another, and containing numbers or letters, according as quantities or operations are to be denoted. This system to be carried out would require special but very simple printing types to be founded for the purpose.

In the next part of this paper an easy and *symmetrical* mode will be given of representing any polynomial either in its developable or expanded form.

# 9.

## A METHOD OF DETERMINING BY MERE INSPECTION THE DERIVATIVES FROM TWO EQUATIONS OF ANY DEGREE.

[*Philosophical Magazine*, XVI. (1840), pp. 132—135.]

LET there be two equations, one of the $n$th, the other of the $m$th degree in $x$; let the coefficients of the first equation be $a_n$, $a_{n-1}$, $a_{n-2} \dots a_0$, *each* power of $x$ having a coefficient attached to it, $a_n$ belonging to $x^n$ and $a_0$ to the constant term.

In like manner let $b_m, b_{m-1} \dots b_0$ be the coefficients of the second equation.

I begin with

*A Rule for absolutely eliminating $x$.*

Form out of the ($a$) progression of coefficients $m$ lines, and in like manner out of the ($b$) progression of coefficients form $n$ lines in the following manner:

1. ($a$) Attach $(m-1)$ zeros all to the *right* of the terms in the ($a$) progression; next attach $(m-2)$ zeros to the right and carry over to the left; next attach $(m-3)$ zeros to the right and carry over 2 to the left. Proceed in like manner until all the $(m-1)$ zeros are carried over to the left and none remain on the right.

The $m$ lines thus formed are to be written under one another.

1. ($b$) Proceed in like manner to form $n$ lines out of the ($b$) progression by scattering $(n-1)$ zeros between the right and left.

2. If we write these $n$ lines under the $m$ lines last obtained, we shall have a solid square $(m+n)$ terms *deep* and $(m+n)$ terms *broad*.

3. Denote the lines of this square by arbitrary characters, which write down in vertical order and permute in every possible way, but separate the permutations that can be derived from one another by an even number of interchanges (effected between *contiguous* terms) from the rest; there will thus be half of one kind and half of another.

4. Now arrange the $(m+n)$ lines accordingly, so as to obtain

$$\tfrac{1}{2}\{(m+n)(m+n-1)\ldots 2\,.\,1\}$$

squares of one kind which shall be called positive squares, and an equal number of the opposite kind which shall be called negative.

Draw diagonals in the same direction in all the squares; multiply the coefficients that stand in any diagonal line together: take the sum of the diagonal products of the *positive* squares, and the sum of the diagonal products of the *negative* squares; the difference between these two sums is the prime derivative of the zero degree, that is, is the result of elimination between the two given equations reduced to its ultimate state of simplicity, there will be no irrelevant factors to reject, and no terms which mutually destroy.

*Example.* To *eliminate* between

$$ax^2+bx+c=0,$$

$$lx^2+mx+n=0,$$

I write down

$$a,\ b,\ c,\ 0, \qquad (1)$$

$$0,\ a,\ b,\ c, \qquad (2)$$

$$l,\ m,\ n,\ 0, \qquad (3)$$

$$0,\ l,\ m,\ n. \qquad (4)$$

I permute the four characters (1), (2), (3), (4), distinguishing them into positive and negative; thus I write together

*Positive Permutations.*

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 1 | 2 | 3 | 2 | 1 | 3 | 4 | 4 | 4 |
| 2 | 3 | 1 | 4 | 4 | 4 | 1 | 3 | 2 | 2 | 1 | 3 |
| 3 | 1 | 2 | 2 | 3 | 1 | 4 | 4 | 4 | 1 | 3 | 2 |
| 4 | 4 | 4 | 3 | 1 | 2 | 3 | 2 | 1 | 3 | 2 | 1 |

and again

*Negative Permutations.*

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 4 | 4 | 2 | 1 | 3 | 2 | 1 | 3 |
| 2 | 3 | 1 | 1 | 2 | 3 | 4 | 4 | 4 | 1 | 3 | 2 |
| 4 | 4 | 4 | 2 | 3 | 1 | 1 | 3 | 2 | 3 | 2 | 1 |
| 3 | 1 | 2 | 3 | 1 | 2 | 3 | 2 | 1 | 4 | 4 | 4 |

I reject from the permutations of each species all those where 1 or 3 appear in the fourth place, and also those where 2 or 4 appear in the first place, for these will be presently seen to give rise to diagonal products which are zero.

The permutations remaining are

*Positive effectual permutations.*

| | | | |
|---|---|---|---|
| 1 | 3 | 3 | 1 |
| 2 | 1 | 4 | 3 |
| 3 | 2 | 1 | 4 |
| 4 | 4 | 2 | 2 |

*Negative effectual permutations.*

| | | | |
|---|---|---|---|
| 3 | 1 | 1 | 3 |
| 1 | 4 | 3 | 2 |
| 4 | 3 | 2 | 1 |
| 2 | 2 | 4 | 4 |

I now accordingly form four positive squares, which are

$$\begin{array}{cccc} a,\ b,\ c,\ 0, & l,\ m,\ n,\ 0, & l,\ m,\ n,\ 0, & a,\ b,\ c,\ 0, \\ 0,\ a,\ b,\ c, & a,\ b,\ c,\ 0, & 0,\ l,\ m,\ n, & l,\ m,\ n,\ 0, \\ l,\ m,\ n,\ 0, & 0,\ a,\ b,\ c, & a,\ b,\ c,\ 0, & 0,\ l,\ m,\ n, \\ 0,\ l,\ m,\ n, & 0,\ l,\ m,\ n, & 0,\ a,\ b,\ c, & 0,\ a,\ b,\ c. \end{array}$$

Drawing diagonal lines from left to right, and taking the sum of the diagonal products, I obtain $a^2n^2 + lb^2n + l^2c^2 + am^2c$. Again, the four negative squares

$$\begin{array}{cccc} l,\ m,\ n,\ 0, & a,\ b,\ c,\ 0, & a,\ b,\ c,\ 0, & l,\ m,\ n,\ 0, \\ a,\ b,\ c,\ 0, & 0,\ l,\ m,\ n, & l,\ m,\ n,\ 0, & 0,\ a,\ b,\ c, \\ 0,\ l,\ m,\ n, & l,\ m,\ n,\ 0, & 0,\ a,\ b,\ c, & a,\ b,\ c,\ 0, \\ 0,\ a,\ b,\ c, & 0,\ a,\ b,\ c, & 0,\ l,\ m,\ n, & 0,\ l,\ m,\ n, \end{array}$$

give as the sum of the diagonal products

$$lbmc + alnc + ambn + lacn,$$

that is,

$$lbmc + ambn + 2acln.$$

Thus the result of eliminating between

$$ax^2 + bx + c = 0,$$

$$lx^2 + mx + n = 0,$$

ought to be, and is

$$a^2n^2 + l^2c^2 - 2acln + lb^2n + am^2c - lbmc - ambn = 0.$$

*Rule for finding the prime derivative of the first degree, which is of the form $Ax - B$.*

Begin as before, only attach one zero less to each progression; we shall thus obtain *not* a square, but an oblong broader than it is deep, containing $(m + n - 2)$ rows, and $(m + n - 1)$ terms in each row: in a word, $(m + n - 2)$ rows, and $(m + n - 1)$ *columns.*

To find $A$ reject the column at the extreme right, we thus recover a square arrangement $(m + n - 2)$ terms broad and deep.

Proceed with this new square as with the former one; the difference between the sums of the positive and negative diagonal products will give $A$.

To find $B$, do just the same thing, with the exception of striking off not the last column, but the last but *one.*

*Rule for finding the prime derivative of any degree, say the rth, namely,* $A_r x^r - A_{r-1} x^{r-1} + \ldots\ldots \pm A_0$.

Begin with adding zeros as before, but the number to be added to the $(a)$ progression is $(m - r)$ and to the $(b)$ progression $(n - r)$.

There will thus be formed an oblong containing $(m + n - 2r)$ rows, and $(m + n - r)$ terms in each row, and therefore the same number of *columns.*

To find any coefficient as $A_s$, strike off all the last $(r + 1)$ columns except that which is $(s)$ places distant from the extreme right, and proceed with the resulting squares as before.

---

Through the well-known ingenuity and kindly proferred help of a distinguished friend, I trust to be able to get a machine made for working Sturm's theorem, and indeed all problems of derivation, after the method here expounded; on which subject I have a great deal more yet to say, than can be inferred from this or my preceding papers.

# 10.

## NOTE ON ELIMINATION.

[*Philosophical Magazine,* XVII. (1840), pp. 379, 380.]

THE object of this brief note is to generalise Theorem 2 in my paper on Elimination* which appeared in the last December number of this *Magazine.* The theorem so generalised presents a symmetry which before was wanting. Here, as in so many other instances, the whole occupies in the memory a *less* space than the part.

To avoid the ill-looking and slippery negative symbols, I warn my reader that I now use two rows of quantities written one over the other, to denote the product of the terms resulting from *taking away* each quantity in the under from each in the upper row.

Let $h_1, h_2 \dots h_m$ be the roots of one equation of coexistence,

$k_1, k_2 \dots k_n$ of the other,

and let the prime derivative of the degree $r$ be required. Take *any* two integers $p$ and $q$, such that $p+q=r$. The derivative in question may be written

$$\Sigma \left\{ (x-h_1)\dots(x-h_p)(x-k_1)\dots(x-k_q) \frac{\begin{pmatrix} h_1 h_2 \dots h_p \\ k_1 k_2 \dots k_q \end{pmatrix} \cdot \begin{pmatrix} h_{p+1} h_{p+2} \dots h_m \\ k_{q+1} k_{q+2} \dots k_n \end{pmatrix}}{\begin{pmatrix} h_1 \;\; h_2 \;\; \dots h_p \\ h_{p+1} h_{p+2} \dots h_m \end{pmatrix} \cdot \begin{pmatrix} k_1 \;\; k_2 \;\; \dots k_q \\ k_{q+1} k_{q+2} \dots k_n \end{pmatrix}} \right\}.$$

N.B. Whatever $p$ and $q$ be taken, so long only as $p+q=r$, the above expression changes nothing but its sign; which, therefore, upon transcendental grounds, it is easy to see is of one name or another, according as $p$ is odd or even.

In the original paper, I asserted this theorem only for the case of $p=0$, or $q=0$.

[* p. 43 above. ED.]

# 11.

## ON THE RELATION OF STURM'S AUXILIARY FUNCTIONS TO THE ROOTS OF AN ALGEBRAIC EQUATION.

[*Plymouth British Association Report* 1841, (Pt II.), pp. 23, 24.]

THE author availed himself of the present meeting of the British Association to bring under the more general notice of mathematicians his discovery, made in the year 1839, of the real nature and constitution of the auxiliary functions (so-called) which Sturm makes use of in *locating* the roots of an equation: these are obtained by proceeding with the left-hand side of the equation and its first differential coefficient as if it were our object to obtain their greatest common factor; the successive remainders, with their signs *alternately* changed and preserved, constitute the functions in question. Each of these may be put under the form of a fraction, the denominator of which is a perfect square, or in fact the product of *many*: likewise the numerator contains a huge heap of factors of a similar form.

These therefore, as well as the denominator, since they cannot influence the series of *signs*, may be rejected; and furthermore we may, if we please, again make every other function, beginning from the last but one, change its sign, if we consent to use changes wherever Sturm speaks of continuations of sign, and *vice versâ*.

The functions of Sturm, thus modified and purged of irrelevancy, the author, by way of distinction, and still to attribute honour where it is really most due, proposes to call "Sturm's Determinators"; and he proceeds to lay bare the internal anatomy of these remarkable forms.

He uses the Greek letter "$\zeta$" to indicate that the squared product of the differences of the letters before which it is prefixed is to be taken.

Let the roots of the equation be called respectively $a, b, c, e \ldots l$, the determinators taken in the inverse order are as follows:—

$$\zeta(a, b, c, e \ldots l).$$

$$\Sigma\zeta(b, c, e \ldots l)\,x - \Sigma a\zeta(b, c, e \ldots l).$$

$$\Sigma\zeta(c, e \ldots l)\,x^2 - \Sigma(a+b)\,.\,\zeta(c, e \ldots l)\,x + \Sigma ab\,.\,\zeta(c, e \ldots l).$$

*  *  *  *  *  *  *

$$\Sigma\,\{\zeta(k, l)\,(x-a)\,(x-b)\,(x-c)\,(x-e) \ldots (x-h)\}.$$

It may be here remarked, that the work of assigning the total number of real and of imaginary roots falls exclusively upon the coefficients of the leading terms, which the author proposes to call "Sturm's Superiors": these superiors are only *partial* symmetric functions of the *squared differences*, but *complete* symmetric functions of the *roots themselves*, differing in the former respect from those other (at first sight similar-looking) functions of the squared differences of the roots, in which, from the time of Waring downwards, the conditions of reality have been sought for. It seems to have escaped observation, that the series of terms constituting any one of the coefficients in the equation of the squares of the differences (with the exception of the first and last) each admit of being separated and classified into various subordinate groups in such a way, that instead of being treated as a single symmetric function of the *roots*, they ought to be viewed as aggregates of many. In fact, Sturm's superior No. 1 is identical with Waring's coefficient No. 1; Sturm's superior No. 2 is a *part* of Waring's coefficient No. 3; Sturm's superior No. 3 is a *part* of Waring's coefficient No. 6; and so forth till we come to Sturm's final superior, which is again coextensive and identical with the last coefficient in the equation of the squares of the differences. The theory of symmetric functions of forms which are themselves symmetric functions of simple letters, or even of other forms, the author states his belief is here for the first time shadowed forth, but would be beside his present object to enter further into. He would conclude by calling attention to the importance to the general interests of algebraical and arithmetical science that a searching investigation should be instituted for showing, *à priori*, how, when a set of quantities is known to be made up partly of possible and partly of *pairs* of impossible values, symmetrical functions of these, one less in number than the quantities themselves, may be formed, from the signs of the ratios of which to unity and to one another the respective amounts of possible and impossible quantities may at once be inferred: in short, we ought not to rest satisfied, until, from the very *form* of Sturm's Determinators, without caring to know how they have been obtained, we are able to pronounce upon the uses to which they may be applied.

# 12.

## EXAMPLES OF THE DIALYTIC METHOD OF ELIMINATION AS APPLIED TO TERNARY SYSTEMS OF EQUATIONS.

[*Cambridge Mathematical Journal*, II. (1841), pp. 232—236.]

This method is of universal application, and at once enables us to reduce any case of elimination to the form of a problem, where that operation is to be effected between quantities linearly involved in the equations which contain them.

As applied to a binary system, $fx = 0$, $\phi x = 0$, the method furnishes a rule by which we may unfailingly arrive at *the determinant,* free from every species of irrelevancy, whether of a linear, factorial, or numerical kind.

The rule itself is given in the *Philosophical Magazine* (London and Edinburgh, Dec. 1840). The principle of the rule will be found correctly stated by Professor Richelot, of Königsberg, in a late number of *Crelle's Journal,* at the commencement of a memoir in Latin bordering on the same subject ("Nota ad Eliminationem pertinens").

My object at present is to supply a few instances of its application to ternary systems of equations.

Ex. 1. To eliminate $x$, $y$, $z$, between the three homogeneous equations

$$Ay^2 - 2C'xy + Bx^2 = 0, \tag{1}$$

$$Bz^2 - 2A'yz + Cy^2 = 0, \tag{2}$$

$$Cx^2 - 2B'zx + Az^2 = 0. \tag{3}$$

Multiply the equations in order by $-z^2$, $x^2$, $y^2$, add together, and divide out by $2xy$; we obtain

$$C'z^2 + Cxy - A'xz - B'yz = 0. \tag{4}$$

By similar processes we obtain

$$A'x^2 + Ayz - B'yx - C'zx = 0, \tag{5}$$

$$B'y^2 + Bzx - C'zy - A'xy = 0. \tag{6}$$

Between these six, treated as simple equations, the six functions of $x$, $y$, $z$, namely, $x^2$, $y^2$, $z^2$, $xy$, $xz$, $yz$, treated as *independent* of each other, may be eliminated; the results may be seen, by mere inspection, to come out

$$ABC\,(ABC - AB'^2 - BC'^2 - CA'^2 + 2A'B'C') = 0,$$

or rejecting the special (N.B. not *irrelevant*) factor $ABC$, we obtain

$$ABC - AB'^2 - BC'^2 - CA'^2 + 2A'B'C' = 0.$$

I may remark, that the equations (1), (2), (3), or (4), (5), (6), express the condition of

$$Ax^2 + By^2 + Cz^2 + 2A'yz + 2B'zx + 2C'xy,$$

having a factor $\lambda x + \mu y + \nu z$; a general symbolical formula of which I am in possession for determining in general the condition of any polynomial of any degree having a factor, furnishes me at once with either of the two systems indifferently. The aversion I felt to reject *either*, led me to employ both, and thus was the occasion of the Dialytic Principle of Solution manifesting itself.

Ex. 2.
$$Ax^2 + ayz + bzx + cxy = 0, \tag{1}$$
$$My^2 + lyz + mzx + nxy = 0, \tag{2}$$
$$Rz^2 + pyz + qzx + rxy = 0. \tag{3}$$

Multiply equation (1) by $\beta y + \gamma z$, equations (2) and (3) by $\nu z$ and $\kappa y$ respectively, and add the products together, we obtain terms of which $y^2z$ and $yz^2$ are the only two into which $x$ does not enter.

Make now the coefficients of each of these zero, and we have

$$a\gamma + l\nu + R\kappa = 0,$$
$$a\beta + M\nu + p\kappa = 0.$$

Let $\nu = a$, $\kappa = a$, then $\gamma = -(l + R)$, $\beta = -(M + p)$.

Hence, multiplying as directed, and then dividing out by $x$, we obtain

$$(m\nu + b\gamma)\,z^2 + (r\kappa + c\beta)\,y^2 + (b\beta + c\gamma + n\nu + q\kappa)\,yz + A\beta xy + A\gamma xz = 0,$$

or by substitution,

$$\{ra - c\,(M+p)\}\,y^2 + \{ma - b\,(l+R)\}\,z^2 + \{an + aq - b\,(M+p) - c\,(l+R)\}\,yz$$
$$- A\,(M+p)\,xy - A\,(M+p)\,xz = 0. \tag{4}$$

Similarly, by preparing the equations so as to admit in turn of $y$ and $z$ as a divisor, we obtain

$$\{ma - l\,(R+b)\}\,z^2 + \{mr - n\,(A+q)\}\,x^2 + \{mc + mp - n\,(R+b) - l\,(A+y)\}\,xz$$
$$- M\,(R+b)\,yz - A\,(A+q)\,xy = 0, \tag{5}$$

$$\{rm - q\,(A+n)\}\,x^2 + \{ra - p\,(M+c)\}\,y^2 + \{rl + rb - p\,(A+n) - q\,(M+c)\}\,xy$$
$$- R\,(A+n)\,xz - R\,(M+c)\,yz = 0. \tag{6}$$

Between the six equations (1), (2), (3), (4), (5), (6), $x^2$, $y^2$, $z^2$, $xy$, $xz$, $yz$, may be eliminated; the result will be a function of nine letters {three out of each equation (1), (2), (3)} equated to zero. *Perhaps* the determinant may be found to contain a special factor of three letters; and if so, may be replaced by a simpler function of six letters only.

Ex. 3. To eliminate between the three general equations

$$Ax^2 + By^2 + Cz^2 + 2Dyz + 2Ezx + 2Fxy = 0,$$

$$Lx^2 + My^2 + Nz^2 + 2Pyz + 2Qzx + 2Rxy = 0,$$

$$fx + gy + hz = 0.$$

By virtue of *one* of the two canons which limit the forms in which the letters can appear combined in the determinant of a general system of equations, we know that the determinant in this case (freed of irrelevant factors) ought to be made up in every term of eight letters (powers being counted as repetitions), namely, $(A, B, C, D, E, F)$ must enter in binary combinations, $(L, M, N, P, Q, R)$ the same, whereas $f, g, h$ must enter in *quaternary* combinations.

To obtain the determinant, write

$$Ax^2 + By^2 + Cz^2 + Dyz + Ezx + Fxy = 0, \qquad (1)$$

$$Lx^2 + My^2 + Nz^2 + Pyz + Qzx + Rxy = 0, \qquad (2)$$

$$fx^2 + gyx + hzx = 0, \qquad (3)$$

$$fxy + gy^2 + hzy = 0, \qquad (4)$$

$$fxz + gyz + hz^2 = 0. \qquad (5)$$

We want one equation more of *three* letters between $x^2$, $y^2$, $z^2$, $xy$, $xz$, $yz$. To obtain this, write

$$(Ax + Ez + Fy)\,x_1 + (By + Fx + Dz)\,y_1 + (Cz + Dy + Ex)\,z_1 = 0,$$

$$(Lx + Qz + Ry)\,x_1 + (My + Rx + Pz)\,y_1 + (Nz + Py + Qx)\,z_1 = 0,$$

$$fx_1 + \qquad gy_1 + \qquad hz_1 = 0.$$

Forget that $x_1 = x$, $y_1 = y$, $z_1 = z$, and eliminate $x_1$, $y_1$, $z_1$, we obtain

$$h \begin{Bmatrix} (Ax + Ez + Fy)(My + Rx + Pz) \\ -(By + Fx + Dz)(Lx + Qz + Ry) \end{Bmatrix}$$

$$+ g \begin{Bmatrix} (Cz + Dy + Ex)(Lx + Qz + Ry) \\ -(Nz + Py + Qx)(Ax + Ez + Fy) \end{Bmatrix}$$

$$+ f \begin{Bmatrix} (Nz + Py + Qx)(By + Fx + Dz) \\ -(Cz + Dy + Ex)(My + Rx + Pz) \end{Bmatrix} = 0.$$

This may be put under the form

$$\alpha x^2 + \beta y^2 + \gamma z^2 + \alpha' yz + \beta' zx + \gamma' xy = 0, \qquad (6)$$

where the coefficients are of the first order in respect to $f, g, h, L, M, N, P, Q, R, A, B, C, D, E, F$; in all of the third order.

Between the equations marked from (1) to (6), the process of linear elimination being gone through, we obtain as equated to zero a function of $5 + 3$, or of eight letters, two belonging to the first equation, two to the second, and four to the third; so that the determinant is clear of all factorial irrelevancy.

Ex. 4. To eliminate $x, y, z$ between the three equations

$$Ax^2 + By^2 + Cz^2 + 2A'yz + 2B'zx + 2C'xy = 0,$$
$$Lx^2 + My^2 + Nz^2 + 2L'yz + 2M'zx + 2N'xy = 0,$$
$$Px^2 + Qy^2 + Rz^2 + 2P'yz + 2Q'zx + 2R'xy = 0.$$

Call these three equations $U = 0$, $V = 0$, $W = 0$, respectively. Write

$$\begin{array}{llllll} xU = 0, & (1) & yU = 0, & (2) & zU = 0, & (3) \\ xV = 0, & (4) & yV = 0, & (5) & zV = 0, & (6) \\ xW = 0, & (7) & yW = 0, & (8) & zW = 0. & (9) \end{array}$$

We have here nine unilateral equations: one more is wanted to enable us to eliminate *linearly* the ten quantities

$$x^3, \quad y^3, \quad z^3, \quad x^2y, \quad x^2z, \quad xy^2, \quad xz^2, \quad xyz, \quad y^2z, \quad yz^2.$$

This tenth may be found by eliminating $x, y, z$ between the three equations

$$x(Ax + B'z + C'y) + y(By + C'x + A'z) + z(Cz + A'y + B'x) = 0,$$
$$x(Lx + M'z + N'y) + y(My + N'x + L'z) + z(Nz + L'y + M'x) = 0,$$
$$x(Px + Q'z + R'y) + y(Qy + R'x + P'z) + z(Rz + P'y + Q'x) = 0;$$

for, by forgetting the relations between the bracketed and unbracketed letters, we obtain

$$(Ax + B'z + C'y)\left\{\begin{array}{l} (My + N'x + L'z)(Rz + P'y + Q'x) \\ -(Qy + R'x + P'z)(Nz + L'y + M'x) \end{array}\right\}$$
$$+ \&c. + \&c. = 0,$$

which may be put under the form

$$\alpha x^3 + \beta y^3 + \gamma z^3 + \delta x^2 y + \ldots\ldots = 0^*. \qquad (10)$$

* We might dispense with a 10th equation, using the nine above given, to determine the ratios of the ten quantities involved to one another; and then by means of any such relations as

$$x^3y \times xy^3 = x^2y^2 \times x^2y^2, \text{ or } x^3 \times y^3 = x^2y \times xy^2, \ \&c.$$

obtain a determinant. But it is easy to see that this would be made up of terms, each containing literal combinations of the 18th order.

Again, we might use five out of the nine equations to obtain a new equation free from $y^3, y^2z, yz^2, z^3$; that is, containing $x$ in every term: which being divided by $x$, and multiplied

By eliminating linearly between the equations marked from (1) to (10), we obtain as zero a quantity of the twelfth order in all, being of the fourth order in respect to the coefficients of each of the three equations, which is therefore the determinant in its simplest form.

I have purposely, in this brief paper, avoided discussing any theoretical question. I may take some other opportunity of enlarging upon several points which have hitherto been little considered in the theory of elimination, such as the Canons of Form,—the Doctrine of Special Factors,—the Method of Multipliers as extended to a system of any order,—the Connexion between the method of Multipliers and the Dialytic Process,—the Idea of Derivations and of Prime Derivatives extended to ultra-binary Systems. For the present I conclude with the expression of my best wishes for the continued success of this valuable Journal.

by $y$, or by $z$, would furnish a 10th equation no longer linearly involved in the 9 already found. The determinant, however, found in this way, would consist of 14-ary combinations of letters.

Finally, we might, instead of a system of ten equations, employ a system of 15, obtained by multiplying each of the given three by any 5 out of the 6 quantities $x^2$, $y^2$, $z^2$, $xy$, $xz$, $yz$; but the determinant, besides being not *totally* symmetrical, would contain combinations of the 15th order.

I may take this opportunity of just adverting to the fact, that the method in the text does in fact contain a solution of the equation

$$\lambda U + \mu V + \nu W = x^r y^s z^t,$$

where $r+s+t=4$, and $\lambda$, $\mu$, $\nu$ are functions of the second degree in regard to $x$, $y$, $z$ to be determined.

# 13.

## INTRODUCTION TO AN ESSAY ON THE AMOUNT AND DISTRIBUTION OF THE MULTIPLICITY OF THE ROOTS OF AN ALGEBRAIC EQUATION.

[*Philosophical Magazine*, XVIII. (1841), pp. 136—139.]

I USE the word *multiplicity* to denote a number, and distinguish between the total and partial multiplicities of the roots of an algebraic equation.

There may be $r$ different roots repeated respectively $h_1, h_2 \ldots h_r$ times.

$r$ is the index of distribution.

$h_1, h_2 \ldots h_r$ are the partial multiplicities, and if $h = h_1 + h_2 + \ldots + h_r$

$h$ is the *total* multiplicity.

The total multiplicity it is clear may be defined as the difference between the index of the equation and the number of its roots distinguishable from one another.

In this Introduction, I propose merely to consider how existing methods may be applied to determine the amount and distribution of multiplicity in a given equation, and conversely, how equations of condition can be formed which shall imply a *given* distribution and amount.

Let the greatest common factor between $fx$ (the argument of the proposed equation) and $\frac{dfx}{dx}$ be called $f_1x$.

And in like manner, let the greatest common factor of $f_1x$ and $\frac{df_1x}{dx}$ be called $f_2x$ and so on, till in the end we come to $f_rx$, which has no common factor with $\frac{df_rx}{dx}$.

Let $k_1, k_2 \ldots k_r$ denote the degrees in $x$ of $fx, f_1x \ldots f_rx$ respectively.

It is easy to see that

$k_1 - k_2$, partial multiplicities, are less than 2, that is, are each units.

$k_2 - k_3$, partial multiplicities, will be less than 3, and therefore either 1 or 2 in value respectively, and so on till we come to

$k_{r-1} - k_r$ which will severally be between zero and $r - 1$, and

$k_r - 0$ of values intermediate between zero and $r$.

Hence there will be

$k_1 - 2k_2 + k_3$ multiplicities each of the value 1,

$k_2 - 2k_3 + k_4$ „ „ „ 2,

..................................................

$k_{r-1} - 2k_r$ ... of the value $r - 1$,

and $k_r$ ............ of the value $r$.

In place of $fx$ with $\dfrac{dfx}{dx}$ we might employ $\dfrac{dfx}{dx}$ with $\dfrac{d^2fx}{dx^2}$ and so on for the rest; the values of $k_2, k_3 \ldots k_r$ will remain unaffected by this change; but the former method would be more expeditious in practice.

The total multiplicity is, of course, $= k_1$.

Suppose now that we propose to ourselves the converse problem to determine the conditions that an algebraic equation may have a given amount of multiplicity distributed in a given manner.

If $h_1, h_2, h_3 \ldots h_r$ be used to denote the given number of partial multiplicities which are respectively of the values $1, 2, 3 \ldots r$, it is easy to see that the quantities derived above by $k_1, k_2 \ldots k_r$ are respectively equal to

$$h_1 + 2h_2 + \ldots\ldots + rh_r,$$
$$h_2 + 2h_3 + \ldots\ldots + rh_{r-1},$$
$$h_3 + 2h_4 + \ldots\ldots + rh_{r-2},$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$h_r.$$

Now from $\dfrac{dfx}{dx}$ having a factor of the degree $k_1$ common with $fx$ we obtain $k_1$ conditions, from $\dfrac{df_1x}{dx}$ having a factor of the degree $k_2$ common with $f_1x$ we obtain $k_2$ more, and so on. So that altogether we obtain in this way

$$k_1 + k_2 + \ldots\ldots + k_r \text{ conditions.}$$

But it may easily be seen that the total multiplicity being $k_1$, the number of conditions *need* never to exceed $k_1$ in number, no matter what its distribution may be. Hence, besides the enormous labour of the process, and the extreme complexity of the results, we obtain by this method more equations by far than are necessary, and it requires some caution to know which to reject.

In my forthcoming paper (to appear in *Philosophical Magazine* of next month) I shall show, by a most simple means, how without the use of derived or other subsidiary functions, to obtain the simplest equations of condition which correspond to a given distribution of a given amount of multiplicity.

The total multiplicity, say $m$, being given in as many ways as that number can be broken into parts, so many different systems of $m$ equations can be formed differing each from the other in the dimensions of the terms.

These systems may be arranged in order so that each in the series shall imply all those that follow it, and be implied in all those that go before, without the converse being satisfied.

The subject of the unreciprocal implication of systems of equations is a very curious one, upon which the limits assigned to me prevent me from enlarging at present. It is closely connected with a part of the theory of elimination, which, as far as I am aware, has either been overlooked, or has not met with the attention which it deserves; I mean the theory of *Special Factors.*

An *example* may make what I mean by these clear.

Let $C$ be a function (if my reader please) void of $x$, which equivalent to zero implies two given equations in $x$ having a common root.

Let $C$ be rid of all irrelevant factors, that is, let $C$ be the simplest form of the determinant, when the coefficients of the two equations are perfectly independent qualities. Now suppose, as is *quite possible in a variety of ways,* that such relations are instituted between the coefficients alluded to as make $C$ split up into factors, so that $C = L \times M \times N = 0$.

Only one of the factors $L$, $M$, $N$ will satisfy the condition of the co-existence of the two given equations: the others are clearly, however, not to be confounded with factors of solution, or irrelevant factors, as they are termed, but are of quite a different nature, and enjoy remarkable properties, which point to an enlarged theory of elimination, and constitute what I call special or singular factors.

I shall feel much obliged to any of the readers of your widely circulated Journal, interested in the subject of this paper, who would do me the honour of communicating with me upon it, and especially if they would (between now and the next coming out of the *Magazine*) inform me whether anything, and if so how much, different from what is here stated has been done in the matter of determining the relations between the coefficients of an equation corresponding to a given amount and distribution of multiplicity in its roots.

I ought to add, that my method enables me not merely to determine the conditions of multiplicity, but also to decompose the equations containing multiple roots into others free of multiplicity, that is, to find, *à priori*, the values of the several quantities

$$\frac{fx f_2x}{(f_1x)^2}, \; \frac{f_1x f_3x}{(f_2x)^2}, \; \ldots\ldots, \; \frac{f_{r-1}x}{(f_rx)^2}, \; f_rx.$$

Moreover, other decompositions, not necessary to be enlarged upon in this place, may be obtained with equal facility.

# 14.

## A NEW AND MORE GENERAL THEORY OF MULTIPLE ROOTS.

[*Philosophical Magazine,* XVIII. (1841), pp. 249—254.]

I SHALL begin with developing the theory of polynomials containing perfect *square factors,* one or more.

First, let us proceed to determine the relations which must exist between the coefficients of such polynomials, and afterwards show how they may be broken up into others of an inferior degree.

A parallelogram filled with letters standing in *one* row is intended to express the product of the squared difference of the quantities contained. Thus $\left(\underline{\overline{ab}}\right)$ indicates $(a-b)^2$, $\left(\underline{\overline{abc}}\right)$ is used to indicate $(a-b)^2(a-c)^2(b-c)^2$, and so forth.

Suppose now that two of the roots $e_1, e_2 \ldots e_n$ belonging to the equation $fx=0$ are equal to one another, it is clear that $\left(\underline{\overline{e_1, e_2 \ldots e_n}}\right)=0$; and moreover is a symmetric function, and can be calculated in terms of the coefficients of $fx$.

Next let us suppose that we have two couples of equals (as for instance $a$ and $b$, two of the roots equal, as also $c$ and $d$ two others), it is clear, that on leaving any one of the roots out, the $(n-1)$ that are left will still contain one equality, and therefore we have

$$\left(\underline{\overline{e_2, e_3 \ldots e_n}}\right)=0, \quad \left(\underline{\overline{e_1, e_3 \ldots e_n}}\right)=0 \ldots \left(\underline{\overline{e_1, e_2 \ldots e_{n-1}}}\right)=0.$$

None of the parallelogrammatic functions above taken *singly,* are symmetric functions of the coefficients, but their sum is; so also is the sum of the product of each into the quantity left out.

Now in general, suppose that the polynomial $fx$ contains $r$ perfect square factors, so that we have $r$ couples of equal roots belonging to the equation $fx=0$, it is clear that $\left(\underline{\overline{e_r, e_{r+1} \ldots e_n}}\right)$ and all the other $\dfrac{n(n-1)\ldots(n-r+2)}{1.2\ldots(r-1)}$ functions of which it is the type are severally zero. Moreover, the sum of these or the sum of the products of each by *any* symmetrical function of the $(r-1)$ letters left out will be a symmetrical function of the coefficients of the powers of $x$ in $fx$. To express now the *affirmative** conditions corresponding to the case of there being $r$ pairs of equal roots, we *might* employ the $r$ equations,

$$\left(\overline{e_1, e_2 \ldots e_n}\right) = 0,$$

$$\Sigma \left(e_2, \overline{e_1 \ldots e_n}\right) = 0,$$

$$\Sigma \left(\overline{e_3 \ldots e_n}\right) = 0,$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$\Sigma \left(e_r, \overline{e_{r+1} \ldots e_n}\right) = 0.$$

But these, except the last, are not the *simplest* that can be employed; that is to say, we can write down $r$ others, the terms of which shall be of lower dimensions in respect to the roots.

Let $\int_\mu$ denote that any rational symmetrical function of the $\mu$th degree is to be taken of the quantities which it precedes.

Then the $r$ equations in question are all contained in the general equation

$$\Sigma \left\{\int_\mu (e_1, e_2 \ldots e_{r-1}) \times \left(e_r, \overline{e_{r+1} \ldots e_n}\right)\right\} = 0;$$

$\mu$ being taken from 0 up to $(r-1)$ we obtain $r$ equations, which in respect to the roots are respectively of all degrees between

$$\frac{n(n-1)\ldots(n-r+2)}{1.2\ldots(r-1)} \text{ and } \frac{n(n-1)\ldots(n-r+2)}{1.2\ldots(r-1)} + (r-1)$$

reckoned inclusively.

Now at this stage it is important to remark that the above $r$ equations, although *necessary*, are not *sufficient*; and indeed, no mere affirmations of equality can be sufficient to ensure there being $r$ pairs of equal roots.

To make this manifest, suppose $r=2$. Then in order that an equation *may* have two pairs of equal roots, we must have by the above formula

$$\Sigma \left(e_2, \overline{e_3 \ldots e_n}\right) = 0, \quad \Sigma \left\{e_1 \left(e_2, \overline{e_3 \ldots e_n}\right)\right\} = 0.$$

But if instead of there being two perfect square factors there be one perfect *cube* factor in $fx$, it may be shown by the same reasoning as above, that the very same two equations apply. In fact, it may be shown in general that no such equations as those given above can be *affirmed* in consequence of there being an amount $r$ of multiplicity consisting of unit parts which may not be affirmed with equal truth as necessary consequences of the same

* The importance of the restriction hinted at by the use of the word affirmative will appear hereafter.

amount distributed *in any other manner* whatever. How to obtain affirmative equations sufficient as well as necessary (under certain limitations) will appear at the close of this present paper.

It is worthy of being remarked, that if we make $\int_\mu$ denote the sum of the products of the quantities to which it is prefixed, taken $\mu$ and $\mu$ together, the equations of affirmation become identical with those obtained by eliminating between $fx$ and $\frac{dfx}{dx}$ *.

It can scarcely be doubted that the illustrious Lagrange, had he chosen to perfect the incomplete theory of equal roots given in the *Résolution Numérique*, by applying to it his own favourite engine of symmetric functions, could scarcely have failed of stumbling by a back passage upon Sturm's memorable theorem.

Let us now proceed to show how a polynomial known to contain one or more perfect square factors may be decomposed.

Let us begin with supposing that it contains but one such factor; so that $fx = \phi x (x-a)^2$.

I shall show how to obtain the equations

$$C(x-a)=0, \quad D\phi x (x-a)=0, \quad E(x-a)^2=0, \quad F(\phi x)=0,$$

each in its lowest terms.

1. To form the equation $Lx + M = 0$, where $x=a$, it is easy to see that if we write down in general the expression $(x-e_1)\left(\overline{e_2, e_3 \dots e_n}\right)$ this will become zero whenever the root $e_1$ left out is not one of the equal roots $(a)$: so that in fact (calling the two equal roots $e_1$, $e_2$ respectively)

$$\Sigma\left\{(x-e_1)\times\left(\overline{e_2, e_3 \dots e_n}\right)\right\} = (x-e_1)\times\left(\overline{e_2, e_3 \dots e_n}\right) + (x-e_2)\times\left(\overline{e_1, e_3 \dots e_n}\right),$$

or simply $$= 2(x-a)\left(\overline{e_2, e_3 \dots e_n}\right).$$

Hence by making

$$x\Sigma\left(\overline{e_2, e_3 \dots e_n}\right) - \Sigma\left\{e_1 \times \left(\overline{e_2, e_3 \dots e_n}\right)\right\} = 0,$$

we have an equation for finding the equal roots $e_1$, $e_2$.

Again, it is easily seen upon the same hypothesis, that

$$\Sigma\left\{(x-e_2)(x-e_3)(x-e_4)\dots(x-e_n)\times\left(\overline{e_2, e_3 \dots e_n}\right)\right\}$$
$$= 2(x-e_2)(x-e_3)\dots(x-e_n)\times\left(\overline{e_2, e_3 \dots e_n}\right).$$

* See my note on Sturm's Theorem, *Phil. Mag.*, December, 1839 [p. 45 above. Ed.].

Hence, to form the equation having the same roots as $(x-a)\,\phi x$, we have only to make

$$x^{n-1}\Sigma\left(\overline{\underline{e_2, e_3 \dots e_n}}\right) - x^{n-2}\Sigma\left\{(e_2+e_3+\dots e_n)\times\left(\overline{\underline{e_2, e_3 \dots e_n}}\right)\right\}\dots\dots$$
$$\pm\,\Sigma\left\{(e_2e_3\dots e_n)\times\left(\overline{\underline{e_2, e_3 \dots e_n}}\right)\right\}=0.$$

Suppose now in general that we have $r$ perfect square factors, so that

$$fx = \phi x\,(x-a_1)^2(x-a_2)^2\dots(x-a_r)^2.$$

To form the equation $C(x-a_1)(x-a_2)\dots(x-a_r)=0$, we have only to make

$$\Sigma\left\{(x-e_1)(x-e_2)\dots(x-e_r)\times\left(\overline{\underline{e_{r+1}, e_{r+2}\dots e_n}}\right)\right\}=0.$$

And to obtain

$$D\phi x\times(x-a_1)(x-a_2)\dots(x-a_r)=0,$$

we must make

$$\Sigma\left\{(x-e_{r+1})(x-e_{r+2})\dots(x-e_n)\times\left(\overline{\underline{e_{r+1}, e_{r+2}\dots e_n}}\right)\right\}=0.$$

The theory of perfect square factors is not yet complete until it has been shown how to obtain constructively $\phi x$, and, as analogy suggests, the complementary part $D'(x-a_1)^2(x-a_2)^2\dots(x-a_r)^2$, each in its lowest terms. To effect the latter it might be said that it is only necessary to take the square of $C(x-a_1)(x-a_2)\dots(x-a_r)$. It is true the polynomial so formed would contain every pair of equal factors, but not in the lowest terms as regards the coefficients (as we shall presently show).

To solve this last part of the problem, let it be agreed that two rows of letters inclosed in a parenthesis shall indicate the product of the squares of the differences got by subtracting each in the row from each in the other, so that

$$\begin{pmatrix}a\\b\end{pmatrix}=(a-b)^2,\quad \begin{pmatrix}a & \\ b & c\end{pmatrix}=(a-b)^2(a-c)^2,\quad \begin{pmatrix}a & b\\ c & d\end{pmatrix}=(a-c)^2(a-d)^2(b-c)^2(b-d)^2.$$

Let us begin with supposing that $fx$ has one pair only of equal roots; to form the simplest quadratic equation containing this pair, write down

$$(x-e_1)(x-e_2)\times\left(\overline{\underline{e_3, e_4\dots e_n}}\right)\times\begin{pmatrix}e_1, e_2 & \\ e_3, e_4 & \dots e_n\end{pmatrix}.$$

Now if $e_1$ and $e_2$ are the two equal roots in question neither of the multipliers of $(x-e_1)(x-e_2)$ vanishes.

If $e_1$ and $e_2$ are neither of them equal roots $\left(\overline{\underline{e_3, e_4\dots e_n}}\right)=0$.

If one of the two only belong to the pair of equal roots

$$\begin{pmatrix}e_1, e_2 & \\ e_3, e_4 & \dots e_n\end{pmatrix}=0.$$

Hence it is clear that

$$\Sigma \left\{(x-e_1)(x-e_2)\times\left(\overline{e_3, e_4 \ldots e_n}\right)\times\begin{pmatrix} e_1, e_2 & \\ e_3, e_4 \ldots e_n \end{pmatrix}\right\}=0$$

is the equation desired.

In like manner if there be $r$ pairs of equal roots the equation of the $(2r)$th degree which contains them all may be written

$$\Sigma \left\{(x-e_1)(x-e_2)\ldots(x-e_{2r})\times\left(\overline{e_{2r+1} \ldots e_n}\right)\times\begin{pmatrix} e_1, e_2 \ldots e_{2r} \\ e_{2r+1} \ldots e_n \end{pmatrix}\right\}=0.$$

The coefficient of $x^{2r}$ in this equation is clearly of

$$(n-2r)(n-2r-1)+4r(n-2r),$$

that is, of $(n+2r-1)(n-2r)$ dimensions. The coefficient of $x^r$ in the equation which contains the $r$ equal roots unyoked together is of $(n-r)(n-r-1)$ dimensions, and consequently the coefficient of $x^{2r}$ in the square of this equation would be of $2(n-r)(n-r-1)$ dimensions, that is, would be $n^2+6r^2-(4r+1)n$ dimensions higher than needful.

Finally, to obtain an equation clear of *simple* as well as double appearances of the equal roots, we have only to write the complementary form

$$\Sigma \left\{(x-e_{2r+1})(x-e_{2r+2})\ldots(x-e_n)\times\left(\overline{e_{2r+1}+e_n}\right)\times\begin{pmatrix} e_1, e_2 \ldots e_{2r} \\ e_{2r+1} \ldots e_n \end{pmatrix}\right\}=0.$$

Let us, now that we are more familiarized with the notation essential to this method, revert to the question with which we set out, and endeavour to obtain $r$ such equations as shall imply *unambiguously* the existence of $r$ pairs of equal roots.

The existence of $r$ such pairs enables us to assert the following disjunctive proposition, which cannot be asserted when the *same amount* of multiplicity is distributed in any other way.

To wit, on selecting any $r$ roots out of the entire number, either these $r$ will all be found again in those that are left, *or* those that are left will contain *inter se*, one repetition at least; so that except on the latter supposition any $(r-1)$ may be absolutely sunk out of those that are left, and there will still be *one* root common to the $(n-2r+1)$ remaining, and to the $r$ originally selected to be left out.

Wherefore calling the roots $e_1, e_2 \ldots e_n$, and giving $\mu$ any value whatever, we have

$$\Sigma \left\{\int_\mu (e_1, e_2 \ldots e_r)\times\left(\overline{e_{r+1}, e_{r+2} \ldots e_n}\right)\times\Sigma\begin{pmatrix} e_1, e_2 \ldots e_r & \\ e_{2r}, e_{2r+1} \ldots e_n \end{pmatrix}\right\}=0.$$

Hence the simplest distinctive equations indicative of the existence of $r$ pairs of equal roots are to be found by putting $\mu$ equal in succession to all values from 0 up to $(r-1)$.

For instance, if we require that an equation of the seventh degree shall have three pairs of equal roots, we need only to call the seven roots respectively $a, b, c, d, e, f, g$, and then our type equation becomes

$$\Sigma\left\{\int_\mu (a\,b\,c) \times (\overline{d\,e\,f\,g}) \times \left\{\begin{matrix} \begin{pmatrix} d\;e \\ a\;b\;c \end{pmatrix} + \begin{pmatrix} d\;f \\ a\;b\;c \end{pmatrix} + \begin{pmatrix} d\;g \\ a\;b\;c \end{pmatrix} \\ + \begin{pmatrix} e\;f \\ a\;b\;c \end{pmatrix} + \begin{pmatrix} e\;g \\ a\;b\;c \end{pmatrix} + \begin{pmatrix} f\;g \\ a\;b\;c \end{pmatrix} \end{matrix}\right\}\right\} = 0.$$

From this it appears that the $r$ distinctive equations for $r$ pairs of equal roots are of different dimensions from the $r$ general or overlying ones corresponding to the multiples $r$, anyhow distributed; the lowest of the latter being of $(n-r+1)(n-r)$, the lowest of the former of

$$(n-r)(n-r-1) + 2r(n-2r+1),$$

that is, of $n(n-1) - 3r(n-1)$ dimensions. In general we shall find that the more unequally distributed the multiplicity may be the lower are the dimensions of the distinctive equations, and are accordingly lowest when the multiplicity is absolutely undistributed*.

* It must not, however, be overlooked, that the equations above given, although decisive as to the existence of $r$ pairs of equal roots *when* the multiplicity is known to be not greater than $r$, do not enable us to affirm with certainty their existence when this limitation is absent: for should the multiplicity exceed $r$, then inevitably (no matter how it may be distributed) $(\overline{e_{r+1}, e_{r+2} \dots e_n})$ is always zero, and consequently nullifies each term of every one of the equations in question. In fact (repugnant as it may appear to be to the ordinary assumptions of analytical reasoning), it is not possible to express with *absolute* unambiguity the conditions of there being a multiplicity $(r)$ distributed in any assigned manner by means of $r$ *affirmative equations alone.*

# 15.

## ON A LINEAR METHOD OF ELIMINATING BETWEEN DOUBLE, *TREBLE*, AND OTHER SYSTEMS OF ALGEBRAIC EQUATIONS.

[*Philosophical Magazine*, XVIII. (1841), pp. 425—435.]

### Part I. Binary Systems.

Let $U$ and $V$ be two integer complete homogeneous functions of $x$ and $y$, one of the $m$th, the other of the $n$th degree; and let it be required to express the condition of the coexistence of the two equations $U=0$, $V=0$ by means of the equation $C=0$, where $C$ is free from all appearances of $x$ or $y$.

This equation, according to the system of notation developed in a preceding paper, and which has been since adopted and sanctioned by the high authority of M. Cauchy, I call the final derivative: the quantity $C$ is designated the final derivee: and it is our present object to show how this may be obtained in a *prime* form, that is to say, divested of irrelevant factors: in this state it must consist of terms, each containing $m+n$ letters, of which $n$ belong to the coefficients of $U$, and $m$ to those of $V$.

Of course in applying this rule it is to be understood that every combination of powers in $U$ or $V$ has a single letter prefixed for its coefficient, and that in the final derivee powers are represented by repetitions of the same character.

Every term in $U$ or $V$ being of the form $Cx^p y^q$, $x^p y^q$ is called an argument, $C$ its prefix.

Assume two integer positive numbers $r$ and $r'$, and also two others $s$ and $s'$, such that $r+r'=n-1$, $s+s'=m-1$, and form from $U=0$, $V=0$ two new equations,

$$x^r y^{r'} U=0, \quad x^s y^{s'} V=0.$$

Such equations are termed the augmentatives of the two given ones respectively; also $x^r y^{r'} U$ and its fellow are termed the augmentees of $U$ and $V$.

$r$ and $r'$ are termed the indices of augmentation belonging to $U$, $s$ and $s'$ the same belonging to $V$.

Finally, it will be useful hereafter to call the given polynomials $U$ and $V$ themselves the proposees, and the given equations which assert their nullity, the propositive equations, or, briefly, the propositives.

Now as many augmentees of either proposee can be formed as there are ways of stowing away between two lockers (vacancies admissible) a number of things equal to the index of the other*; hence we shall have $n$ augmentees of $U$, and $m$ of $V$: thus there will be $m+n$ augmentatives each of the degree $m+n-1$, and the number of arguments is clearly $m+n$ also, so that they can be eliminated linearly, and the final derivee thus found, containing $m+n$ letters (properly aggregated) in each term, will be in its prime form, that is, incapable of further reduction, and void of irrelevant factors.

It is worthy of remark, that the final derivee obtained by arranging in square battalion the prefixes of the augmentees, permuting the rows or columns, and reading off diagonal products, affected each with the proper sign (according to the well known rule of Duality), will not only be free from factorial irrelevancy, but also of linear redundancy, which latter term I use to signify the reappearance of the same combination of prefixes, sometimes with positive and sometimes with negative signs: furthermore, it follows obviously from the nature of the process that no numerical quantity in the final derivee will be greater than the higher of the indices of the two given polynomials.

## Part II. Ternary Systems.

### Case A. *Indices all equal.*

#### *Method* 1.

Let there be now three proposees, $U$, $V$, $W$, integer complete homogeneous functions of $x$, $y$, $z$, each of the degree $n$: let

$$r+r'+r''=n-1, \quad s+s'+s''=n-1, \quad t+t'+t''=n-1,$$

$$x^r y^{r'} z^{r''} U, \ x^s y^{s'} z^{s''} V, \ x^t y^{t'} z^{t''} W,$$

will, as above, be called the augmentees of $U$, $V$, $W$, and every other part of the notation previously described is to be preserved.

* "Tot Augmenta utriusvis ex æquationibus propositis formari possunt quot modi sint inter duo receptacula (utrivis vel ambobus omnino vacare licet) rerum, quarum numerus indicem alterius æquat, distributionem faciendi."

Suppose now

$$U = 0, \quad V = 0, \quad W = 0,$$

we shall have as many augmentative equations formed from each proposee as there are ways of stowing away $n$ things between *three* lockers (vacancies admissible)*, that is, $n\frac{n+1}{2}$ of each kind; in all, therefore, $3\frac{n(n+1)}{2}$, and every one of these will be of the degree $2n-1$, so that the number of arguments to be eliminated is equal to the number of ways of stowing away $2n-1$ things between three lockers (empty ones counting), that is

$$\frac{2n(2n+1)}{2}.$$

As yet, then, we have not *enough* equations for eliminating these linearly.

Make, however,

$$\alpha + \beta + \gamma = n + 1,$$

and write

$$U = x^\alpha F + y^\beta F' + z^\gamma F'' = 0,$$
$$V = x^\alpha G + y^\beta G' + z^\gamma G'' = 0,$$
$$W = x^\alpha H + y^\beta H' + z^\gamma H'' = 0,$$

it will always be possible to make the multipliers of $x^\alpha$, $y^\beta$, $z^\gamma$ integer functions: for if we look to any argument in $U$, $V$, or $W$, it is of the form $x^a y^b z^c$, and one of the letters $a$, $b$, $c$ must be *not less* than its correspondent $\alpha$, $\beta$, $\gamma$, for otherwise $a+b+c$ would be not greater than $\alpha+\beta+\gamma-3$, that is, $n$ would be not greater than $(n+1)-3$, or $n-2$, which is absurd: if now any one, as $a$, be equal to or greater than $\alpha$, it may be made to supply an integer part to the multiplier of $x^\alpha$.

Here it may be asked what is to be done with such terms as $Kx^a y^b z^c$, when two letters $a$, $b$ are each not less than their correspondents $\alpha$, $\beta$: the answer is, such terms may be made to enter under the multiplier of $x^\alpha$, or of $x^\beta$, or to supply a part to both in any proportion at pleasure†.

From the equations above we get, by linear elimination,

$$FG'H'' + GH'F'' + HF'G'' - GF'H'' - HG'F'' - FH'G'' = 0.$$

This may be denoted thus: $\Pi(\alpha, \beta, \gamma) = 0$, which equation I call a *secondary* derivative, and the left side of it a *secondary* derivee; $\alpha$, $\beta$, $\gamma$ may likewise be termed the indices of derivation (as $r$, $s$, $t$, &c. are of augmentation).

Now since $\alpha + \beta + \gamma = n + 1$, it is clear that the index of $\Pi(\alpha, \beta, \gamma)$ is always $n + n + n - (n+1)$; that is, $2n - 1$.

* See for Latin translation the preceding note.

† The prefixes of any such terms (say $K$) may be conceived as made up of two parts, an arbitrary constant, as $e$ and $(K-e)$; $e$ will disappear spontaneously from the final derivee.

1st. Let any two of the indices of derivation be taken zero, then it is easily seen that all the terms in $\Pi(\alpha, \beta, \gamma)$ vanish, and consequently the secondary derivative equations obtained upon this hypothesis become mere identities, and are of no use.

2nd. Let any *one* of them become zero.

It is manifest, from the doctrine of simple equations, that $\Pi(\alpha, \beta, \gamma)$ may be made equal to

$$\left\{\lambda U + \mu V + \nu W\right\} \frac{1}{x^{\alpha}},$$

or

$$\left\{\lambda' U + \mu' V + \nu' W\right\} \frac{1}{x^{\beta}},$$

or

$$\left\{\lambda'' U + \mu'' V + \nu'' W\right\} \frac{1}{x^{\gamma}},$$

upon the understanding that

$$\begin{aligned} \lambda &= G'H'' - G''H', & \mu &= H'F'' - H''F', & \nu &= F'G'' - F''G', \\ \lambda' &= G''H - GH'', & \mu' &= H''F - HF'', & \nu' &= F''G - FG'', \\ \lambda'' &= GH' - G'H, & \mu'' &= HF' - H'F, & \nu'' &= FG' - F'G. \end{aligned}$$

The three rows of coefficients will be respectively of the degrees

$$(n-\beta) + (n-\gamma), \quad (n-\gamma) + (n-\alpha), \quad (n-\alpha) + (n-\beta).$$

Thus if any one of the indices $\alpha$, $\beta$, $\gamma$ be zero, $\Pi(\alpha, \beta, \gamma)$ becomes identical with $\lambda^{?} U + \mu^{?} V + \nu^{?} W$, where the multipliers of $U$, $V$, $W$ are of $2n - (\alpha + \beta + \gamma)$ dimensions, that is of $(n-1)$ dimensions, and may accordingly be put under the form

$$\Sigma A x^{r} y^{r'} z^{r''} U + \Sigma B x^{s} y^{s'} z^{s''} V + \Sigma C x^{t} y^{t'} z^{t''} W,$$

that is to say, becomes a *linear function* of the augmentatives, and therefore if combined with them in the process of linear elimination would *give rise* to the identity $0 = 0$.

Hence we must reject all such secondary derivatives as have zero for one of the indices of derivation. But all others, it may be shown, will be *linearly* independent of one another, and of the augmentees previously found. Hence, besides $3\dfrac{n(n+1)}{2}$ equations of augment of the degree $2n-1$, we shall have of the same degree so many equations of derivation as there are ways of stowing away between three lockers $(n+1)$ things, under the condition that no locker shall ever be left *empty*, that is $\dfrac{n(n-1)}{2}$ *.

Thus, then, in all we have $n\dfrac{n-1}{2} + 3\dfrac{n(n+1)}{2} = \dfrac{2n(2n+1)}{2}$ equations, which is exactly equal to the number of arguments to be eliminated. Hence

* Vide page 76 for the Latin version.

the final derivee can be obtained by the usual explicit rule of permutation, and moreover will be *its lowest form,* for it will contain in each term $\frac{n(n+1)}{2}$ prefixes belonging to the augmentatives of $U$, and a like number pertaining to those of $V$ and of $W$, as well as $n\frac{n-1}{2}$ belonging to the secondary derivatives, each prefix in any one of which is triliteral, containing a prefix drawn out of those belonging to each of the proposees.

Thus every member containing $n\frac{n+1}{2}+n\frac{n-1}{2}$, that is $n^2$ of the original prefixes belonging to $U$, $V$, $W$, singly and respectively, the final derivee evolved by this process will be in its lowest terms; as was to be proved.

CASE A. *Indices all equal.*

*Method* 2.

It is remarkable that we may vary the method just given by making

$$r+r'+r''=n-2, \quad s+s'+s''=n-2, \quad t+t'+t''=n-2.$$

The augmentatives will thus be of the degree $2n-2$.

Furthermore, we must make $\alpha+\beta+\gamma=n+2$. It will still be possible to satisfy by integer multipliers the equations

$$U=x^\alpha F+y^\beta F'+z^\gamma F'',$$
$$V=x^\alpha G+y^\beta G'+z^\gamma G'',$$
$$W=x^\alpha H+y^\beta H'+z^\gamma H'',$$

[these it will be useful in future to term the *equations,* $x^\alpha$, $y^\beta$, $z^\gamma$ being the *arguments,* and $F$, $G$, $H$, &c. the *factors* of decomposition] for otherwise calling the indices of $x$, $y$, $z$ in any original argument $a$, $b$, $c$, their sum or $n$ would be not greater than $(n+2)-3$, that is $(n-1)$, which is absurd.

For the same reasons as in the last case no index of augmentation must be made zero: the degree of each will be $(n-\alpha)+(n-\beta)+(n-\gamma)$, that is $(2n-2)$, and their number $\frac{(n+1)n}{2}$; the number of augmentatives will be $\frac{3(n-1)n}{2}$ linearly uninvolved, each of the degree $2n-2$, and therefore containing $\frac{(2n-1)2n}{2}$ arguments.

Now $$\frac{(n+1)n}{2}+\frac{3(n-1)n}{2}=\frac{(2n-1)2n}{2}.$$

Hence the final derivee may be found, and it will be in its *lowest terms*, for every member will contain $\frac{3(n-1)n}{2}$ letters due to the augmentative, and $\frac{3(n+1)n}{2}$ due to the partial derivative equations; in all then there will be $3n^2$ letters in each term.

This second method being applied to three quadratic equations of the most general form, leads to the problem of eliminating between six simple equations which lies within the limits of practical feasibility, and it is my intention to register the final derivee upon the pages of some one of our scientific Transactions as a standing monument for the guidance of hereafter coming explorers*.

## Scholium to Case A.

If we attempt to carry forward these processes to quaternary systems, it becomes necessary to make

$$\alpha+\beta+\gamma+\delta=(r-2)n+1$$

or else

$$\alpha+\beta+\gamma+\delta=(r-2)n+2,$$

where $r$ is the number of proposees.

Now if the factors in the equations of decomposition are all integer, one of the indices of derivation must be not greater than the corresponding index in any of the original arguments, which may easily be shown to be always impossible for a system of equations, *complete* in *all* their terms, whenever their number $r$ is greater than three, if $\alpha+\beta+\gamma+\delta=(r-2)n+2$; but if $\alpha+\beta+\gamma+\delta=(r-2)n+1$ only possible for the case of $n=2$.

## Particular method applicable to *four Quadratics*.

Let $U=0$, $V=0$, $W=0$, $Z=0$, be four quadratic equations existing between $x$, $y$, $z$, $t$.

Make

$$\begin{array}{cccc} xU=0, & xV=0, & xW=0, & xZ=0, \\ yU=0, & yV=0, & yW=0, & yZ=0, \\ zU=0, & zV=0, & zW=0, & zZ=0, \\ tU=0, & tV=0, & tW=0, & tZ=0. \end{array}$$

* Elimination between *two* quadratics leads to a final derivee made up of *seven* terms only; the final derivee of *three* quadratics is made up of at least several *thousand*; nay, I believe I may safely say, several *myriads* of terms!

Also write
$$U = x^2F + yF' + zF'' + tF''' = 0,$$
$$V = x^2G + yG' + zG'' + tG''' = 0,$$
$$W = x^2H + yH' + zH'' + tH''' = 0,$$
$$Z = x^2K + yK' + zK'' + tK''' = 0.$$

By eliminating linearly we get
$$\Sigma\,\{F\Sigma G'(H''K''' - H'''K'')\} = 0,$$
which will be of the third degree, since the factors represented by the *unmarked* letters $F$, $G$, $H$, $K$ are of zero, and all the rest of *unit* dimensions.

Similarly we may obtain other equations, so that besides the *sixteen* augmentatives already written down, we have four secondary derivatives, namely,
$$\Pi(2\ 111) = 0, \quad \Pi(12\ 11) = 0, \quad \Pi(11\ 21) = 0, \quad \Pi(111\ 2) = 0.$$
Thus we have *twenty* equations and as many arguments to eliminate, since a perfect cubic function of four letters contains twenty terms.

The final derivee will contain $16 + 4 . 4$ letters, that is 32, 8 or $2^3$ belonging to each system of original prefixes in each member, and will therefore be in its lowest terms: for one of the canons of form teaches us, *à priori*, that every member of the derivee deduced from any number of assumed equations must contain in each member as many prefixes belonging to one equation of the system as there are units in the product of the indices of all the rest taken together.

## Corollary to Case A.

Either of the two methods given as applicable to this case enables us to determine integer values of $X$, $Y$, $Z$, which shall satisfy the equation
$$XU + YV + ZW = Fx^py^qz^r,$$
where $F$ is the final derivee and $p + q + r = 3n - 2$. For by the doctrine of simple equations we know how to express $F$ in terms of the linear functions, out of which it is obtained by permutation, that is we are able to assign values of $A$, $B$, $C$, and their antitypes, as also of $L$ and its antitype, which shall satisfy the equation
$$\Sigma\,(Ax^ry^{r'}z^{r''}U) + \Sigma\,(Bx^sy^{s'}z^{s''}V) + \Sigma\,(Cx^ty^{t'}z^{t''}W) + \Sigma\,\{L\Pi(\alpha, \beta, \gamma)\} = Fx^fy^gz^h, \qquad (1)$$
where $A$, $B$, $C$, as well as $L$ and all the quantities formed after them, are made up of integer combinations of the original prefixes.

Now the functions $\Pi(\alpha, \beta, \gamma)$ may be expressed in three ways in terms of $U$, $V$, $W$, as has been already shown.

We may therefore suppose these functions to be divided into three groups, and make

$$\Sigma L \Pi (\alpha\beta\gamma) = \Sigma \frac{QU + Q'V + Q''W}{x^\alpha} + \Sigma \frac{RU + R'V + R''W}{x^\beta} + \Sigma \frac{SU + S'V + S''W}{x^\gamma}. \qquad (2)$$

And it is evident that the equations (1) and (2) lead immediately to the equation

$$XU + YV + ZW = Fx^{a+f} y^{b+g} z^{c+h},$$

if we call $a$, $b$, $c$ the *greatest values* attributed respectively to $\alpha$, $\beta$, $\gamma$.

Now if we suppose the first method to be followed,

$$f + g + h = 2n - 1.$$

And it will always be possible to make $a$, $b$, $c$ of what values we please *subject* to the condition of $a + b + c = n - 1$; for *one* at least of the indices of derivation in $\Pi (\alpha, \beta, \gamma)$ must be not greater than its correspondent among $a, b, c$; otherwise $\alpha + \beta + \gamma$ would be not less than $(a + b + c) + 3$; but

$$\alpha + \beta + \gamma = n + 1$$

$$a + b + c = n - 1,$$

which is absurd.

Hence we can satisfy $XU + YV + ZW = Fx^p y^q z^r$, $p$, $q$, $r$ being subject to the condition of $p + q + r = 3n - 2$, but otherwise arbitrary.

Moreover, we can *not* do so if $p + q + r$ be *less* than $3n - 2$, for that would require $a + b + c$ to be less than $n - 1$. Now if two of the indices of derivation, as $\alpha$ and $\beta$, be made equal to $a + 1$, $b + 1$ respectively, the third $\gamma = (n + 1) - (a + b + 2) = (n - 1) - (a + b)$, and is therefore greater than $c$: so that $\alpha + \beta + \gamma$ for this case becomes greater than $a + b + c$, and the method falls to the ground.

In fact, I have discovered a theorem which lets me know this, *à priori*, a law which serves as a staff to guide my feet from falling into error in devising linear methods of solution, and the importance of which all candid judges who have studied the general theory of elimination cannot fail to recognize. To wit, if $X_1$, $X_2$, $X_3 \ldots X_n$ be $n$ integer complete polynomial functions of $n$ letters $x_1$, $x_2 \ldots x_n$, and severally of the degree $b_1$, $b_2$, $b_3 \ldots b_n$; then it is always possible to satisfy the identity

$$P_1X_1 + P_2X_2 + P_3X_3 + \ldots + P_nX_n = Fx_1^{a_1}x_2^{a_2}x_3^{a_3} \ldots x_n^{a_n},$$

if $\alpha_1 + \alpha_2 + \alpha_3 + \ldots + \alpha_n$ be equal to or greater than $b_1 + b_2 + b_3 + \ldots + b_n - n + 1$, but otherwise *not**.

This again is founded immediately upon a simple proposition, of which I have obtained a very interesting and instructive demonstration, shortly to appear, and which may be enumerated thus: "*The number of augmentees of the same degree that can be formed, linearly independent of one another, out of any number of polynomial functions of as many variables, may be either equal to or less than the number of distinct arguments contained in such augmentees, but* never *greater. The latter will be the case when the index of the augmentees diminished by unity is less than the sum of the indices of the original unaugmented polynomials each so diminished; the former, when the aforesaid index is equal to or greater than the aforesaid sum.*"

To return to the particular case of finding $X$, $Y$, $Z$ to satisfy

$$XU + YV + ZW = Fx^p y^q z^r.$$

This has been already done according to the first method; if we employ the *second* method of elimination we shall have

$$f + g + h = 2n - 2.$$

But, now since $\alpha + \beta + \gamma = n + 2$, we shall easily see by the same method as above, that the least value of $a + b + c$ {where $a$, $b$, $c$ denote respectively the greatest values of $\alpha$, $\beta$, $\gamma$, appearing in the denominator of the fractional *forms* used to express $\Pi(\alpha, \beta, \gamma)$}, will be one greater than before, or $n$; so that $f + g + h + a + b + c$ will still be equal to $3n - 2$, as we might, *à priori*, by virtue of our rule, have been assured.

## Ternary Systems.

Case B. *Two of the indices equal; the third* less *by a unit.*

Let $U = 0$, $V = 0$, $W = 0$, be the three given equations severally of the degree $n$, $n$, $(n - 1)$.

* Hence it is apparent, that in applying the method of multipliers, a curious and important distinction exists between the cases of there being two equations, and there being a greater number to eliminate from: for in the first case the element of arbitrariness needs never to appear; in the *latter* it cannot possibly be excluded from appearing in the multipliers.

This will explain how it comes to pass that the method of the text may be employed to give *various* solutions of the $XU + YV + ZW = Fx^p y^q z^r$; thus not only can $p$, $q$ and $r$ be variously made up of $(f + a)$, $(g + b)$, $(h + c)$, but also $\Pi(\alpha, \beta, \gamma)$ when two of the indices ($\alpha$, $\beta$ suppose) are each not greater than the assigned greatest values $a$, $b$ may be made to figure indifferently either under the form

$$\frac{\lambda U + \mu V + \nu W}{x^\alpha} \text{ or that of } \frac{\lambda' U + \mu' V + \nu' W}{x^\beta}.$$

Make $r + r' + r'' = n - 2, \quad s + s' + s'' = n - 2, \quad t + t' + t'' = n - 1,$

by multiplying $U$ into $x^r y^{r'} z^{r''}$, $V$ into $x^s y^{s'} z^{s''}$, $W$ into $x^t y^{t'} z^{t''}$, we obtain augmentees each of the same, namely, the $(2n-2)$th degree.

The number of these is

$$\frac{(n-1)n}{2} + \frac{(n-1)n}{2} + \frac{n(n+1)}{2}.$$

Again, make $\alpha + \beta + \gamma = n + 1.$

It will still be possible, as before, to form equations of decomposition in which $x^\alpha$, $y^\beta$, $z^\gamma$ are the arguments, and affected with *integer* factors. For if we look to $W$ *even*, all its arguments are of the form $x^a y^b z^c$, where $a + b + c = (n-1)$, and each of these cannot be less than its correspondent, for that would be to say that $(n-1)$ is not greater $(n+1)-3$, *à fortiori*, $U$ and $V$ can be decomposed in the manner described. Thus, then, we shall obtain as many secondary derivees as in the last case (Method 1), that is, $\frac{n(n-1)}{2}$ {since $\alpha + \beta + \gamma$ is still equal to $(n+1)$}, as before. Moreover, each of these will be of $(n-\alpha) + (n-\beta) + (n-1-\gamma)$, that is of $2n-2$ dimensions.

Altogether, therefore, we have

$$\left\{\frac{(n-1)n}{2} + \frac{(n-1)n}{2} + \frac{n(n+1)}{2}\right\} + \frac{(n-1)n}{2}$$

linear independent equations of the degree $2n-2$, and the number of arguments to eliminate is $\frac{(2n-1)2n}{2}$. Now these two numbers are equal. Thus we obtain a final derivee containing of $U$'s coefficients $\frac{(n-1)n}{2} + \frac{(n-1)n}{2}$, an equal number of $V$'s, but of $W$'s $\frac{n(n+1)}{2} + \frac{(n-1)n}{2}$; now $n(n-1)$, $n(n-1)$ and $n^2$ exactly express the number that ought to appear of each of these respectively: hence the final derivee is clear *of irrelevant factors*.

## Ternary Systems.

Case C. *Two of the indices equal; the third one* greater *by a unit.*

Here, calling $n$ the highest index, the augmentees must each be made of the degree $(2n-3)$, their number will evidently be

$$\frac{(n-2)(n-1)}{2} + \frac{(n-1)n}{2} + \frac{(n-1)n}{2},$$

making the sum of the indices of derivation now, as before, equal to $(n+1)$; it will be still possible to form integer equations of decomposition, which will give rise to augmentatives of the degree $(n-\alpha)+(n-1)-\beta+(n-1)-\gamma$, that is, of $(2n-3)$ dimensions. The total number of equations, what with augmentatives and secondary derivatives, will be

$$\left\{\frac{(n-2)(n-1)}{2}+\frac{(n-1)n}{2}+\frac{(n-1)n}{2}\right\}+\frac{n(n-1)}{2}=\frac{4n^2-4n+2}{2}=\frac{(2n-2)(2n-1)}{2},$$

that is, is equal to the exact number of distinct arguments contained between them.

Also the final derivative will contain in each member

$$\frac{(n-2)(n-1)}{2}+\frac{n(n-1)}{2},$$

that is, $(n-1)(n-1)$, letters belonging to the first equation, and

$$\frac{(n-1)n}{2}+\frac{n(n-1)}{2},$$

that is, $n(n-1)$ belonging to those of the second and of the third, and will therefore be in its *lowest terms.*

### Corollary to Cases B and C.

It is not necessary, after all that has been already said, to do more than just point out that the processes applicable to these cases enable us to determine $X$, $Y$, $Z$, which satisfy the equation

$$XU+YV+ZW=Fx^fy^gz^h,$$

where $$f+g+h=3n-3 \text{ for Case B},$$

and $$f+g+h=3n-4 \text{ for Case C}.$$

# 16.

## MEMOIR ON THE DIALYTIC METHOD OF ELIMINATION. PART I.

[*Philosophical Magazine*, XXI. (1842), pp. 534—539*.]

THE author confines himself in this part to the treatment of two equations, the final and other derivees of which form the subject of investigation.

The author was led to reconsider his former labours in this department of the general theory by finding certain results announced by M. Cauchy in *L'Institut*, March Number of the present year, which flow as obvious and immediate consequences from Mr Sylvester's own previously published principles and method.

Let there be two equations in $x$,

$$U = ax^n + bx^{n-1} + cx^{n-2} + ex^{n-3} + \&\text{c.} = 0,$$

$$V = \alpha x^m + \beta x^{m-1} + \gamma x^{m-2} + \&\text{c.} \qquad = 0,$$

and let $n = m + \iota$, where $\iota$ is zero or any positive value (as may be).

Let any such quantities as $x^r U$, $x^s V$, be termed augmentatives of $U$ or $V$.

To obtain the derivee of a degree $s$ units lower than $V$, we must join $s$ augmentatives of $U$ with $s + \iota$ of $V$. Then out of $2s + \iota$ equations

$$x^0 U = 0, \quad x^1 U = 0, \quad x^2 U = 0, \ldots\ldots\, x^{s-1} U = 0,$$

$$x^0 V = 0, \quad x^1 V = 0, \quad x^2 V = 0, \ldots\ldots\, x^{\iota+s-1} V = 0,$$

we may eliminate linearly $2s + \iota - 1$ quantities.

Now these equations contain no power of $x$ higher than $m + \iota + s - 1$; accordingly, all powers of $x$, superior to $m - s$, may be eliminated, and the derivee of the degree $(m - s)$ obtained in its prime form.

Thus to obtain the final derivee (which is the derivee of the degree zero), we take $m$ augmentatives of $U$ with $n$ of $V$, and eliminate $(m + n - 1)$ quantities, namely,

$$x, \ x^2, x^3, \ldots\ldots \text{ up to } x^{m+n-1}.$$

* Reprinted from *Proc. Roy. Irish Acad.*, Vol. II. (1840—1844), p. 130.

This process, founded upon the dialytic principle, admits of a very simple modification. Let us begin with the case where $\iota = 0$, or $m = n$. Let the augmentatives of $U$ be termed $U_0$, $U_1$, $U_2$, $U_3$, ... and of $V$, $V_0$, $V_1$, $V_2$, $V_3$, ..., the equations themselves being written

$$U = ax^n + bx^{n-1} + cx^{n-2} + \&c.$$

$$V = a'x^n + b'x^{n-1} + c'x^{n-2} + \&c.$$

It will readily be seen that

$$a'U_0 - aV_0,$$

$$(b'U_0 - bV_0) + (a'U_1 - aV_1),$$

$$(c'U_0 - cV_0) + (b'U_1 - bV_1) + (a'U_2 - aV_2), \&c.$$

will be each linearly independent functions of $x$, $x^2$, ... $x^{m-1}$, no *higher* power of $x$ remaining. Whence it follows, that to obtain a derivee of the degree $(m - s)$ in its prime form, we have only to employ the $s$ of those which occur first in order, and amongst them eliminate $x^{m-1}$, $x^{m-2}$, ... $x^{m-s+1}$. Thus, to obtain the *final* derivee, we must make use of $n$, that is, the entire number of them.

Now, let us suppose that $\iota$ is not zero, but $m = n - \iota$. The equation $V$ may be conceived to be of $n$ instead of $m$ dimensions, if we write it under the form

$$0x^n + 0x^{n-1} + 0x^{n-2} + \dots + 0x^{m+1} + \alpha x^m + \beta x^{m-1} + \&c. = 0,$$

and we are able to apply the same method as above; but as the first $\iota$ of the coefficients in the equation above written are zero, the first $\iota$ of the quantities

$$(a'U_0 - aV_0), \quad (b'U_0 - bV_0) + (a'U_1 - aV_1), \&c.$$

may be read simply

$$-aV_0, \quad -bV_0 - aV_1, \quad -cV_0 - bV_1 - aV_2, \&c.$$

and evidently their office can be supplied by the simple augmentatives themselves,

$$V_0 = 0, \quad V_1 = 0, \quad V_2 = 0 \dots V_{\iota-1} = 0;$$

and thus $\iota$ letters, which otherwise would be *irrelevant*, fall out of the several derivees.

The author then proceeds with remarks upon the general theory of simple equations, and shows how by virtue of that theory his method contains a solution of the identity

$$X_r U + Y_r V = D_r$$

where $D_r$ is a derivee of the $r$th degree of $U$ and $V$, and accordingly, $X_r$ of the form

$$\lambda + \mu x + \nu x^2 + \dots + \theta x^{m-r-1},$$

and $Y_r$ of the form

$$l + mx + \dots + tx^{n-r-1},$$

and accounts *à priori* for the fact of not more than $(n-r)$ simple equations being required for the determination of the $(m+n-2r)$ quantities $\lambda$, $\mu$, $\nu$, &c. $l$, $m$, $n$, &c., by exhibiting these latter as *known* linear functions of no more than $(n-r)$ unknown quantities left to be determined.

Upon this remarkable relation may be constructed a method well adapted for the expeditious computation of numerical values of the different derivees.

He next, as a point of curiosity, exhibits the values of the secondary functions,

$$a'U_0 - aV_0,$$

$$b'U_0 - bV_0 + a'U_1 - aV_1,$$

$$c'U_0 - cV_0 + b'U_1 - bV_1 + a'U_2 - aV_2, \text{ \&c.}$$

under the form of symmetric functions of the roots of the equations $U=0$, $V=0$, by aid of the theorems developed in the London and Edinburgh *Philosophical Magazine*, December 1839*, and afterwards proceeds to a more close examination of the final derivee resulting from two equations each of the same (any given) degree.

He conceives a number of cubic blocks each of which has two numbers, termed its *characteristics*, inscribed upon one of its faces, upon which the value of such a block (itself called an *element*) depends.

For instance, the value of the *element*, whose *characteristics* are $r$, $s$, is the difference between two products: the one of the coefficient $r$th in order occurring in the polynomial $U$, by that which comes $s$th in order in $V$; the other product is that of the coefficient $s$th in order of the polynomial $U$, by that $r$th in order of $V$; so that if the degree of each equation be $n$, there will be altogether $\frac{1}{2}n(n+1)$ such elements.

The blocks are formed into squares or flats (*plafonds*) of which the number is $\frac{n}{2}$ or $\frac{n+1}{2}$, according as $n$ is even or odd. The first of these contains $n$ blanks in a side, the next $(n-2)$, the next $(n-4)$, till finally we reach a square of four blocks or of one, according as $n$ is even or odd. These flats are laid upon one another so as to form a regularly ascending pyramid, of which the two diagonal planes are termed the planes of separation and symmetry respectively. The former divides the pyramid into two halves, such that no element on the one side of it is the same as that of any block in the other. The plane of symmetry, as the name denotes, divides the pyramid into two exactly *similar* parts; it being a rule, that *all elements lying in any given line of a square* (*plafond*) *parallel to the plane of separation are identical*; moreover, the sum of the characteristics is the same, for *all* elements lying *anywhere* in a *plane* parallel to that of separation.

[* p. 40 above. Ed.]

All the terms in the final derivee are made up by multiplying $n$ elements of the pile together, under the sole restriction, that no two or more terms of the said product shall lie in any one plane out of the two *sets* of planes perpendicular to the sides of the squares. The *sign* of any such product is determined by the places of either set of planes parallel to a side of the squares and to one another, in which the elements composing it may be conceived to lie.

The author then enters into a disquisition relating to the *number* of terms which will appear in the final derivee, and concludes this first part with the statement of two general canons, each of which affords as many tests for determining whether a prepared combination of coefficients can enter into the final derivee of any number of equations as there are units in that number, but so connected as together only to afford double that number, *less* one, of independent conditions.

The first of these canons refers simply to the number of letters drawn out of *each of the given equations* (supposed homogeneous); the second to what he proposes to call the *weight* of every term in the derivee in respect to *each of the variables* which are to be eliminated.

The author subjoins, for the purpose of conveying a more accurate conception of his Pyramid of derivation, examples of the mode in which it is constructed.

When $n = 1$ there is one flat, viz.

| |
|---|
| 1, 2 |

When $n = 2$ there is one flat, viz.

| | |
|---|---|
| 2, 3 | 2, 4 |
| 2, 4 | 3, 4 |

Let $n = 3$, there will be two flats:

| |
|---|
| 2, 3 |

| | | |
|---|---|---|
| 1, 2 | 1, 3 | 1, 4 |
| 1, 3 | 1, 4 | 2, 4 |
| 1, 4 | 2, 4 | 3, 4 |

Let $n = 4$, there will still be two flats only:

| | |
|---|---|
| 2, 3 | 2, 4 |
| 2, 4 | 3, 4 |

| | | | |
|---|---|---|---|
| 1, 2 | 1, 3 | 1, 4 | 1, 5 |
| 1, 3 | 1, 4 | 1, 5 | 2, 5 |
| 1, 4 | 1, 5 | 2, 5 | 3, 5 |
| 1, 5 | 2, 5 | 3, 5 | 4, 5 |

Let $n = 5$, there will be three flats:

| |
|---|
| 3, 4 |

| | | |
|---|---|---|
| 2, 3 | 2, 4 | 2, 5 |
| 2, 4 | 2, 5 | 3, 5 |
| 2, 5 | 3, 5 | 4, 5 |

| | | | | |
|---|---|---|---|---|
| 1, 2 | 1, 3 | 1, 4 | 1, 5 | 1, 6 |
| 1, 3 | 1, 4 | 1, 5 | 1, 6 | 2, 6 |
| 1, 4 | 1, 5 | 1, 6 | 2, 6 | 3, 6 |
| 1, 5 | 1, 6 | 2, 6 | 3, 6 | 4, 6 |
| 1, 6 | 2, 6 | 3, 6 | 4, 6 | 5, 6 |

Let $n = 6$, there will be three flats:

| | |
|---|---|
| 3, 4 | 3, 5 |
| 3, 5 | 4, 5 |

| | | | |
|---|---|---|---|
| 2, 3 | 2, 4 | 2, 5 | 2, 6 |
| 2, 4 | 2, 5 | 2, 6 | 3, 6 |
| 2, 5 | 2, 6 | 3, 6 | 4, 6 |
| 2, 6 | 3, 6 | 4, 6 | 5, 6 |

| | | | | | |
|---|---|---|---|---|---|
| 1, 2 | 1, 3 | 1, 4 | 1, 5 | 1, 6 | 1, 7 |
| 1, 3 | 1, 4 | 1, 5 | 1, 6 | 1, 7 | 2, 7 |
| 1, 4 | 1, 5 | 1, 6 | 1, 7 | 2, 7 | 3, 7 |
| 1, 5 | 1, 6 | 1, 7 | 2, 7 | 3, 7 | 4, 7 |
| 1, 6 | 1, 7 | 2, 7 | 3, 7 | 4, 7 | 5, 7 |
| 1, 7 | 2, 7 | 3, 7 | 4, 7 | 5, 7 | 6, 7 |

Thus the work of computation reduces itself merely to calculating $n\frac{n+1}{2}$ elements, or the $n(n+1)$ cross-products out of which they are constituted, and combining them factorially after that law of the pyramid, to which allusion has been already made.